

BCH 301: ENZYMOLOGY

**B.Sc BIOCHEMISTRY AND MICROBIOLOGY
STUDENTS LECTURE**

2019/2020 SESSION

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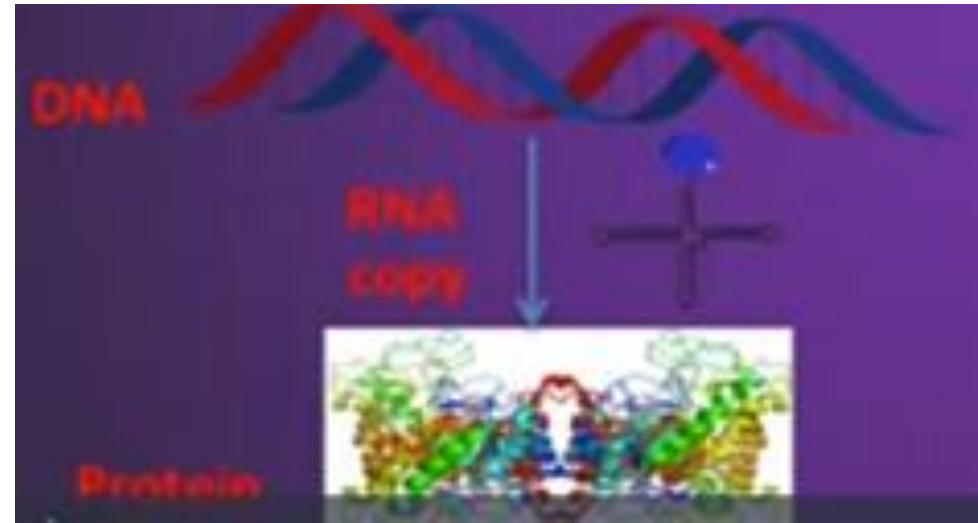
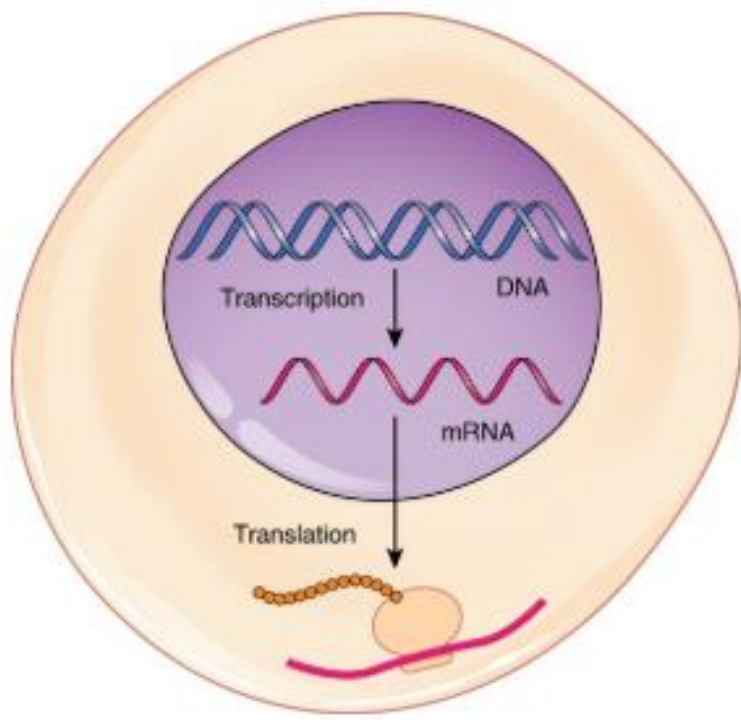
28TH JANUARY 2021

ENZYME OVERVIEW AND GENERAL INTRODUCTION

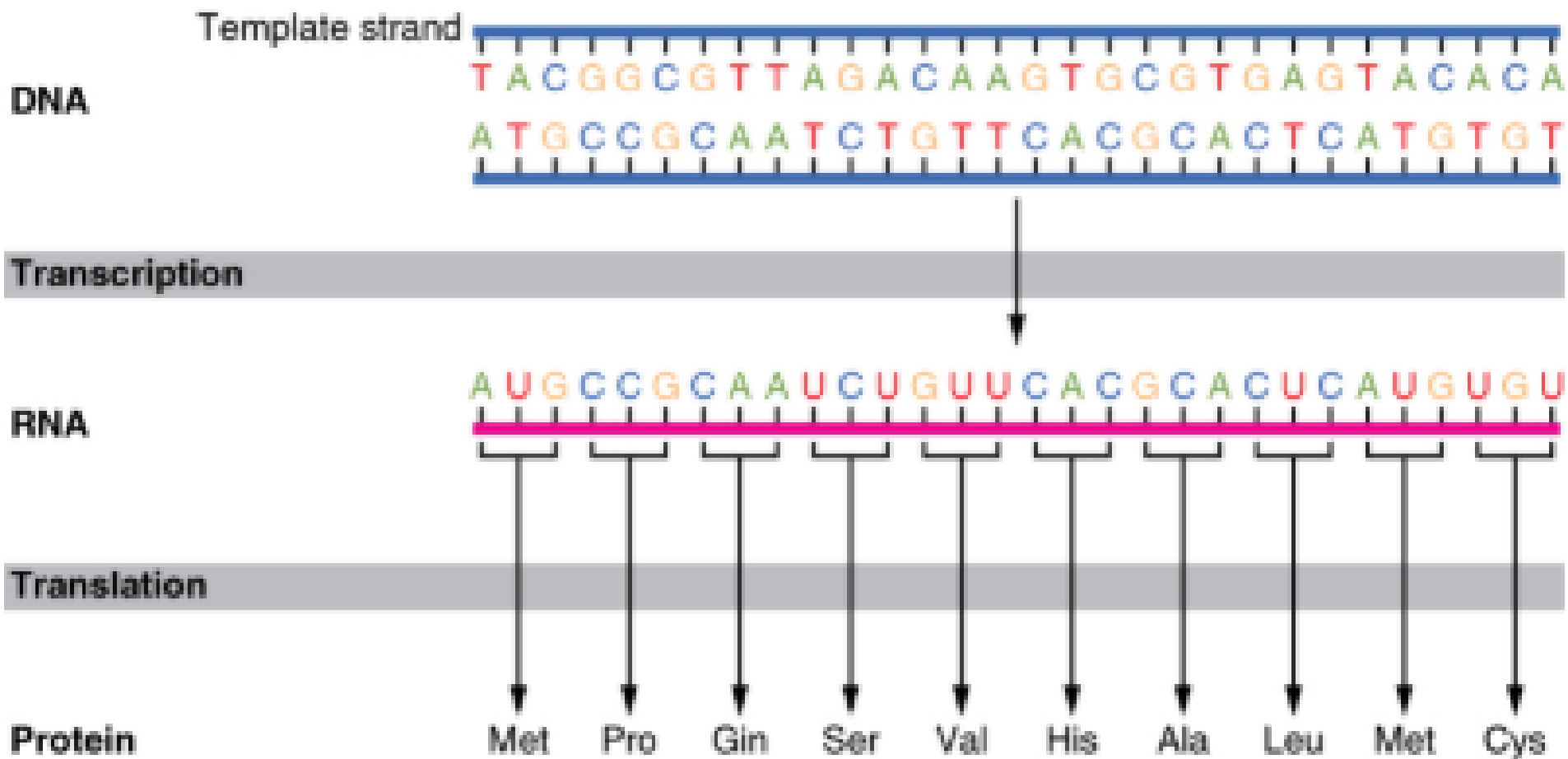
- Enzymology is the study of enzyme activities or functions including their structure, kinetics, and their interaction with other factors.
- Enzymes are biological catalysts that increases or speed up the rate of biochemical reactions taking place within living cells without themselves undergoing any permanent or overall change in the process.
- Biochemical reactions are the processes used for the formation, breakdown and rearrangement of molecules in living organisms leading to:

- build-up of new tissues,
 - replacement of old tissues,
 - conversion of food into energy,
 - disposal of toxic materials,
- Enzymes catalyze nearly all the chemical reactions taking place in the cells of the body.
- They accelerate the rate of both forward and backward reactions.
- Impairment of enzymes and their activities are reflected by some changes in the cell or even death. There can be no life without enzymes.

- Enzymes are synthesized or produced by living organisms:
 - animal,
 - plant
 - microorganism
- This is done through transcription and translation.
- Transcription in the nucleus produces mRNA copy of the DNA segment coding for a specific protein using RNA polymerase which is then transported into the cytoplasm for translation to proteins.
- Translation in the ribosome reads the transcript and produce amino acids in the correct sequence for each specific protein.



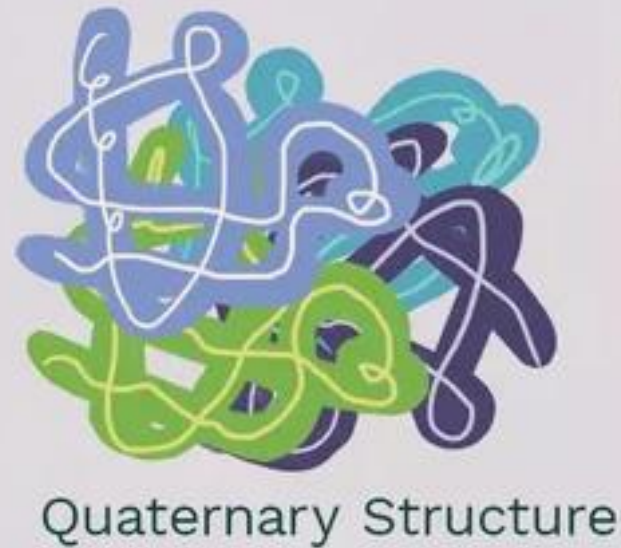
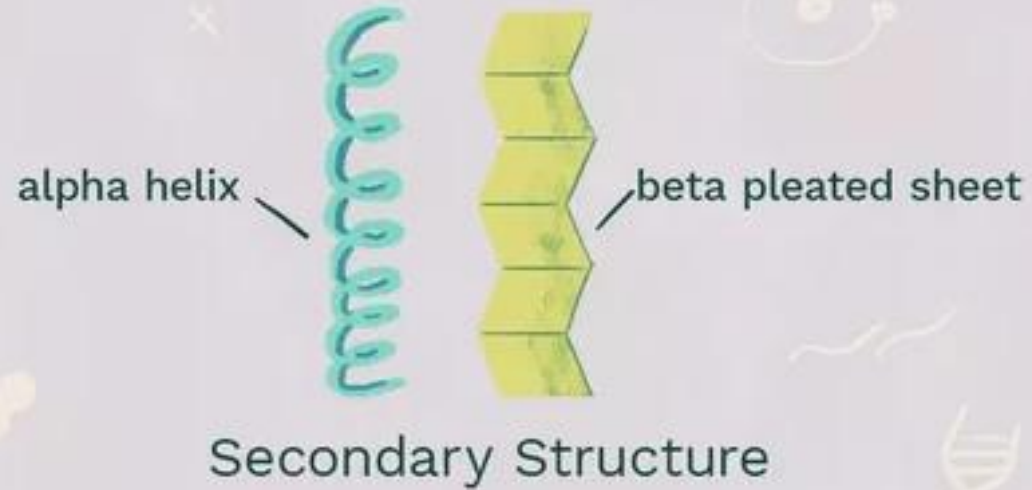
From DNA to Protein through transcription and translation.
The genetic code is universal such that a gene from one organism can be transcribed and translated in another organism.



DNA holds all of the genetic information necessary to build a cell's proteins. The nucleotide sequence of a gene is ultimately translated into amino acid sequence of the gene's corresponding protein.

- The DNA encoded sequence of amino acids gives four different structure of proteins:
 - Primary structure
 - Secondary structure
 - Tertiary structure
 - Quaternary
- Enzymes are proteins, they are at the quaternary level of protein structure
- This is obtained through folding

Types of Protein Structures



- Ordinary chemical reactions convert reactants to products but enzyme catalyzed reactions convert a **SUBSTRATE** or **SUBSTRATES** to products.
- That is the molecule that is bound by the enzyme and converted to product is called a **substrate**.
- Some enzymes require no chemical groups for activity other than their amino acid residues in their active site.
 - Such enzymes are referred to as simple enzymes because they consists only of protein in their active form.
- Usually only a few of the 20 amino acid residues or side chains participate directly in enzyme catalysis.
- The best examples of polar amino acids participants in catalysis are Cys, His, Ser, Asp, Glu and Lys.

- Some enzymes may require an additional chemical compound called **cofactor for proper function**.
 - That is such enzymes are active only when they combine with their **cofactors**.
- A cofactor is a small non-protein molecules that is bound (either tightly or loosely) to an enzyme and is required for catalysis.
- **Cofactors** are either
 - inorganic ions or
 - small organic molecules
- Therefore we can also define a cofactor as an inorganic ion or a small organic molecule that aid in enzyme catalysis.

Introduction cont...

- The common metal ions cofactors required for complete enzyme activity include Mg^{2+} , Cu^{2+} , Fe^{2+} , Fe^{3+} , Ca^{2+} , Zn^{2+} , Mn^{2+} , Ni^{2+} etc.
- Metal ions in the active site are attached to one or more amino acid side-chains.
- The metal ions have various functions, such as electron exchange and substrate stabilization.

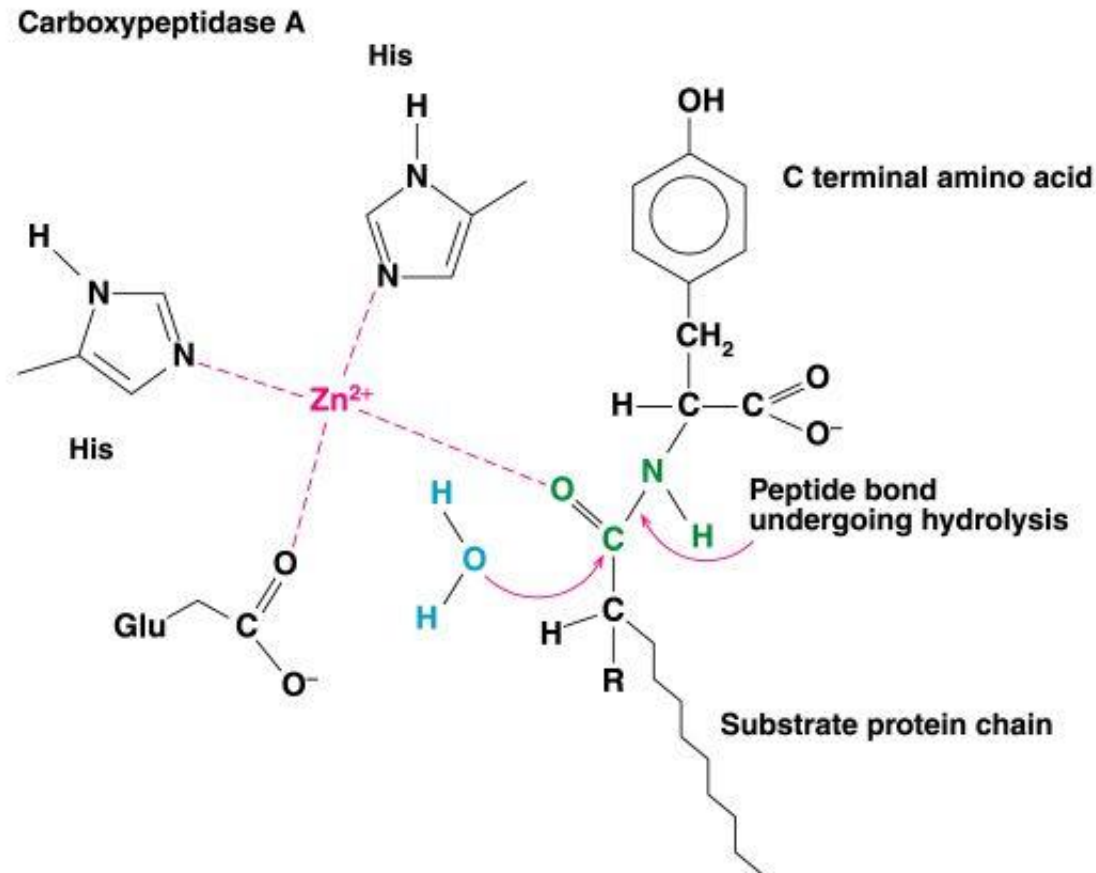
Table 21.8 Enzymes and the Metal Ions Required as Cofactors

Metal Ion Cofactor	Function	Enzyme
Cu^{2+}	Oxidation-reduction	Cytochrome oxidase
$\text{Fe}^{2+}/\text{Fe}^{3+}$	Oxidation-reduction	Catalase
	Oxidation-reduction	Cytochrome oxidase
Zn^{2+}	Used with NAD^{+}	Alcohol dehydrogenase
		Carbonic anhydrase
		Carboxypeptidase A
Mg^{2+}	Hydrolyzes phosphate esters	Glucose-6-phosphatase
Mn^{2+}	Removes electrons	Arginase
Ni^{2+}	Hydrolyzes amides	Urease

Introduction cont...

- Essential metal ions that serve as cofactors in enzymatic reactions are either loosely or tightly bound to the enzyme molecule.
- Essential metal ions that bind reversibly to enzyme and often participate in substrate binding are referred to as **Activator ions**.
- Whereas essential metal ions that are tightly bound to the enzyme molecule lead to the formation of **metalloenzymes**.
 - That is metalloenzymes contain firmly bound metal ions in their active sites (examples: Fe, Zn, Cu, Co).
 - The cations in metalloenzymes participate directly in catalysis.

- For example, a Zn^{2+} ion in the active site of carboxypeptidase A promotes hydrolysis of a C-terminal amino acid from a polypeptide by interacting with the carbonyl oxygen
- The Zn^{2+} activates the carbonyl in a similar way as an acid catalyst



- Small organic molecules that serve as cofactors in enzymatic reactions are called **Coenzymes**.
 - **Coenzymes are** either loosely or tightly bound to the enzyme molecule.
- The chemical groups transported by coenzymes are
 - hydride ion ($\text{H}^+ + 2e^-$) carried by NAD,
 - mole of hydrogen carried by FAD; and
 - amide ($-\text{NH}_2$) carried by pyridoxal phosphate.
- **Coenzymes** prepare the active site for proper substrate binding and/or participate in catalysis.
- They are required in small quantities and are not destroyed during the reaction.
- A coenzyme or a metal ion that is covalently bound to the enzyme protein is called a prosthetic group

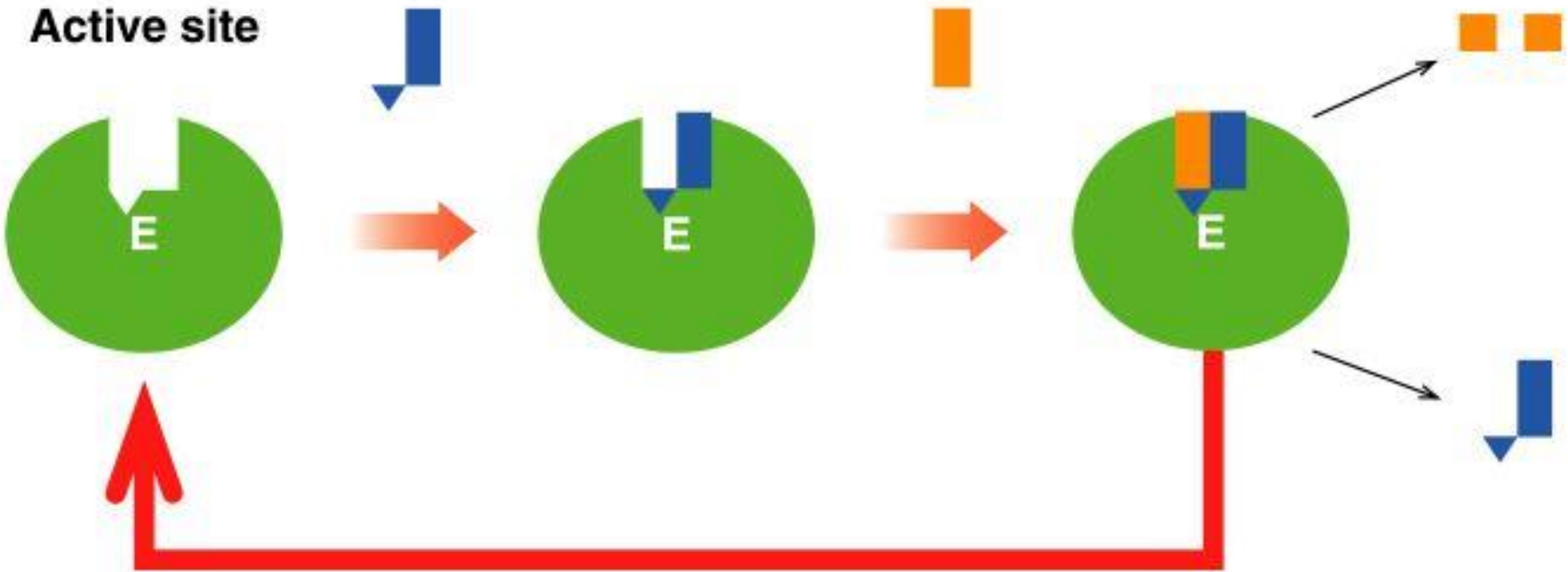
- The protein component of an enzyme without its necessary prosthetic group is called an APOENZYME or apoprotein.
- Whereas the active enzyme with its needed prosthetic group bound or covalently attached is called the HOLOENZYME
 - i.e. a completely catalytically active enzyme together with its cofactor.
- Holoenzyme = Apoenzyme + Cofactor
 (active) (inactive) (inactive)
- The functional unit of an enzyme is referred to as a holoenzyme. It is often made up of an apoenzyme (the protein part) and a coenzyme (the non-protein part).
- A zymogen or proenzyme is the precursor form of the enzyme in an inactive form
- An activator is any substance that increases the rate of an enzyme-catalyzed reaction.

Coenzyme

Substrate

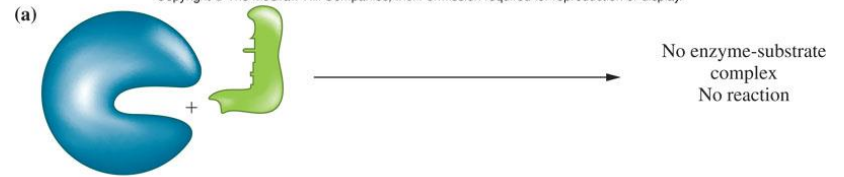
Products

Active site

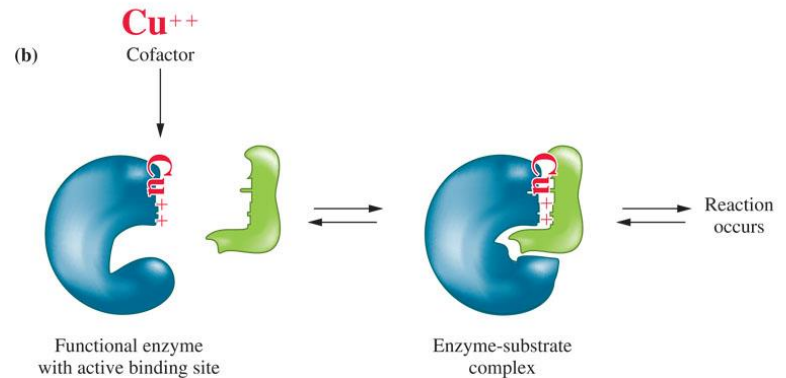


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Apoenzyme + substrate



Functional enzyme
with active binding site

Enzyme-substrate
complex

- Coenzymes use in enzyme catalyzed reactions are metabolically active forms of vitamins.
- The word vitamin comes from the Greek word “VITAMINE” which means vital for life
 - vita = life
 - amine = containing nitrogen (the first vitamins discovered contained nitrogen)
- Vitamins are organic compounds or molecules required in the diet in **very small quantities** for metabolism, normal growth and maintenance of health but **cannot be synthesized** in the body either at all or in adequate quantities.

VITAMIN

- This implies that vitamins must be synthesized by some organisms
 - Plants and
 - Bacteria
- Although the body is able to produce part or even all of its requirements for some of the vitamins examples:
 - vitamin D from cholesterol and
 - niacin from tryptophan.
- However, vitamins must be provided in the diet.

Classification of vitamins

Water soluble

- Thiamin (B_1)
- Riboflavin (B_2)
- Niacin (B_3)
- Pantothenic acid (B_5)
- Pyridoxine (B_6)
- Biotin (B_7) or Vitamin H
- Folic acid B_9 or Vitamin M
- Cobalamin (B_{12})
- Ascorbic acid(Vitamin C)

Fat soluble

- Vitamin A
- Vitamin D
- Vitamin E
- Vitamin K

Vitamins

Fat soluble

- Vitamin A
- Vitamin D
- Vitamin E
- Vitamin K

Water soluble

Non B-complex

Vitamin C

B-complex

Energy-releasing

- Thiamine (B₁)
- Riboflavin (B₂)
- Niacin (B₃)
- Pyridoxine (B₆)
- Biotin (B₇)
- Pantothenic acid (B₅)

Hematopoietic

- Folic acid (B₉)
- Vitamin B₁₂
(cyanocobalamin)

General properties of water soluble vitamins

- They differ chemically from each other
- They are soluble in water
- Subject to cooking losses
- They are readily excreted in the urine
 - Therefore toxicity as a result of overdose is rare
- They are not stored in large quantities in the body (except B12)
 - therefore they must be supplied continually in the diet

- The are usually converted to the metabolically active form called coenzyme which participate in biochemical reactions i.e.
 - many coenzymes are derived from **water-soluble vitamins**
- All of the water-soluble vitamins and two of the fat-soluble vitamins, A and K, function as cofactors or **coenzymes**.
- **Coenzymes** participate in numerous biochemical reactions involving energy release or catabolism, as well as the accompanying anabolic reactions.

Sources

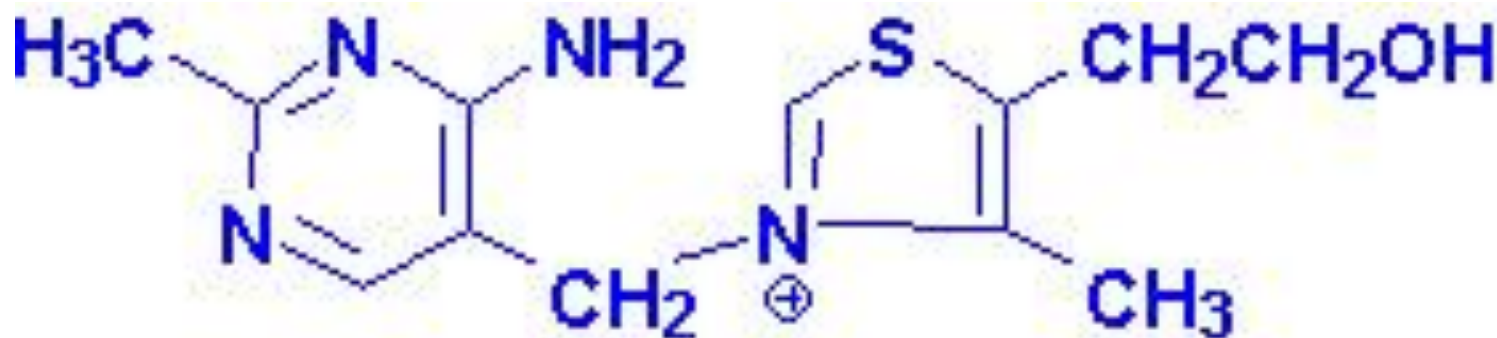
Thiamine (B₁)	Unpolished Rice, Cereals and Whole wheat (brown) bread.
Riboflavin (B₂)	All types of food (No deficiency due to diet)
Niacin (B₃)	Cereals, nuts, milk, meat, fish (Synthesized endogenously from Tryptophan)
Pyridoxine (B₆)	Meat, Fish, Poultry, Cereals, non-citrus fruits, Potatoes etc.
Ascorbic Acid (Vit. C)	Citrus Fruits (Lemon and Oranges)
Folic Acid	Green Leafy Vegetables.
Vitamin B₁₂	Meat and Meat Products.

Thiamin

- Thiamin is also known as vitamin B1 .
- It was the first B vitamin to be identified
- It is a derivative of substituted pyrimidine and a thiazole linked by a methylene bridge.
- Its biologically active form is thiamin pyrophosphate, TPP
- TPP is formed in the brain and liver by the enzyme, Thiamin diphosphotransferase.
- Dietary sources include whole grains, milk products and yeast

Structure of thiamin

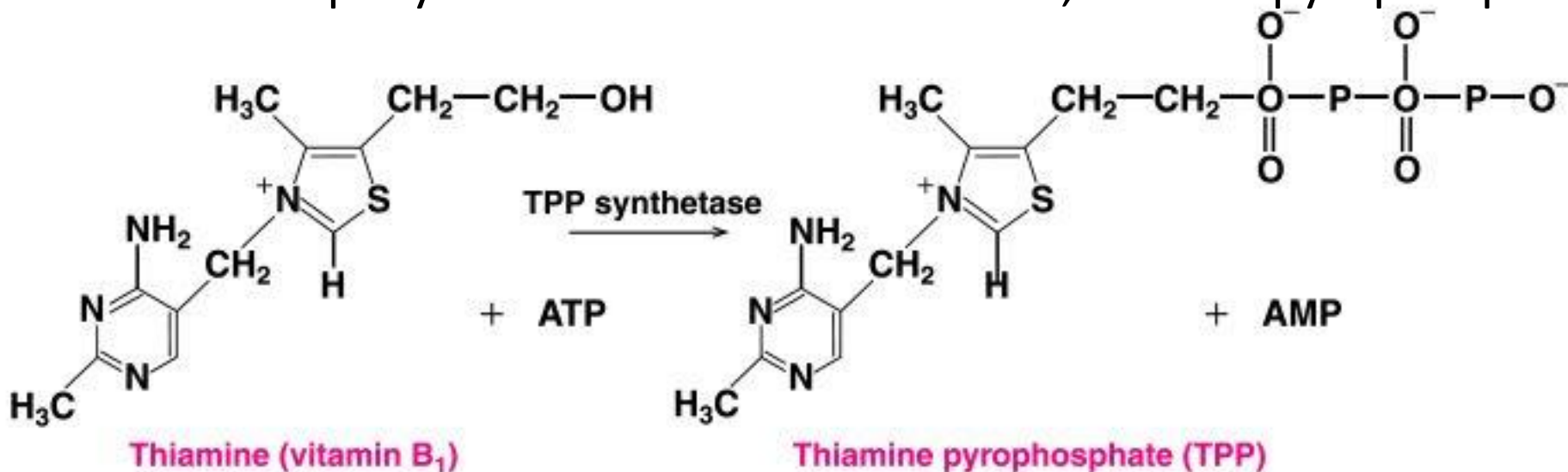
ring methyl bridge



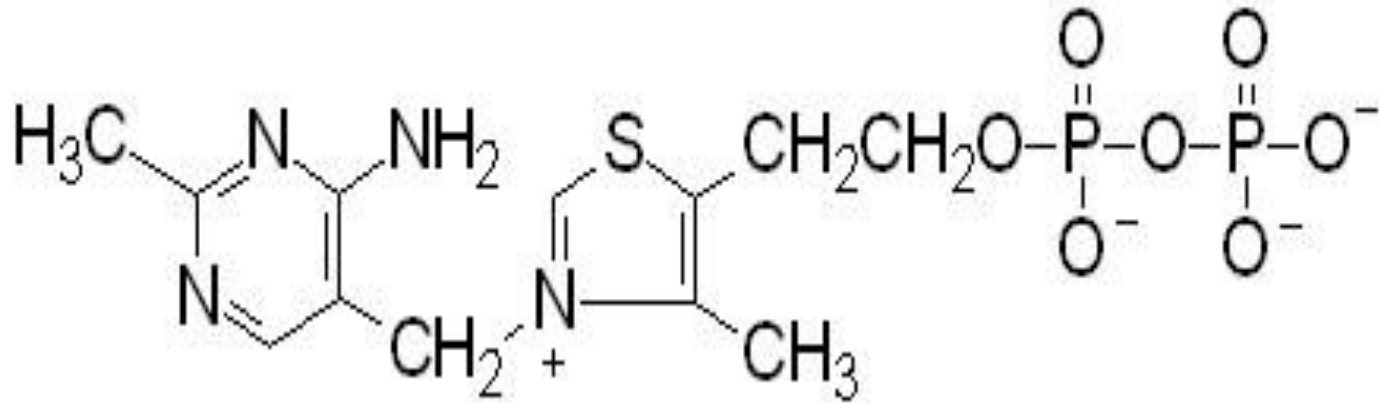
Pyrimidine ring

Thiazole ring

Thiamin is rapidly converted to its active form, thiamine pyrophosphate



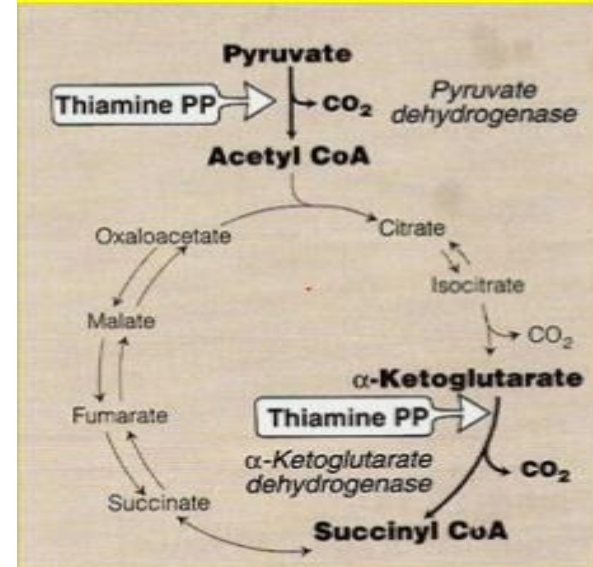
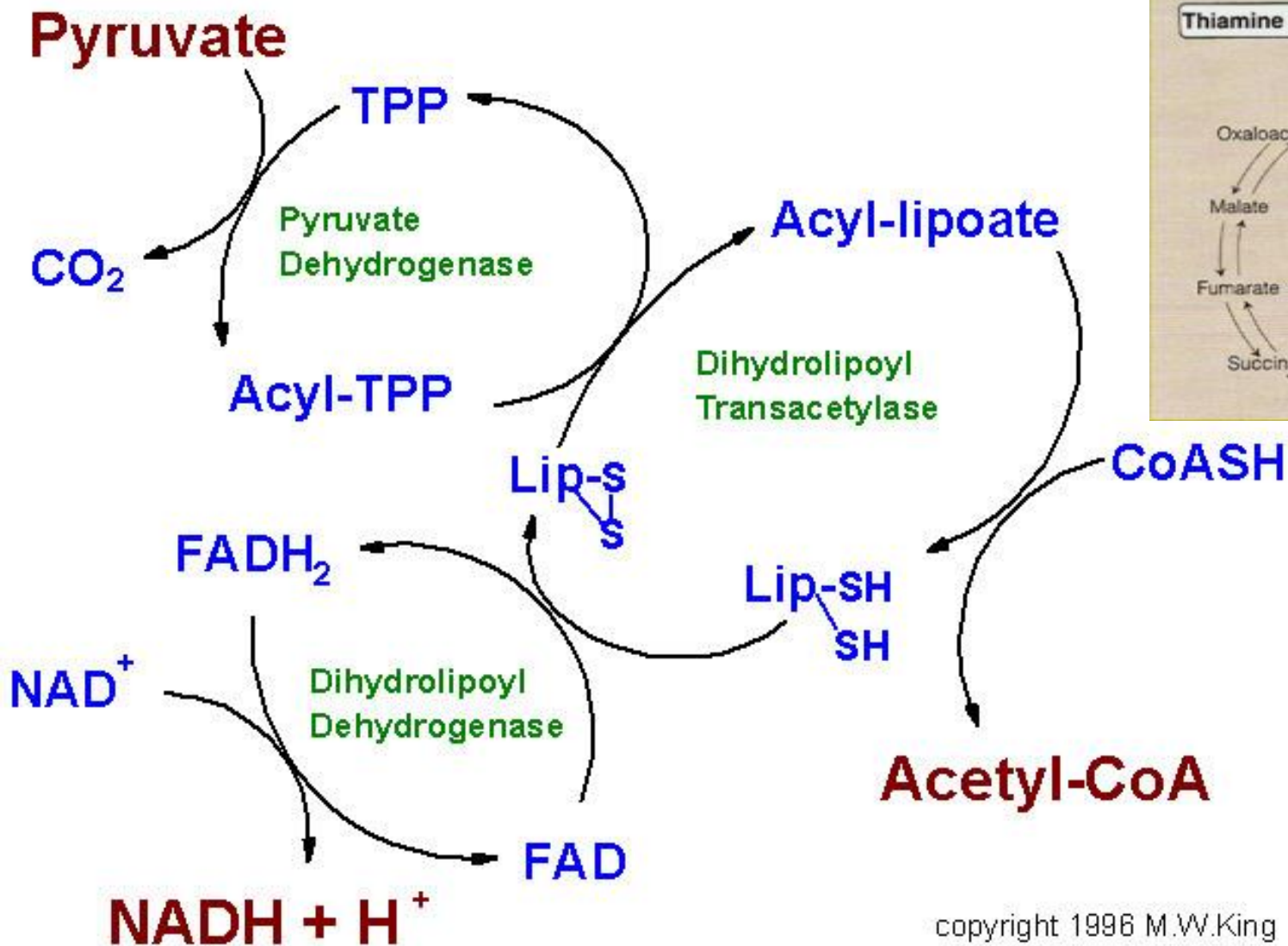
Structure of the coenzyme of thiamin



Thiamin pyrophosphate (TPP): coenzyme form

Biochemical functions of B1

- In form of TPP, vitamin B1 is involved with the energy releasing reactions of carbohydrate metabolism
- **Oxidative Decarboxylation Reactions**
 - Pyruvate Dehydrogenase
 - Where it removes carbon from pyruvate to make acetyl CoA
 - $\text{Pyruvate} + \text{CoA} + \text{NAD} \rightarrow \text{AcetylCoA} + \text{CO}_2 + \text{NADH}$
 - α -Ketoglutarate Dehydrogenase
 - Where it participates in Krebs cycle
 - $\text{A-KG} + \text{CoA} + \text{NAD} \rightarrow \text{SuccCoA} + \text{CO}_2 + \text{NADH}$

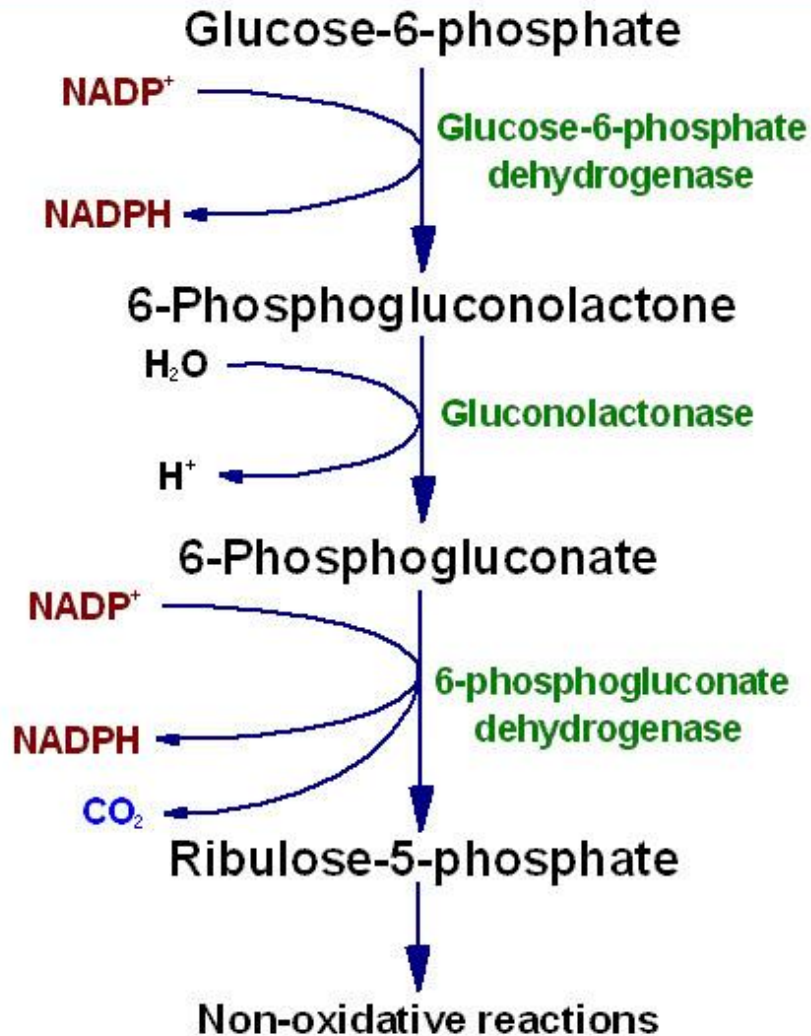


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Biochemical Functions of B1

- **Transketolation**
 - HMP pathway
- **Peripheral Nerve Function**
- TPP or TPPP
- non-cofactor function
- mechanism?

Oxidative Stage of Pentose Phosphate Pathway



Xylose-5 phosphate

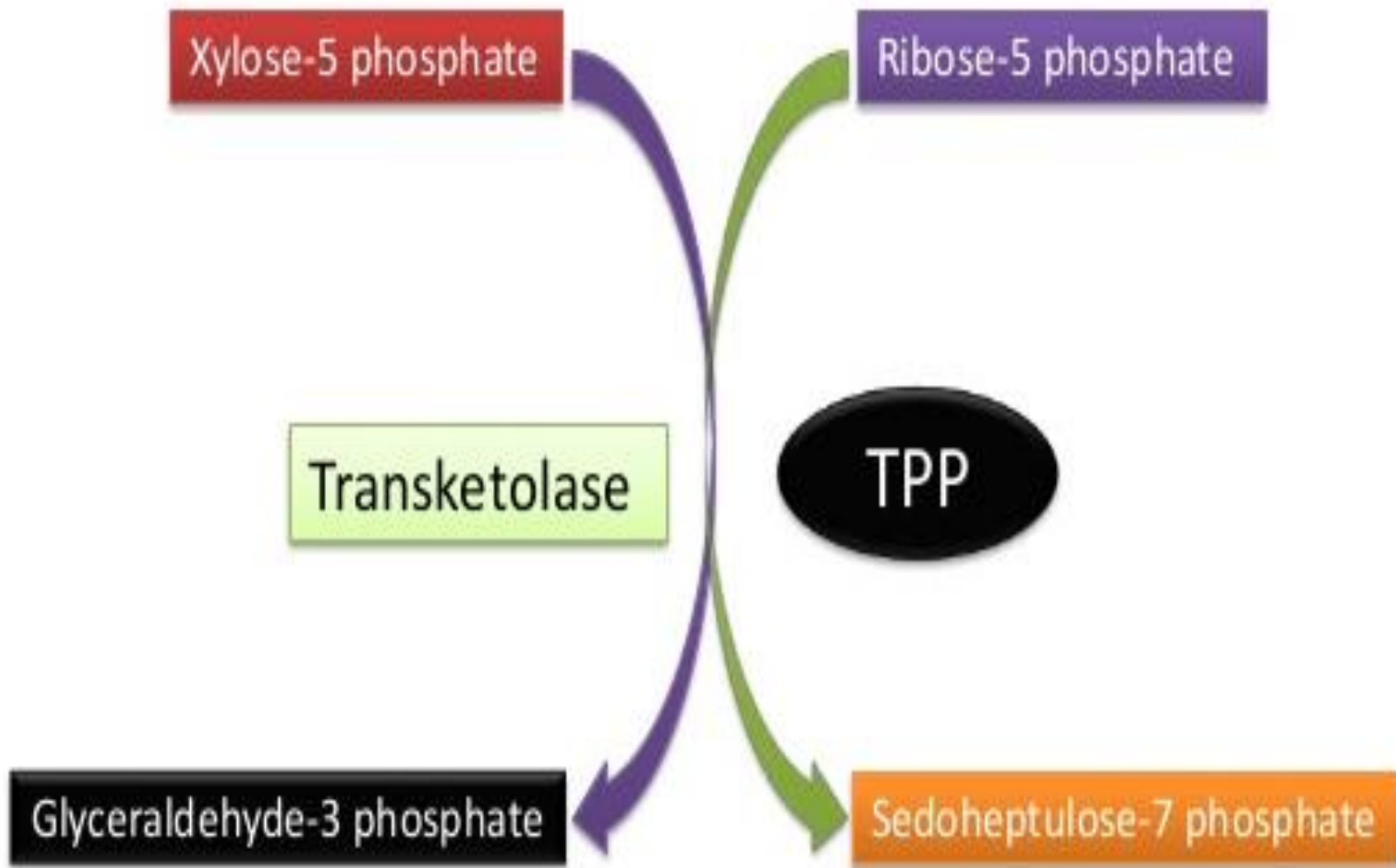
Ribose-5 phosphate

Transketolase

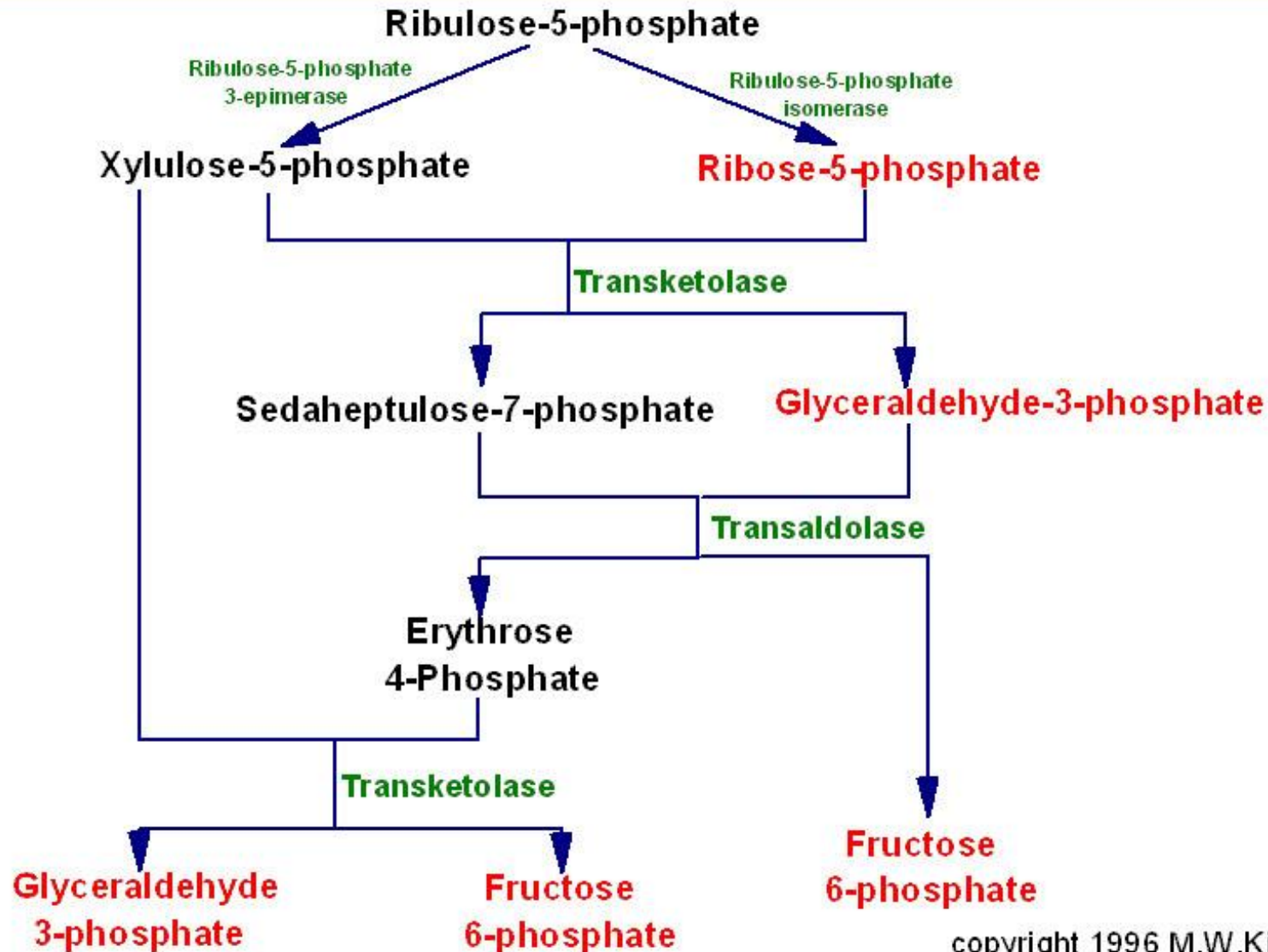
TPP

Glyceraldehyde-3 phosphate

Sedoheptulose-7 phosphate



Non-Oxidative Stage of Pentose Phosphate Pathway



Thiamin Deficiency

- Thiamin deficiency leads to a disease known as Beri-beri
- There are three types
- Wet Beri-beri: Edema and weak heart muscle
- Dry Beri-beri: degeneration of peripheral nerve and weak muscle
- Infantile Beri-beri: Affects infants

Causes of Deficiency Manifestation

- Inadequate diet i.e. consumption of polished rice.
 - Thiamin is present in the outer layer of rice grains.
- Alcoholism
- Pregnancy and lactation: Leads to increase demand
- The presence of the enzyme thiaminase (in raw fish) inactivates thiamin by breaking the thiazole ring

RIBOFLAVIN

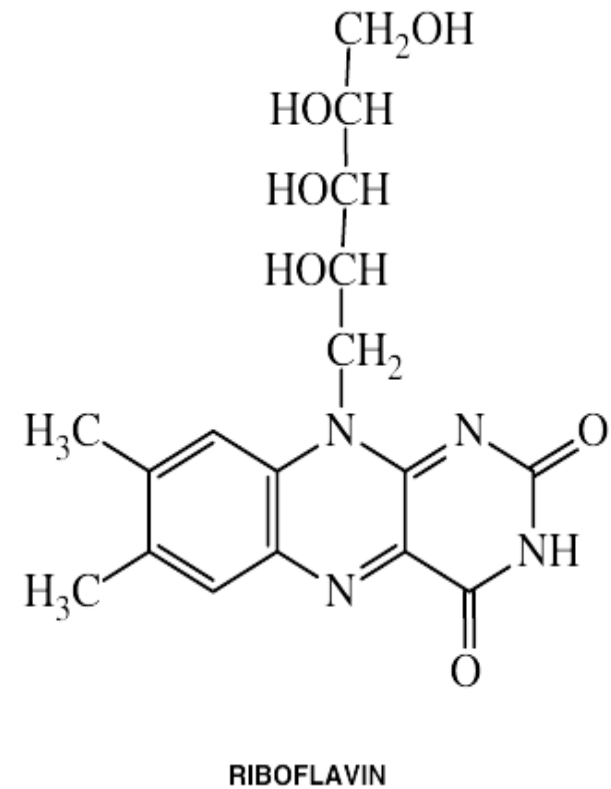
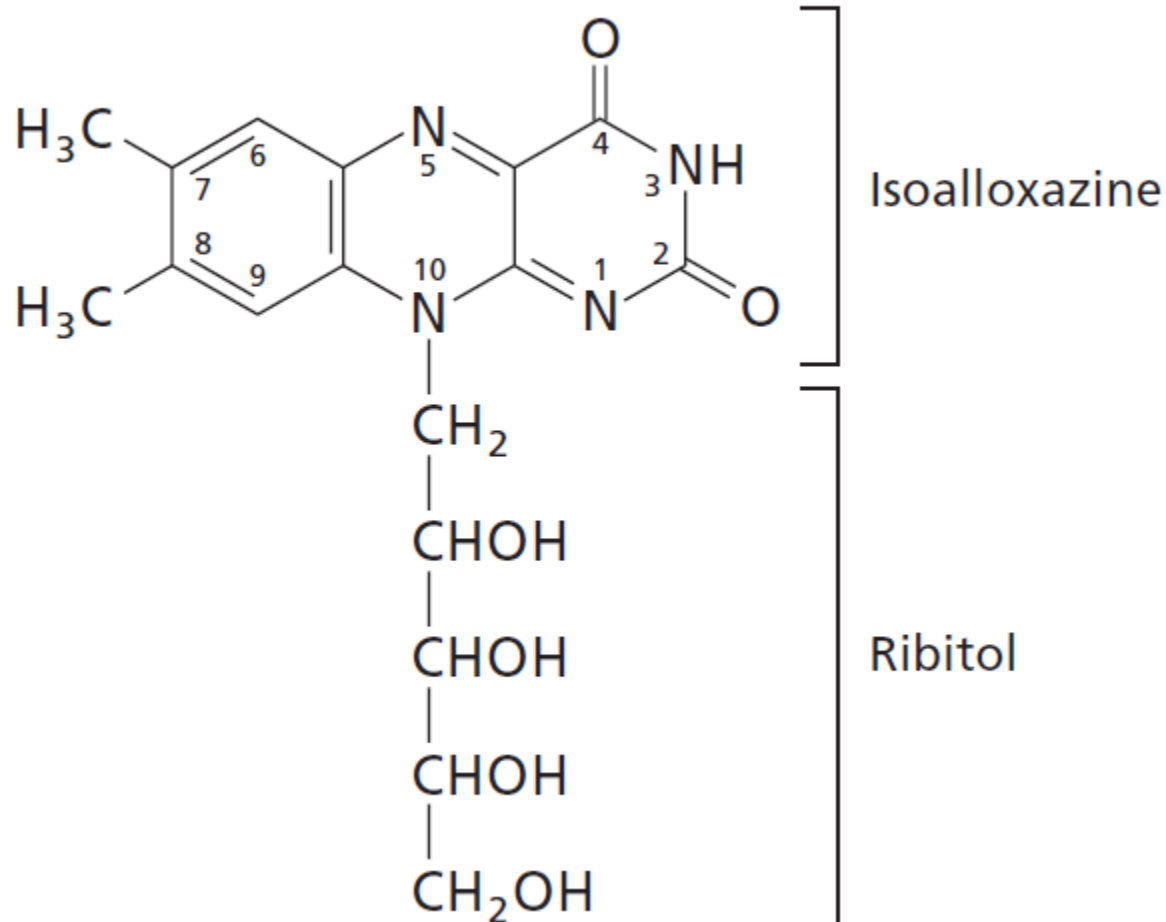
- The name "**riboflavin**" comes from
 - **Ribose** (the sugar whose reduced form, ribitol, forms part of its structure) and
 - **Flavin** (the ring-moiety which imparts the yellow colour to the oxidized molecule)
- Riboflavin is also known as vitamin B₂
- It is a heterocyclic isoalloxazine ring attached to a sugar alcohol, ribitol.

RIBOFLAVIN

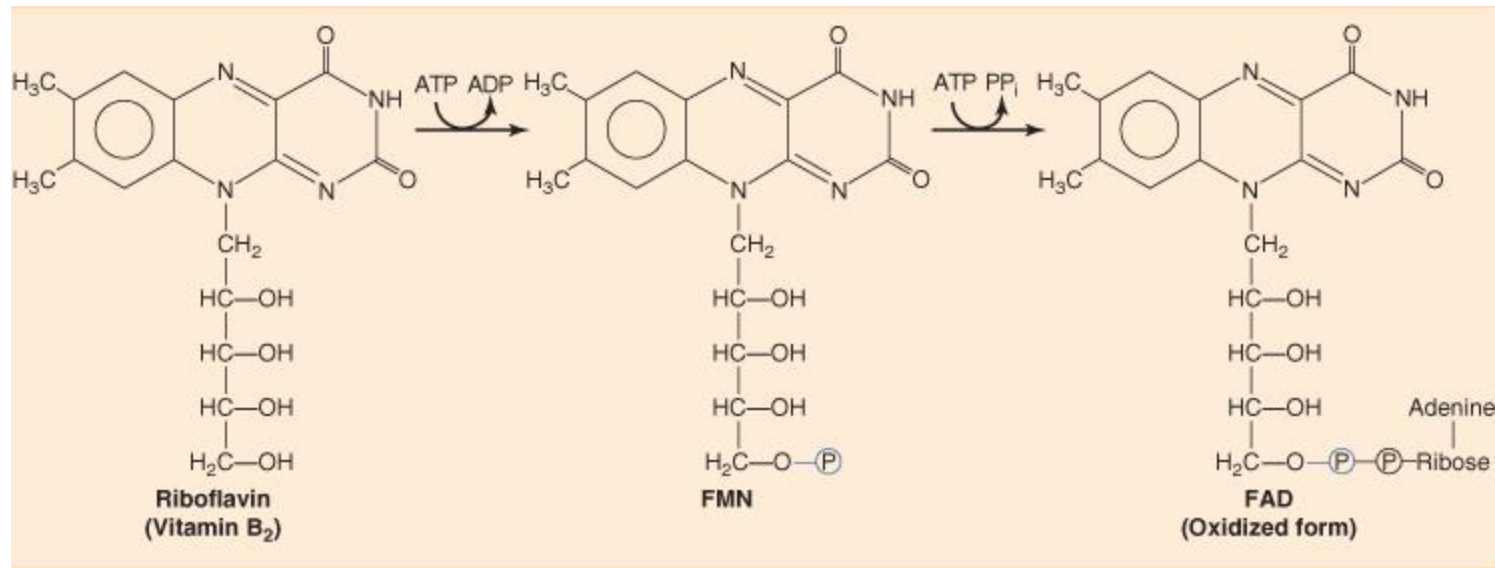
- The coenzymes forms of riboflavin are:
 - flavin mononucleotide (FMN)
 - Flavin adenine dinucleotide (FAD)Both coenzymes are formed in the intestine and liver
- The reduced forms of FMN and FAD are
 - FMNH₂ and
 - FADH₂, respectively.
- The enzymes that require FMN or FAD as cofactors are termed flavoproteins.
- Flavoproteins are involved in a wide range of redox reactions, e.g. succinate dehydrogenase and xanthine oxidase.

STRUCTURE OF RIBOFLAVIN

(a)



CONVERSION TO THE COENZYME FORM



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FMN → ATP-dependent phosphorylation of riboflavin

FAD → further reaction with ATP in which its AMP moiety is transferred to FMN.

FMN, FAD

Coenzyme:

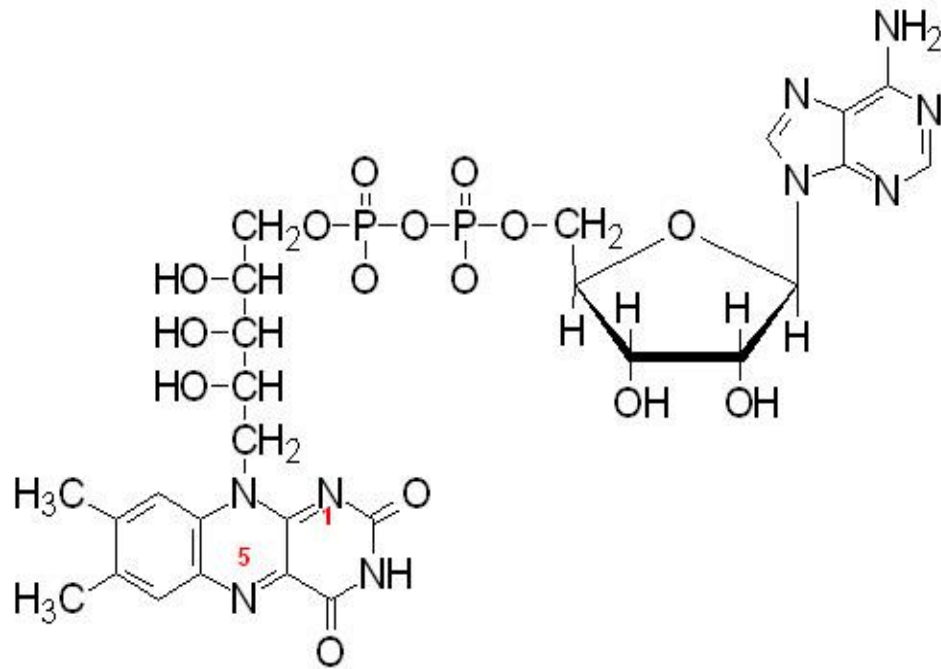
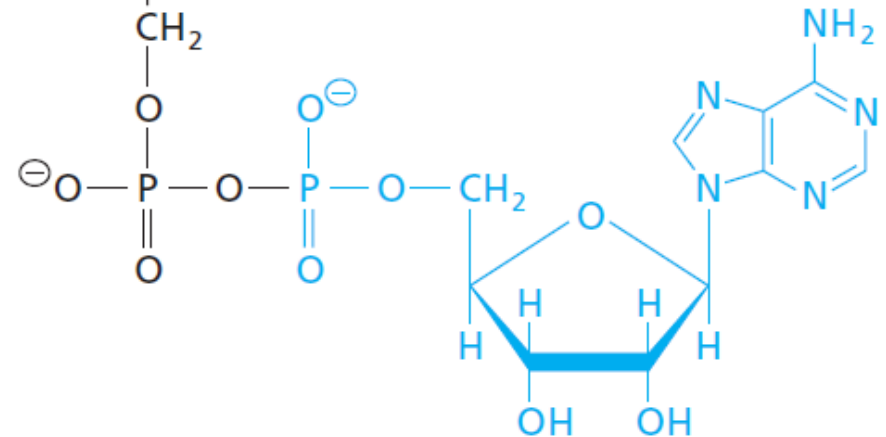
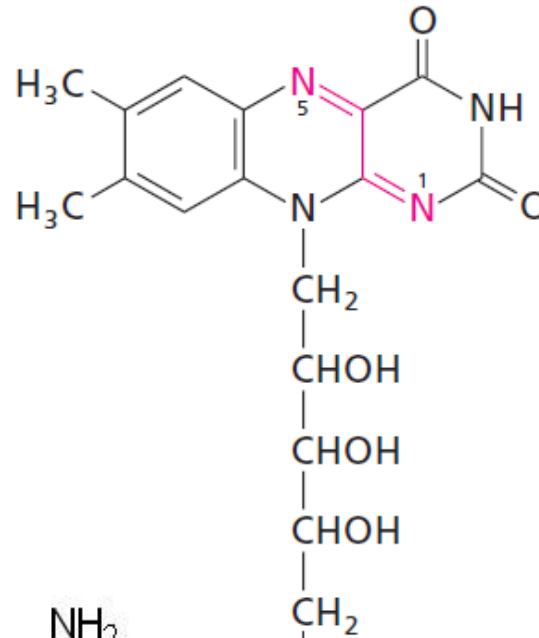
Flavin mononucleotide

FMN

Flavin adenine dinucleotide

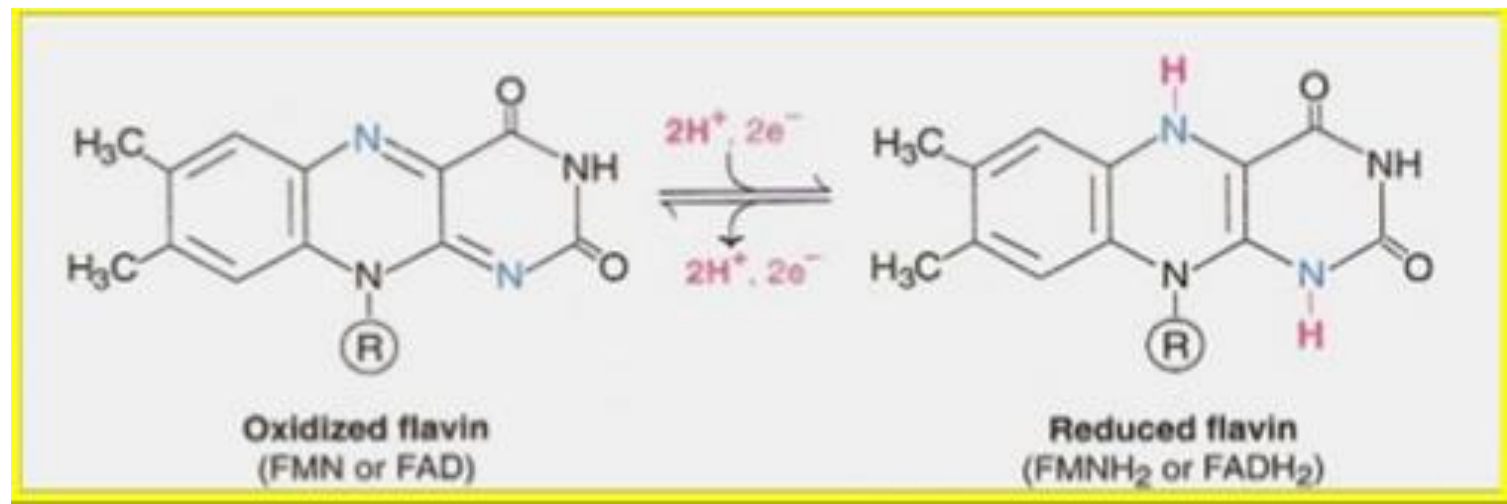
FAD

(b)



Oxidised and Reduced form of flavin ring

- The heterocyclic isoalloxazine ring serves as an acceptor of two hydrogen atoms with electrons
 - $\text{FMN} \rightarrow \text{FMNH}_2$
 - $\text{FAD} \rightarrow \text{FADH}_2$
- The coenzymes FMN(FMNH₂) and FAD(FADH₂)
- Participate in many oxidation reduction reaction in the Electron Transport Chain



Biochemical functions of riboflavin

- FMN and FAD serve as prosthetic groups of oxidoreductase enzymes
- Oxidative degradation of fatty acids (FAD is the prosthetic group of acylCoA the prosthetic group of acylCoA dehydrogenase.
- Oxidative deamination of α -amino acids: 1.FMN: Prosthetic group of L-a.a.oxidase. 2.FAD: Prosthetic group of D-a.a. oxidase
- Riboflavin is also needed to help the body convert vitamin B6 and folate into active forms

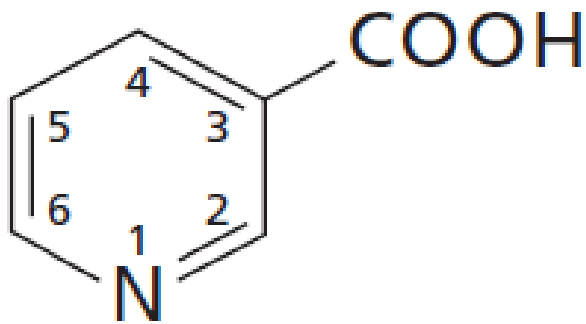
Niacin

- Niacin consist of two vitamers:
 - nicotinic acid
 - nicotinamide
- It is also known as vitamin B₃
- It is unique among the B vitamins in that it can be synthesized from Tryptophan in the body.
 - So it is not strictly considered a vitamin.
- However, the conversion is inefficient and most people required dietary sources of both tryptophan and niacin.
 - Where 60 mg of Tryptophan is required to produce 1mg of Niacin and occurs only after all the body requirements for Tryptophan is met.

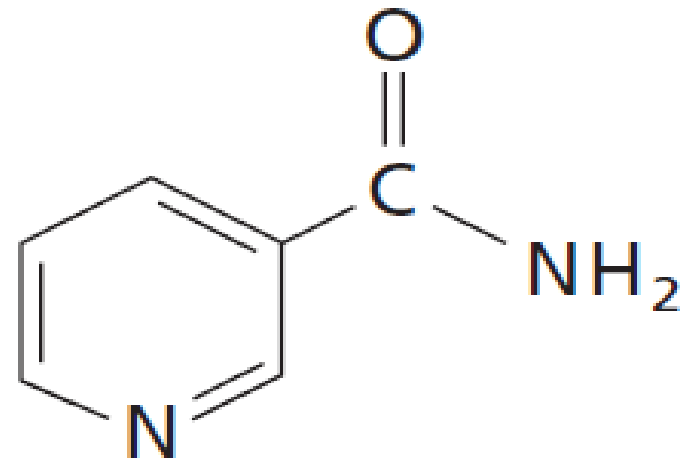
Niacin

- Both nicotinic acid and nicotinamide can serve as the dietary source of vitamin B₃.
- Niacin is required for the synthesis of the active forms of vitamin B₃
 - Nicotinamide adenine dinucleotide (NAD⁺)
 - Nicotinamide adenine dinucleotide phosphate (NADP⁺).
- Both NAD⁺ and NADP⁺ function as cofactors for numerous dehydrogenases
- In the structure of coenzymes, nitrogen atom of nicotinamide carries a positive charge due to formation of an extra bond
- Hence coenzymes are NAD⁺ & NADP⁺

Structures of nicotinic acid and nicotinamide



Nicotinic acid
(Niacin)



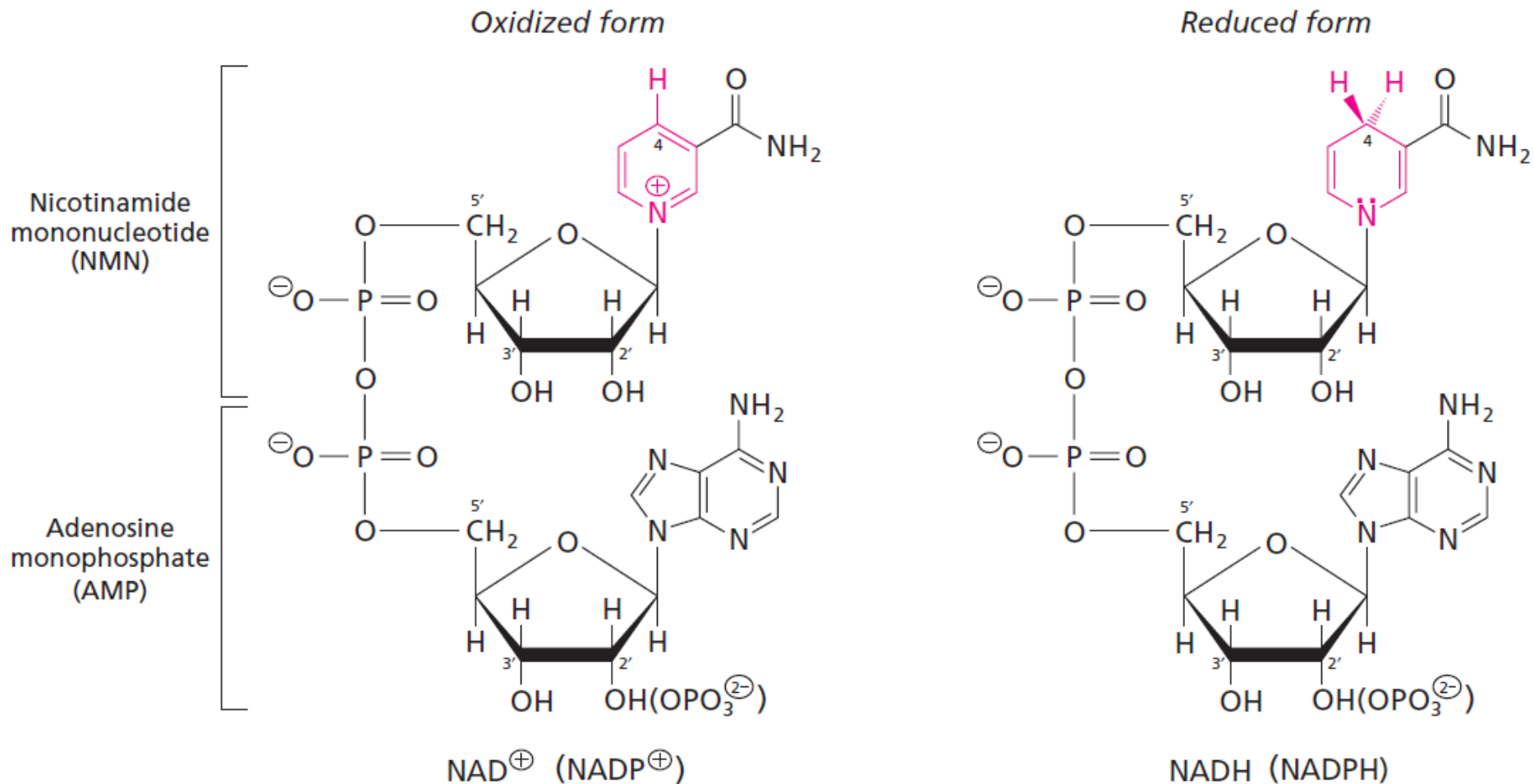
Nicotinamide

Niacin is the generic descriptor for the two compounds that have the biological action of the vitamin

The amide form of niacin is known as Niacinamide or Nicotinamide

- Dietary **nicotinamide, niacin and tryptophan** contribute to the synthesis of the **coenzymes NAD⁺ and NADP⁺**
- *1 mg of niacin is formed from 60 mg of tryptophan*
- Phosphoribosyl pyrophosphate and ATP provides ribose phosphate and AMP moiety for synthesis of NAD⁺
- *Glutamine* donates amide group

Niacin Coenzyme: NADH and NADPH



BIOCHEMICAL FUNCTIONS

- The coenzymes:
 - NAD⁺ (NADH) and
 - NADP⁺ (NADPH)are involved in oxidation-reduction in
- Carbohydrate,
- Lipid
- protein metabolism

Carbohydrate Metabolism

1. Glyceraldehyde 3P-Dehydrogenase

It catalyses the conversion of Glyceraldehyde 3P to 1,3 Bisphosphoglycerate

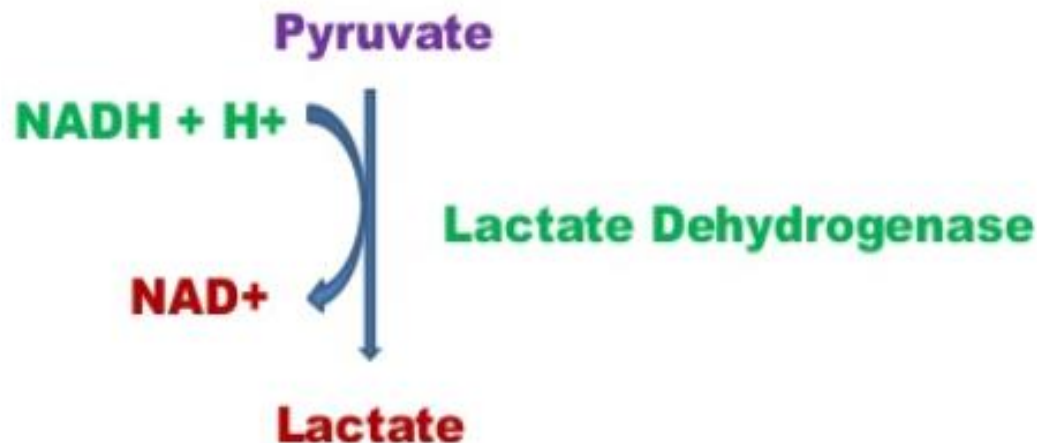


2. Lactate Dehydrogenase

It catalyses the interconversion of lactate to pyruvate

It occurs in anaerobic conditions (pyruvate to lactate in muscle and erythrocytes)

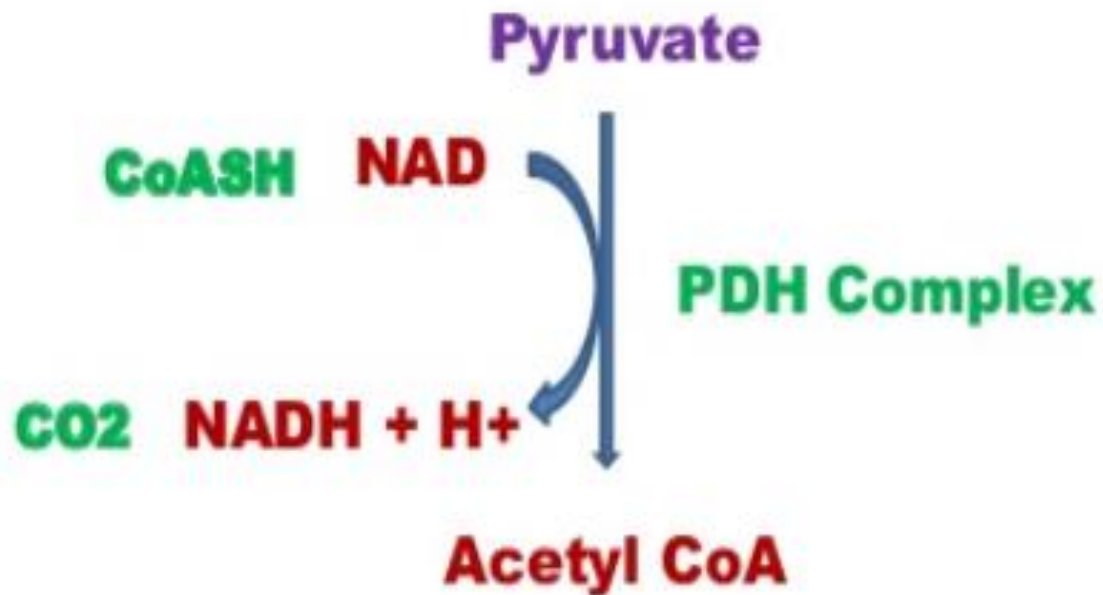
Gluconeogenesis (lactate to pyruvate in the liver)



3. Pyruvate Dehydrogenase Complex

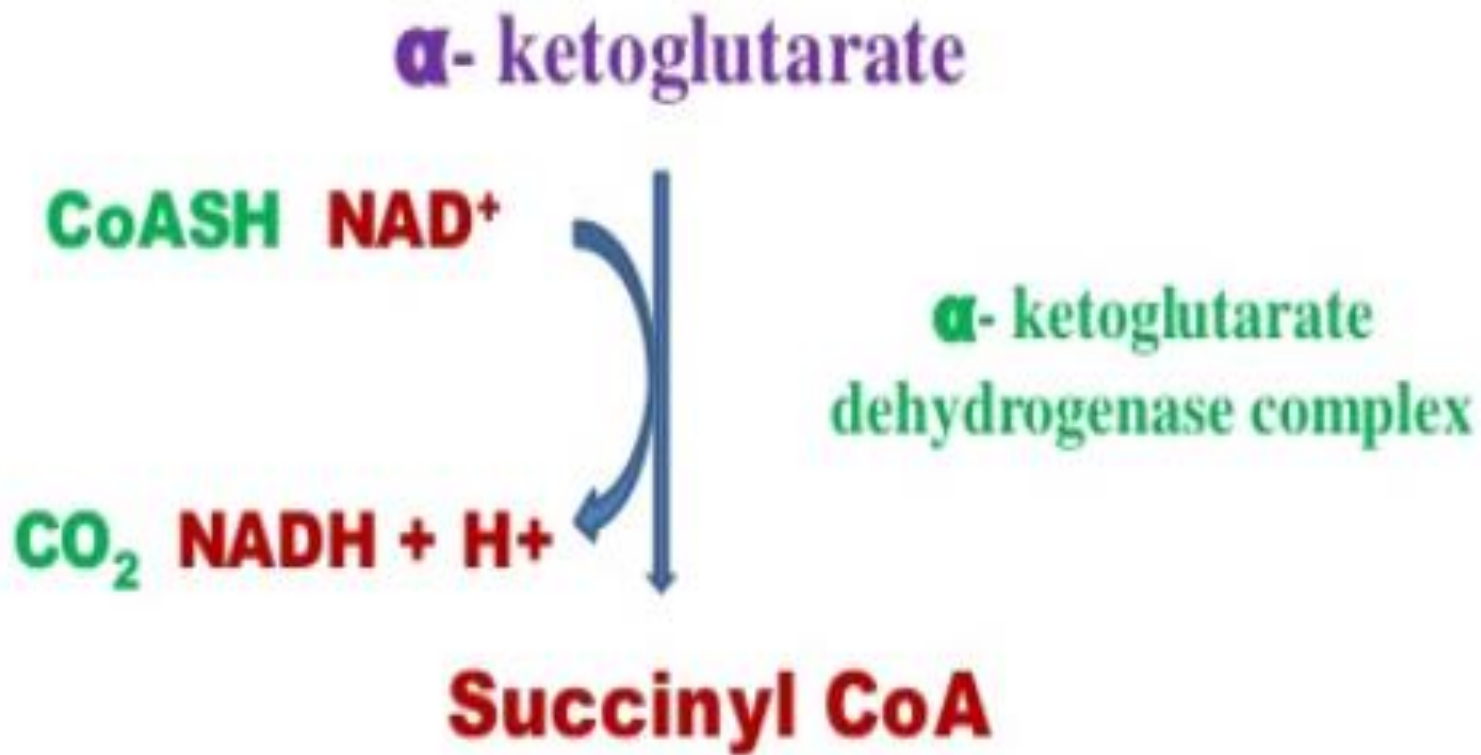
Pyruvate dehydrogenase catalyses the conversion of pyruvate to acetyl CoA

It utilizes NAD which is reduced to form $\text{NADH} + \text{H}^+$



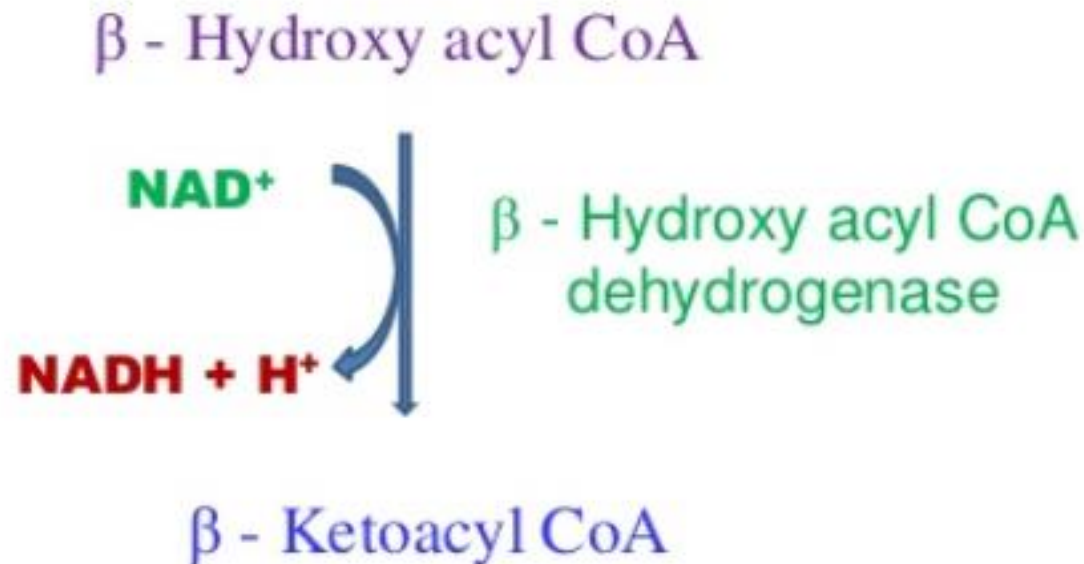
4. α -Ketoglutarate Dehydrogenase Complex

It catalyses the conversion of α -Ketoglutarate to succinyl CoA



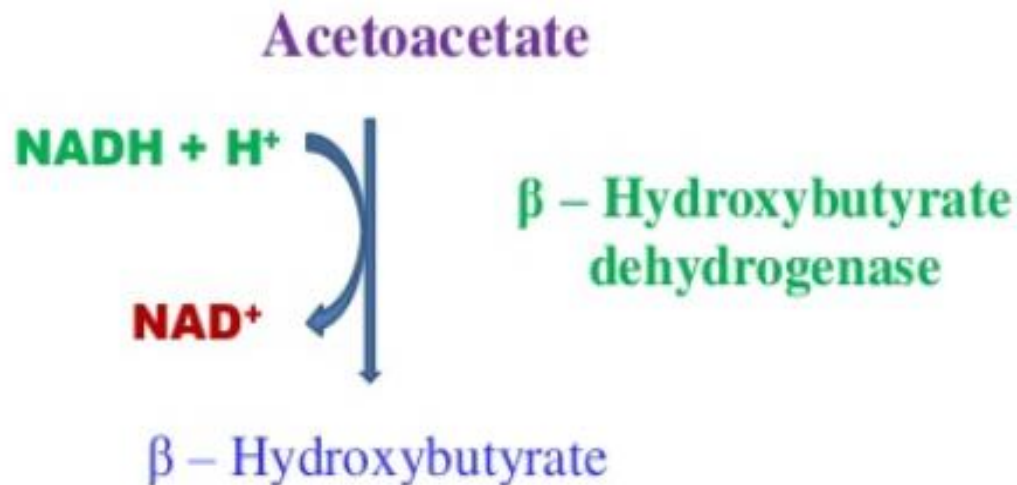
Lipid metabolism

- β - *Hydroxy acyl CoA dehydrogenase*:
- It catalyzes the oxidation of β - *Hydroxy acyl CoA* to β - *Ketoacyl CoA*



β – Hydroxybutyrate dehydrogenase

- It catalyzes the inter conversion of *acetoacetate to β – Hydroxybutyrate*
- Acetoacetate to β – Hydroxybutyrate in **liver** & β – Hydroxybutyrate to acetoacetate **extra hepatic tissues**



Alcohol dehydrogenase

- It catalyzes the *oxidation of alcohol to acetaldehyde*
- NAD is reduced to NADH + H⁺
- *Fatty liver & alcoholism:*
- NADH + H⁺ generated during oxidation of alcohol suppresses *TCA cycle* & diverts citrate for fatty acid synthesis leading to *increased synthesis of TAG*
- Consumption of *alcohol can lead to Fatty liver*

Ethylalcohol (alcohol or ethanol)

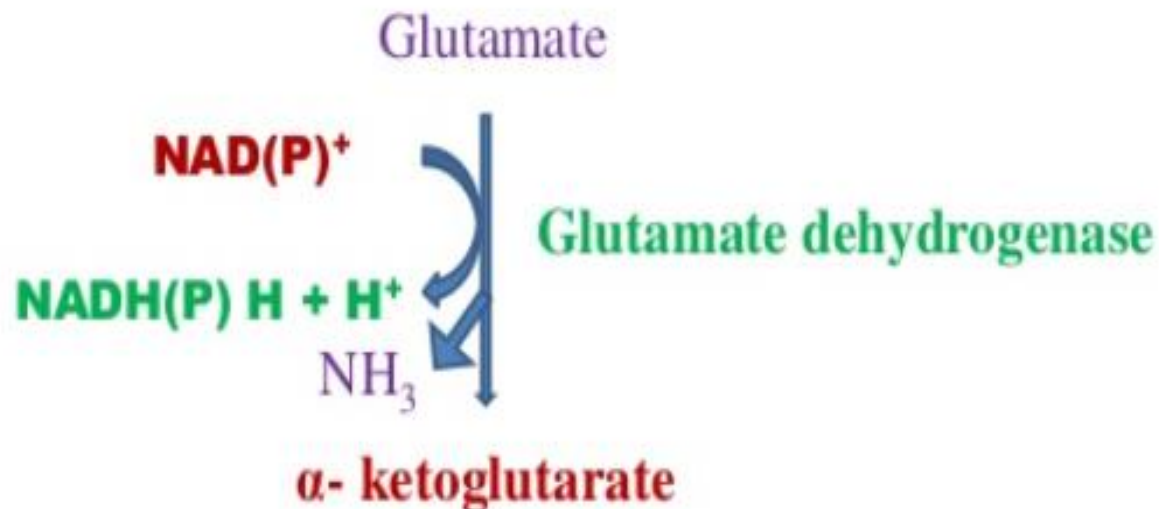


Protein metabolism

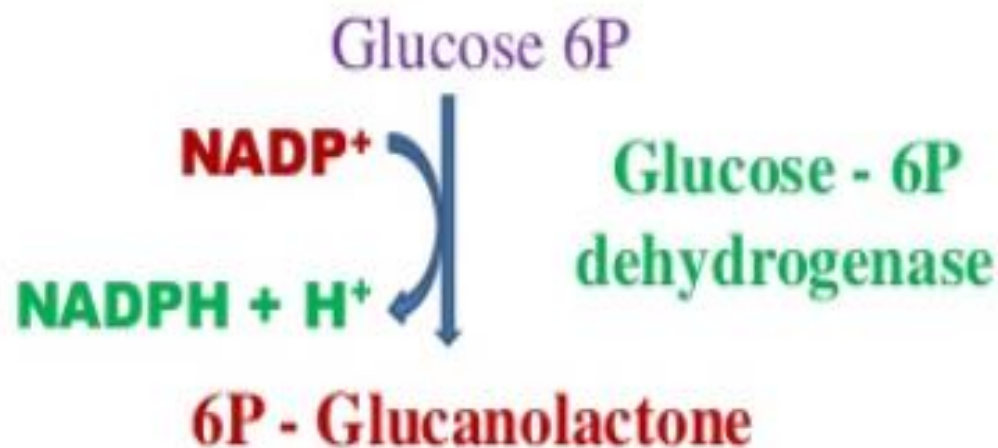
- *Branched chain α - ketoacid dehydrogenase:*
- It catalyzes the oxidation of α - keto acid derived from branched chain amino acids to *corresponding acyl CoA*



- **Glutamate dehydrogenase:**
- It utilizes either NADP or NAD for the conversion of glutamate to α -ketoglutarate & ammonia

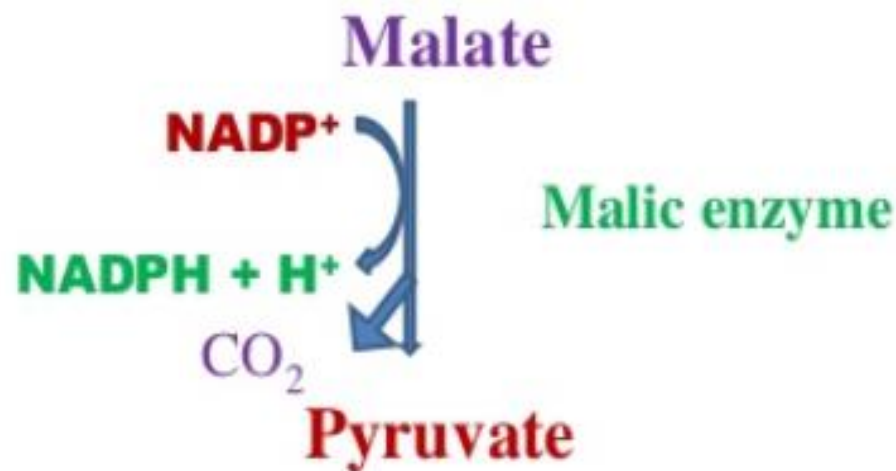


- **NADP⁺ dependent:**
- *Glucose - 6P dehydrogenase:*
- It catalyzes the **oxidation of glucose - 6P to 6P - Glucanolactone**

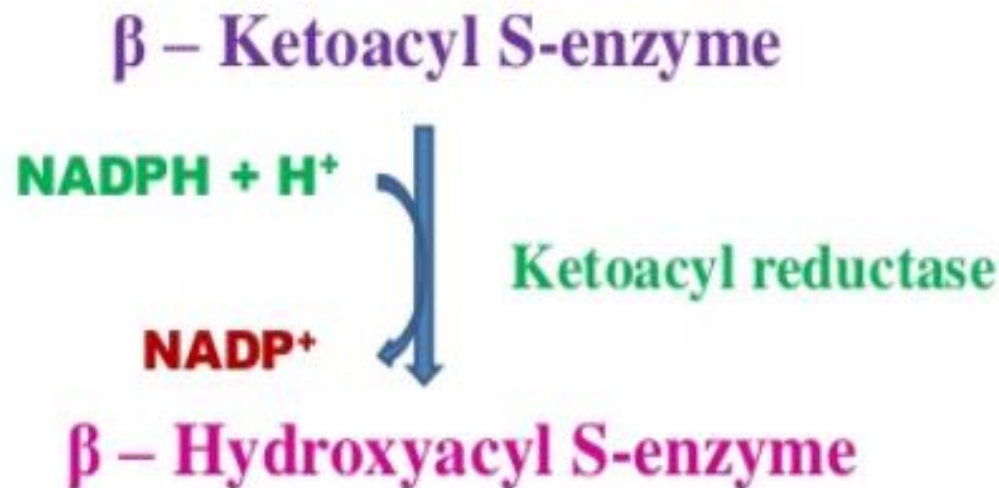


Malic enzyme

- It catalyzes the conversion of *malate to pyruvate & CO₂*
- This reaction provides NADPH for fatty acid synthesis and cholesterol biosynthesis



- *β – Ketoacyl reductase:*
- It is the one of the components of FAS complex
- It catalyzes the *reduction of β – Ketoacyl molecule to β – Hydroxyacyl molecule*



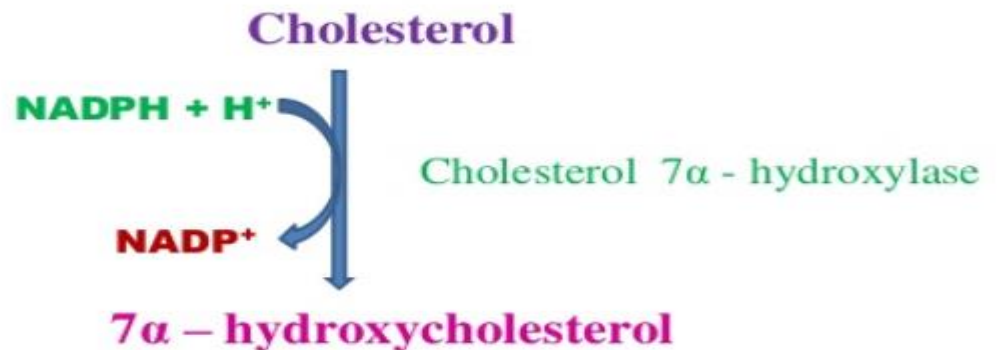
HMG CoA reductase

- It catalyzes the reduction of HMG CoA to mevalonate



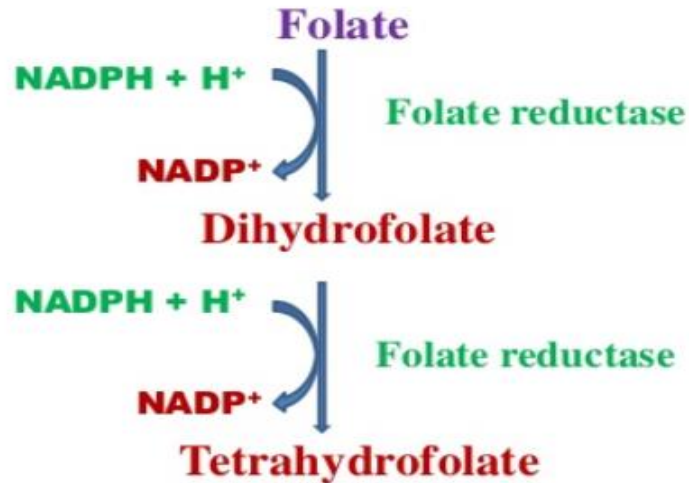
Cholesterol 7 α - hydroxylase

- It catalyzes the hydroxylation of cholesterol to 7 α - hydroxycholesterol
- It is rate limiting step in bile acid synthesis



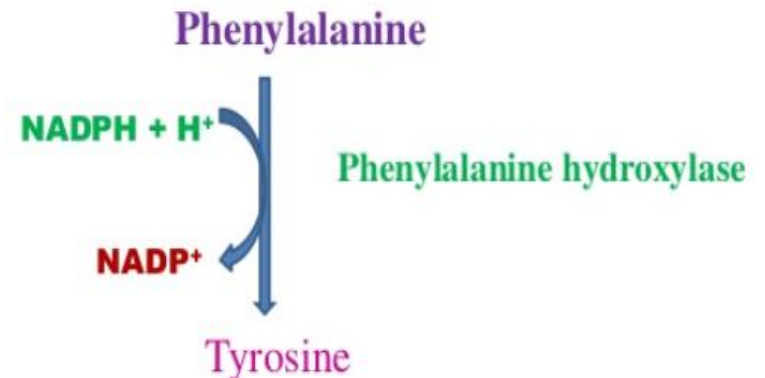
Folate reductase

- It catalyzes the formation of tetrahydrofolate (FH₄) from folate



Phenylalanine hydroxylase

- It catalyzes the conversion of phenylalanine to tyrosine



Deficiency Manifestation

- Deficiency of Niacin leads to the clinical condition called **pellagra**(=rough skin)

Pellagra is characterized 3 D's:

Dermatitis: In early stages, bright red erythema occurs, in feet, ankles and face

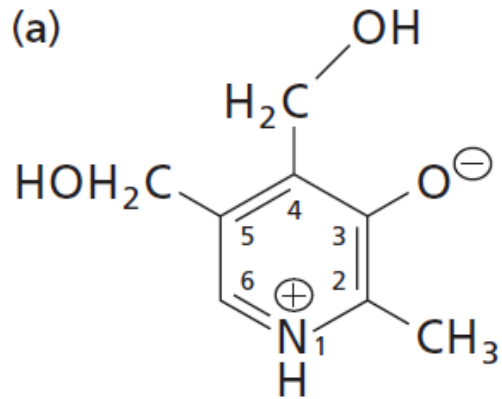
- **Diarrhea:**
- Diarrhea may be mild or severe with blood & mucus
- Leads to weight loss
- Nausea & vomiting
- **Dementia:** Irritability, inability to concentrate & poor memory

Vitamin B₆

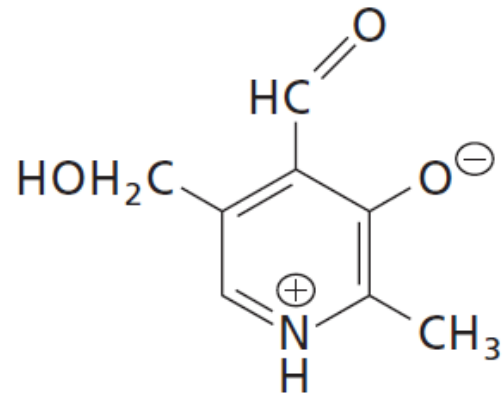
- Vitamin B₆ consists of 3 closely related pyridine derivatives:
 - Pyridoxine,
 - Pyridoxal
 - Pyridoxamine
- All the three compounds are efficiently converted to the biologically active form of vitamin B₆ pyridoxal phosphate in the intestine catalysed by pyridoxal kinase
- Pyridoxamine is mostly present in plants
- Pyridoxine and pyridoxal are present in animal foods

Vitamin B₆

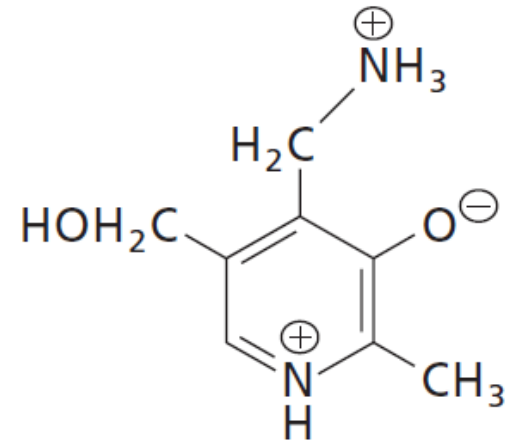
(a)



Pyridoxine

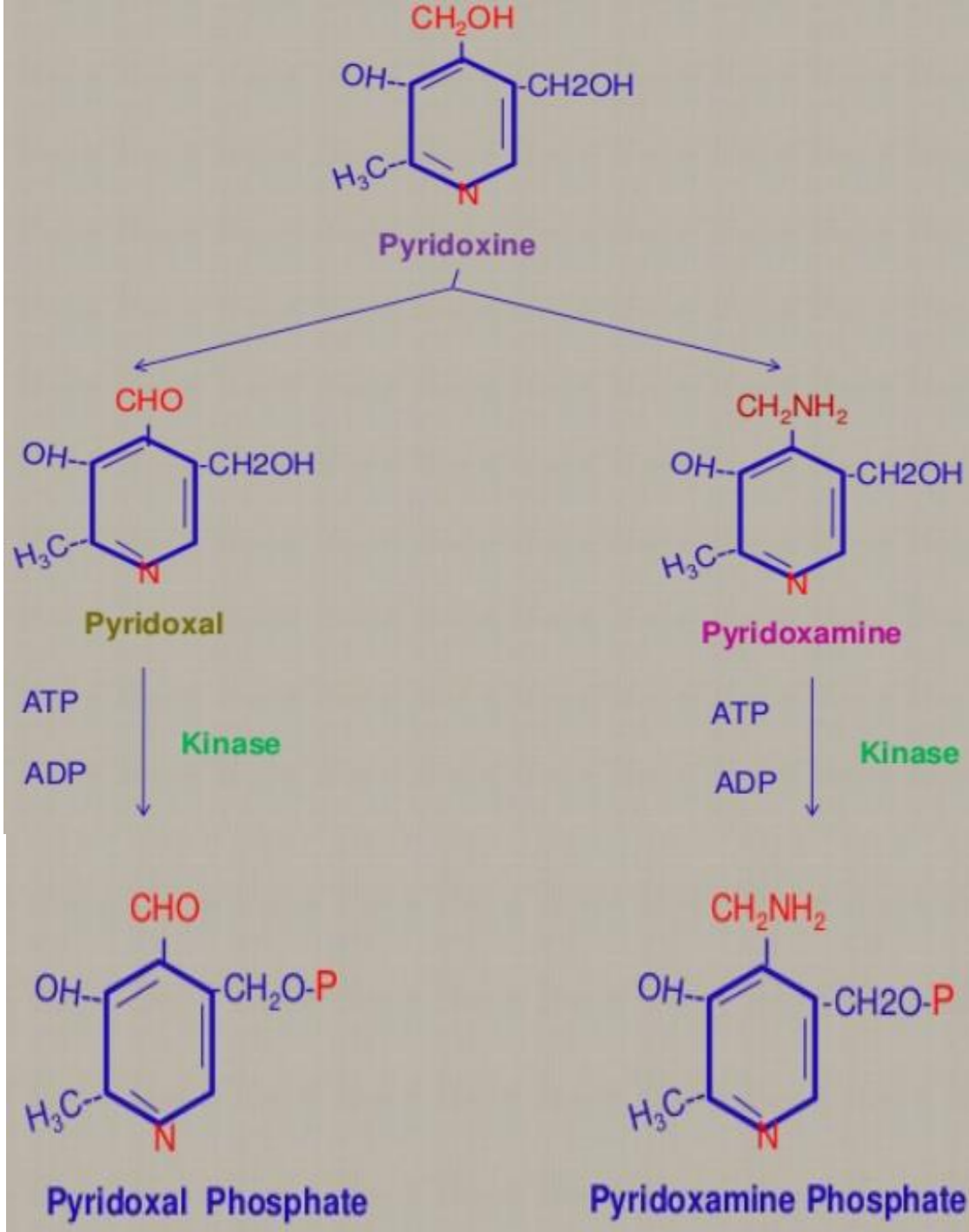
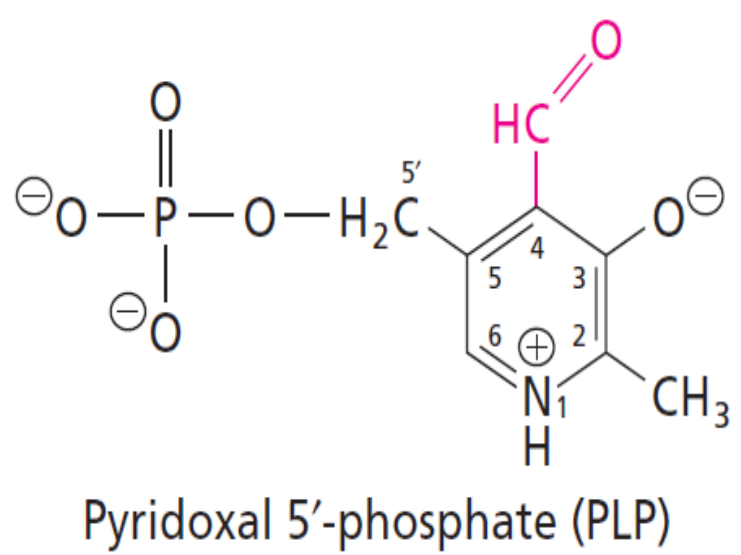


Pyridoxal



Pyridoxamine

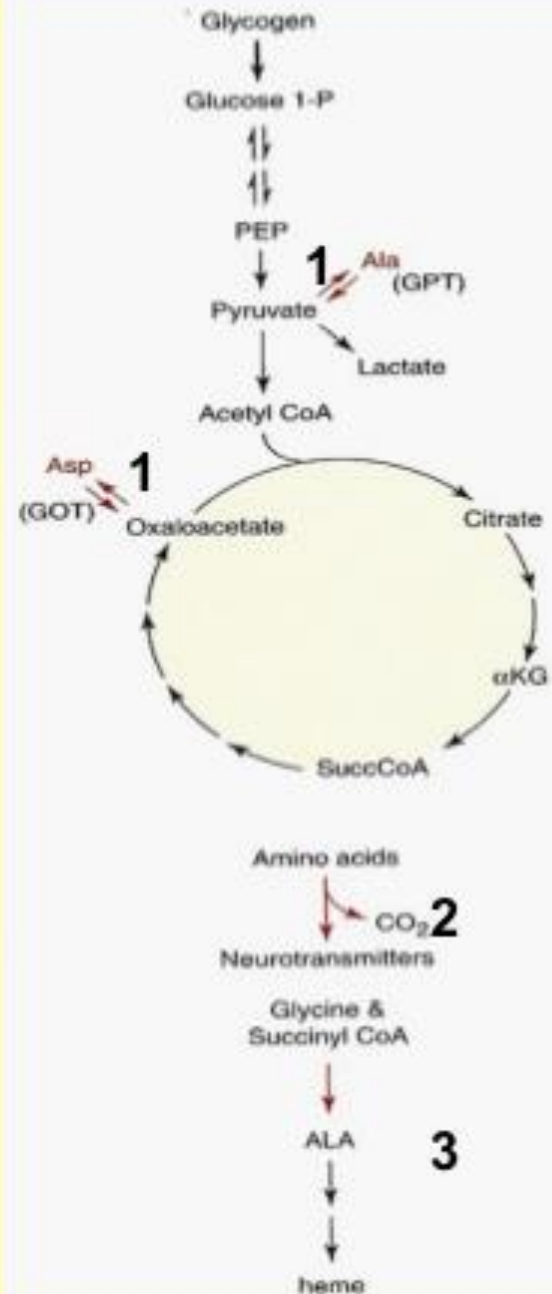
Coenzyme:
pyridoxal phosphate (PLP) and
pyridoxamine phosphate (PMP)



Biochemical Functions

- Pyridoxal Phosphate (PLP) the active coenzyme form of Vitamin B₆ is closely associated with the metabolism of Amino acids.

1. Transamination.
2. Decarboxylation.
3. Heme synthesis.
4. Deamination.
5. Production of Niacin

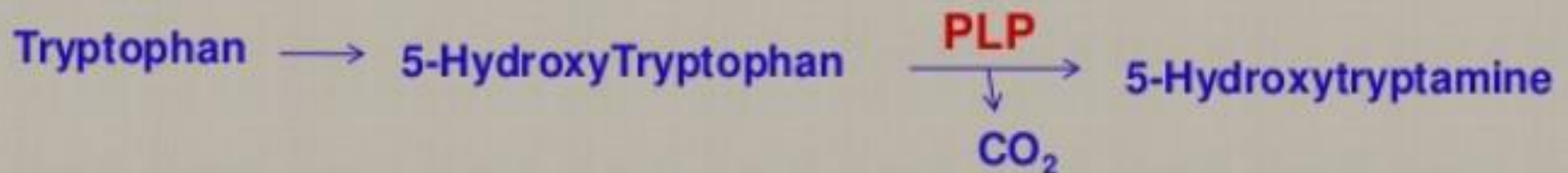


Transamination

- PLP is involved in transamination reaction
converting amino acids to keto acids
- Keto acids enter the TCA cycle & get oxidized to
generate energy
- During transamination, PLP interacts with amino
acids to form Schiff base
- The amino group is handed over to PLP to form
Pyridoxamine phosphate & ketoacid is liberated

Decarboxylation

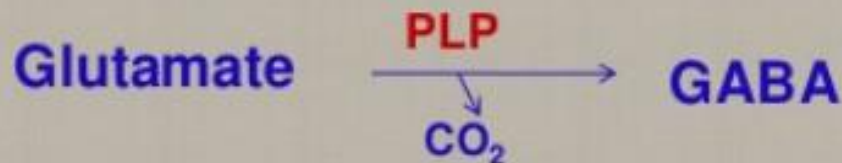
- **α** - Amino acids undergo decarboxylation to form respective amines
- The reaction is carried out by decarboxylases which require PLP
- Serotonin produced from tryptophan is important in nerve impulse transmission
- It regulates sleep, behaviour, blood pressure



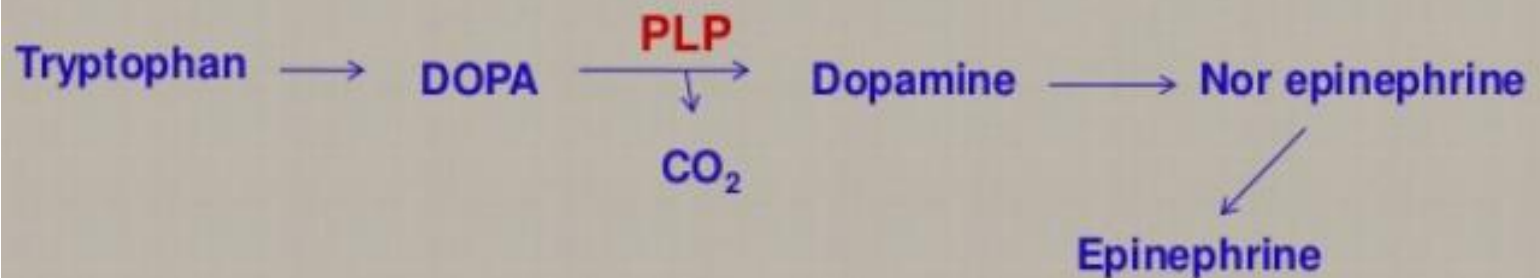
- Histamine is vasodilator & lowers blood pressure
- It stimulates gastric HCl secretion & is involved in inflammation & allergic reactions

- $$\text{Histidine} \xrightarrow[\text{CO}_2]{\text{Decarboxylase, PLP}} \text{Histamine}$$

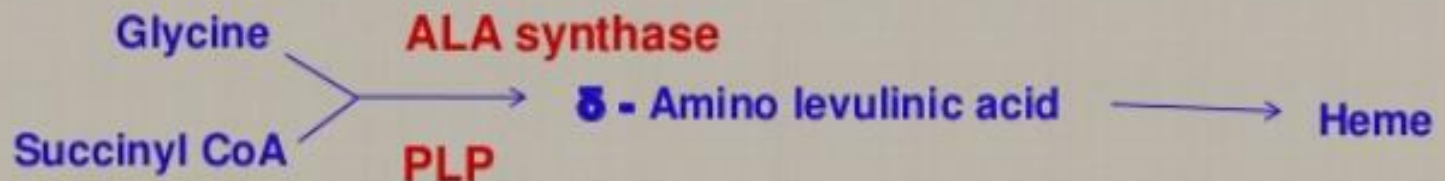
• Glutamate on decarboxylation gives GABA
- GABA inhibits transmission of nerve impulses



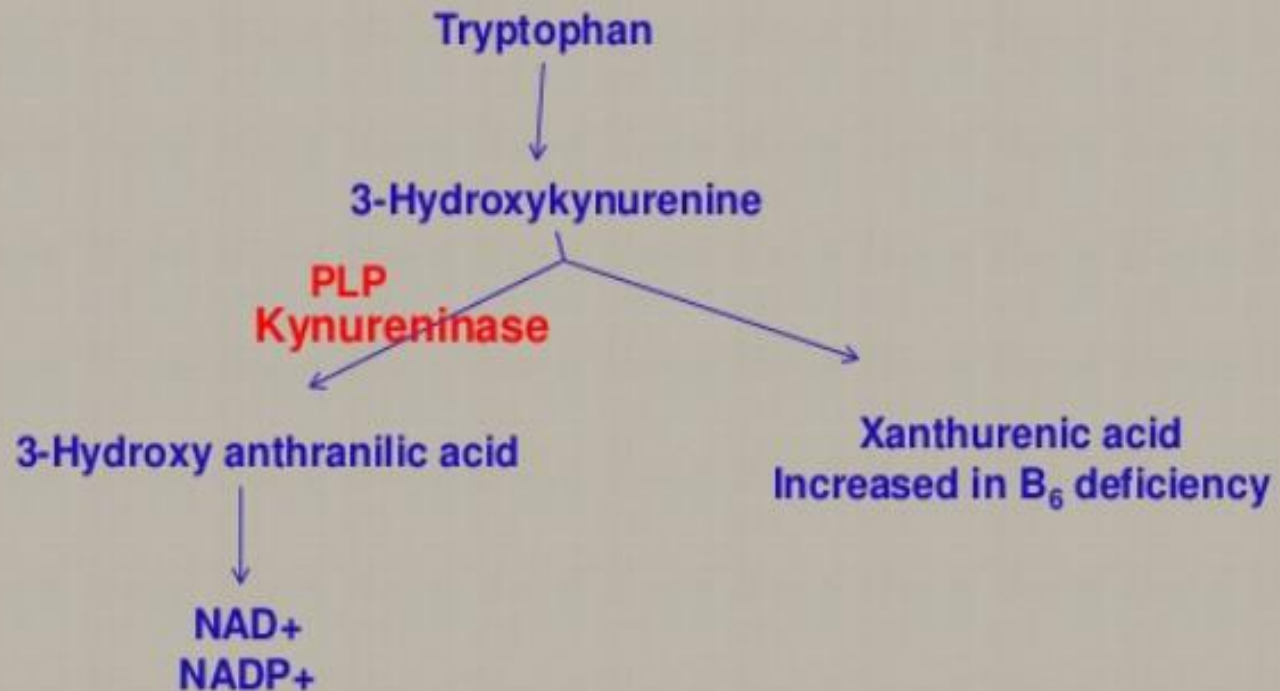
- The synthesis of catecholamines (dopamine, nor epinephrine & epinephrine) from tyrosine require PLP



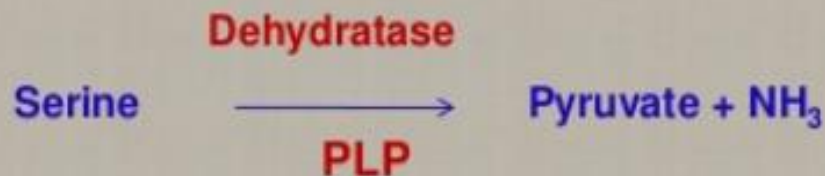
- PLP is required for synthesis of δ - amino levulinic acid, the precursor for heme synthesis

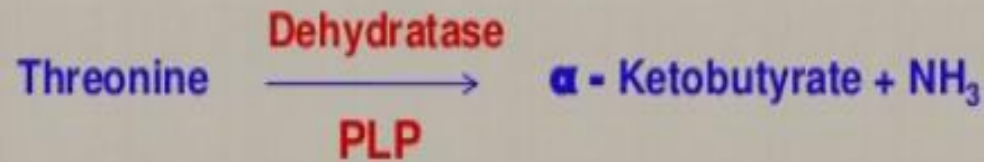


- The synthesis of niacin coenzymes (NAD^+ & NADP^+) from tryptophan require PLP
- Kynureninase require PLP



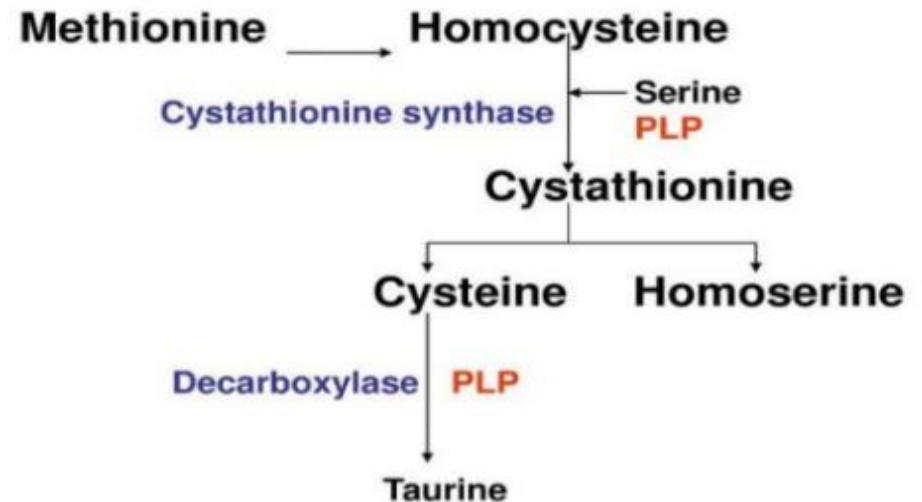
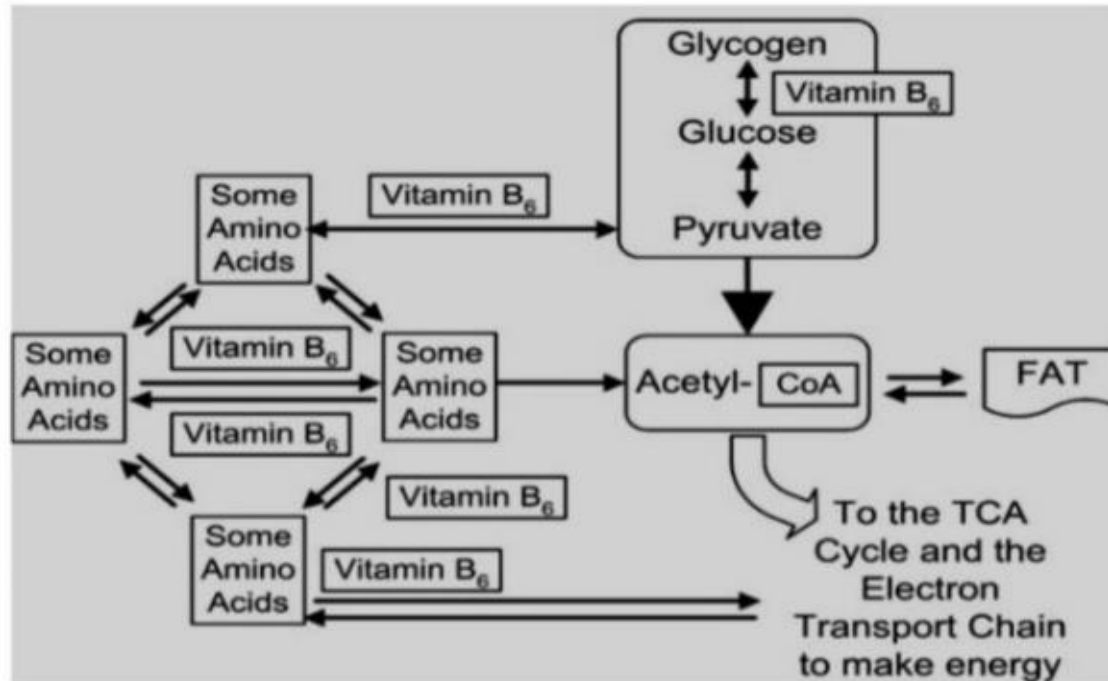
- PLP Plays an important role in **metabolism of sulfur containing amino acids**
- Transsulfuration from homocysteine to serine occurs in the synthesis of cysteine
- PLP dependent enzyme cystathionine synthase
- **Deamination** of hydroxyl group containing amino acids require PLP





- Synthesis of serine from glycine require PLP.
- Glycogen phosphorylase contains PLP
- It converts glycogen to glucose 1-phosphate
- PLP is needed for the absorption of amino acids from intestine
- B6 is useful to prevent urinary stone formation

Vitamin B6 assists energy production from protein and carbohydrates.



RDA

- Adult men – 2 - 2.2 mg/day
- Adult women - 2.0 mg/day
- Pregnancy & lactation - 2.5 mg/day

- **Dietary sources:**

- Animal sources
- Egg yolk, fish, milk, meat
- Vegetable sources
- Wheat, corn, cabbage, roots & tubers

Causes of Deficiency

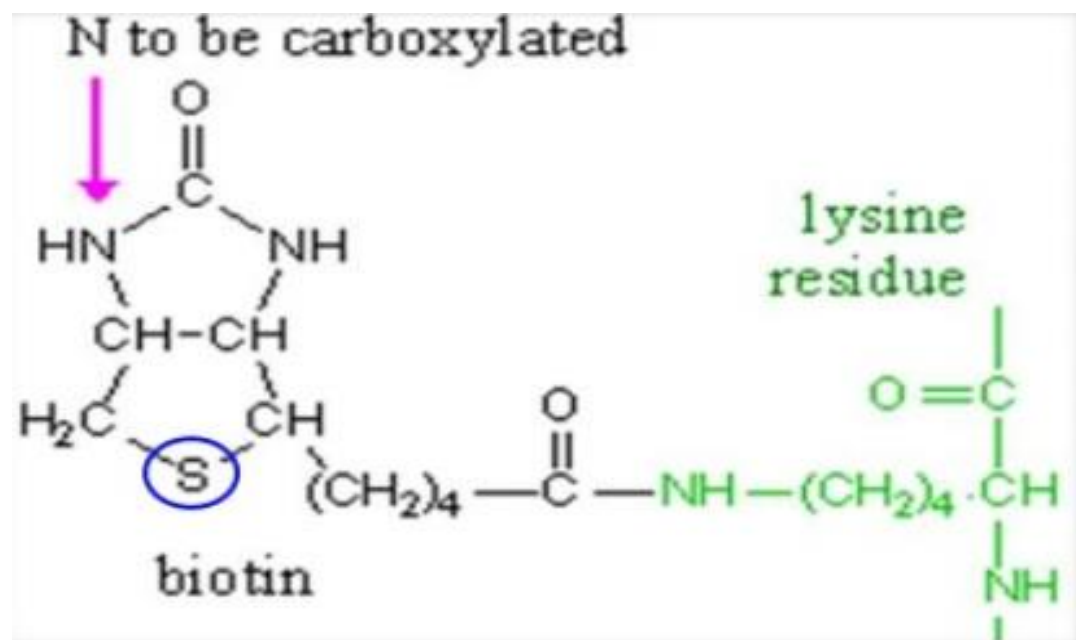
- Decreased dietary intake
- Alcoholism
- Impaired absorption
- Antivitamins: Chronic administration of drugs such as isoniazid and penicillamine
- Deficiency
 - Microcytic anemia

Biotin

- Biotin is also known as vitamin B7 .
- It is composed of a non-aromatic ring containing a sulphur atom and a propanoic acid side chain.
- Biotin occur naturally as D-biotin
- It becomes active only bound to the enzyme
- The biologically active form of biotin is biocytin
- Biotin is the cofactor of enzymes involved in carboxylation reactions, e.g.
 - acetyl-CoA carboxylase and
 - pyruvate carboxylase.

Biotin

- Biotin is found in numerous foods and it is also synthesised by intestinal bacteria
 - as such deficiencies of the vitamin are rare.
- Deficiencies are generally seen only after
 - long antibiotic therapies which deplete the intestinal fauna or
 - following excessive consumption of raw eggs.
- due to the affinity of the egg white protein, avidin, for biotin preventing intestinal absorption of the biotin.



Biochemical functions

- Biotin is the coenzyme for 4 carboxylases.

1.Acetyl coenzyme A carboxylase: found in the mitochondria; catalyzes the carboxylation of Acetyl CoA to Malonyl CoA.

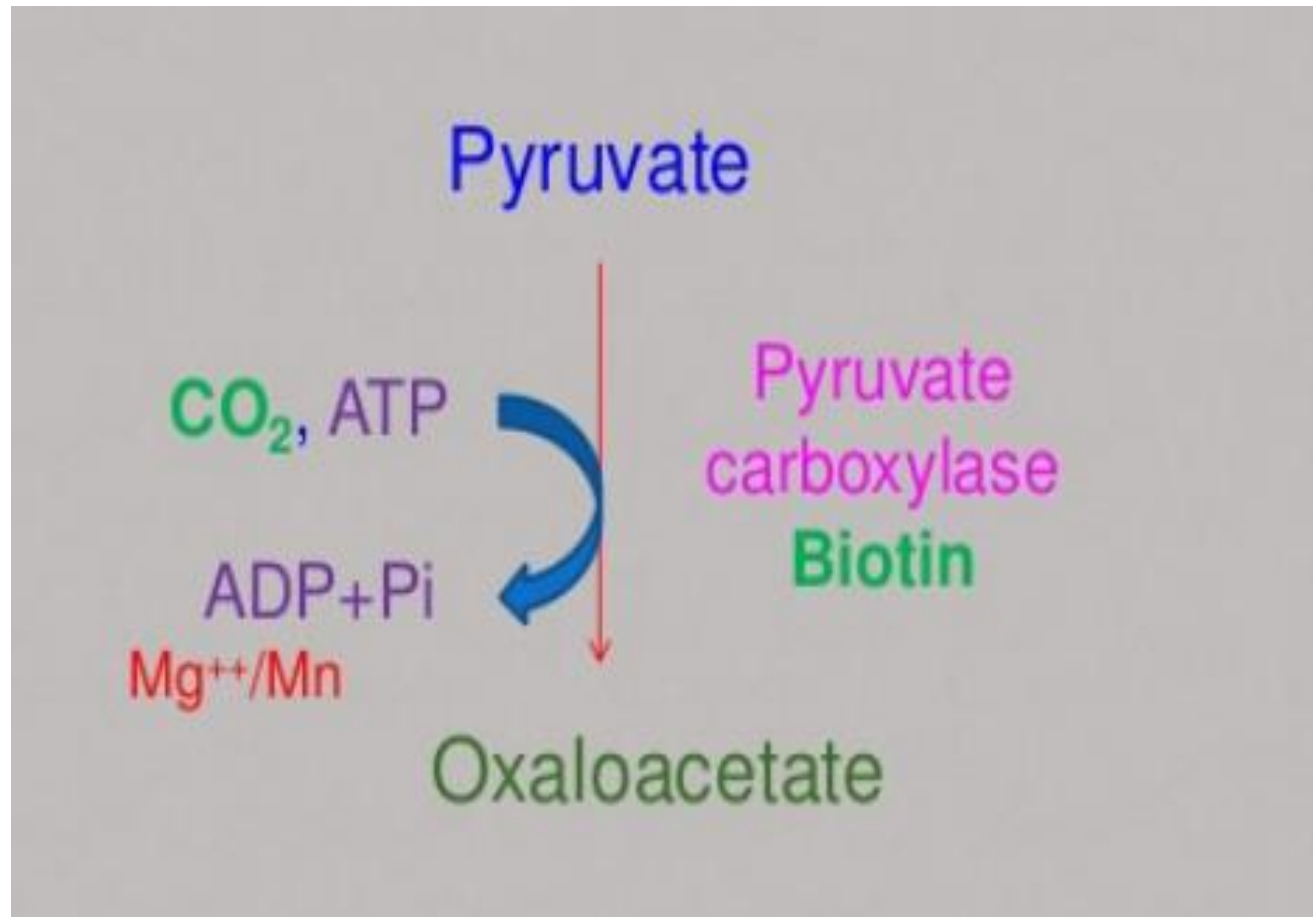
2.Pyruvate carboxylase: found in the mitochondria; catalyzes the carboxylation of pyruvate to form oxaloacetate.

3.Methylcrotonyl-CoA carboxylase: found in the mitochondria; involved in the metabolism of L-leucine.

4.Propionyl-CoA carboxylase: involved in the metabolism of L-isoleucine, L-valine, L-threonine, and L-methionine.

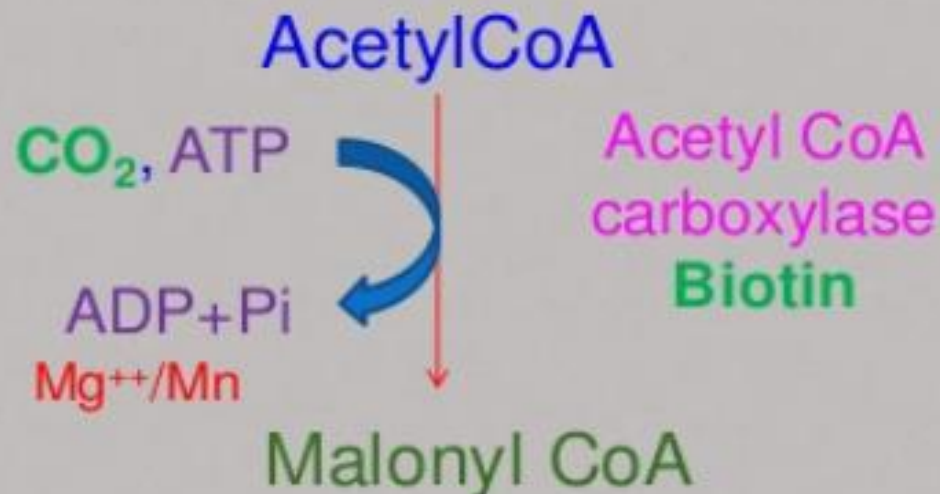
Pyruvate Carboxylase

Pyruvate carboxylase catalyses the conversion of pyruvate to oxaloacetate



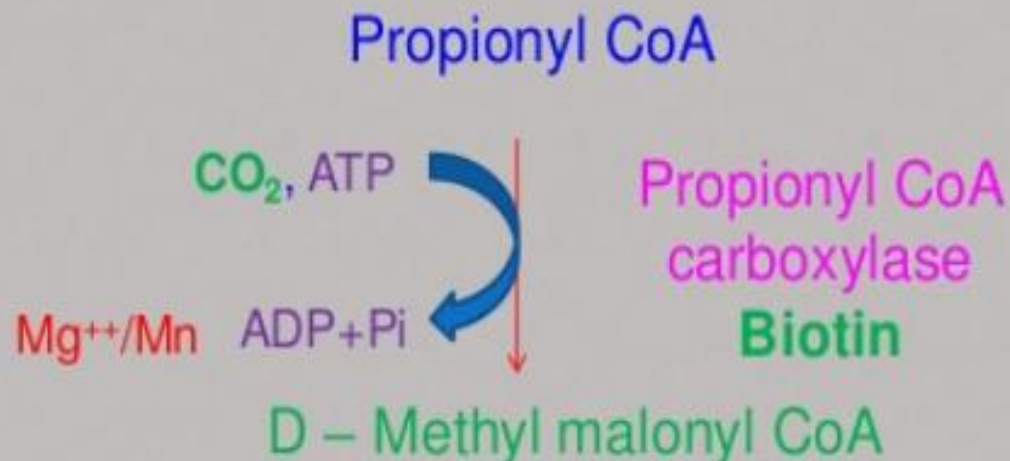
Acetyl CoA carboxylase

- Acetyl CoA carboxylase catalyzes the formation of malonyl CoA from acetyl CoA
- The reaction provides acetate molecule for fattyacid synthesis



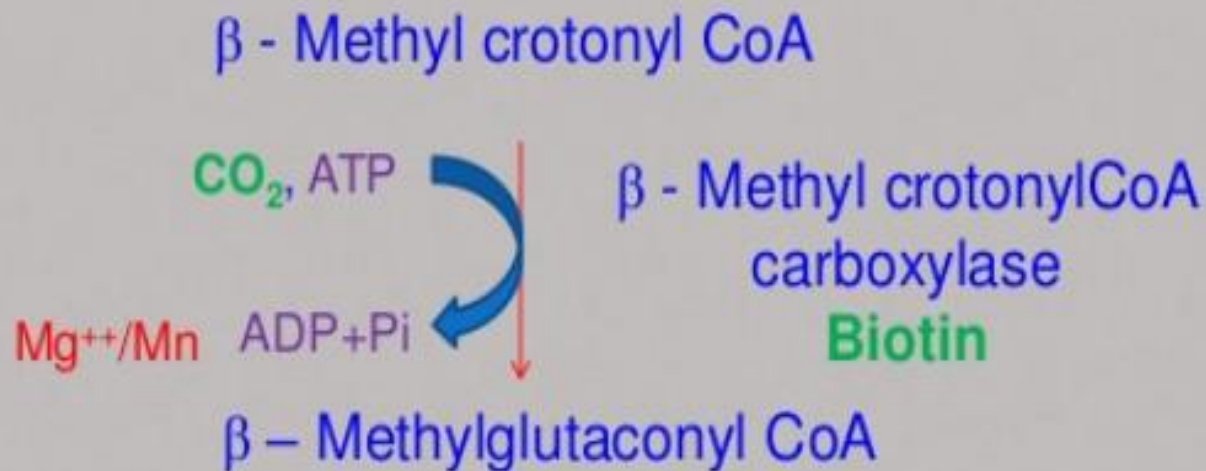
Propionyl CoA Carboxylase

- Propionyl CoA carboxylase catalyzes the formation of *D – Methyl malonyl CoA* from *propionyl CoA* (from odd chain FA & methionine)
- It required for entry of Propionyl CoA to TCA cycle via succinyl CoA



β - Methyl crotonyl CoA carboxylase

- β - Methyl crotonyl CoA carboxylase catalyzes the formation of β - Methylglutaconyl CoA from β - Methyl crotonyl CoA
- It essential for leucine catabolism



- All the carboxylation reactions in the biological system are not dependent on Biotin
- Few carboxylation reactions which do not require biotin
- Formation of carbamoyl phosphate in urea cycle
- Incorporation of CO_2 in purine synthesis

Biotin Antagonists

- Avidin (Raw egg white injury factor)
- Raw egg white injury factor is a heat labile protein known as avidin & is present in raw egg white
- Avidin binds to biotin & makes its unavailable for absorption
- Avidin is inactivated by boiling the eggs & biotin is readily absorbed when boiled eggs are used in the diet

Vitamin	Coenzyme	Coenzyme Function	Deficiency Disease
vitamin B ₁ (thiamine)	thiamine pyrophosphate	decarboxylation reactions	beri-beri
vitamin B ₂ (riboflavin)	flavin mononucleotide or flavin adenine dinucleotide	oxidation-reduction reactions involving two hydrogen atoms	—
vitamin B ₃ (niacin)	nicotinamide adenine dinucleotide or nicotinamide adenine dinucleotide phosphate	oxidation-reduction reactions involving the hydride ion (H ⁻)	pellagra
vitamin B ₆ (pyridoxine)	pyridoxal phosphate	variety of reactions including the transfer of amino groups	—
vitamin B ₁₂ (cyanocobalamin)	methylcobalamin or deoxyadenoxylcobalamin	intramolecular rearrangement reactions	pernicious anemia
biotin	biotin	carboxylation reactions	—
folic acid	tetrahydrofolate	carrier of one-carbon units such as the formyl group	anemia
pantothenic Acid	coenzyme A	carrier of acyl groups	—
vitamin C (ascorbic acid)	none	antioxidant; formation of collagen, a protein found in tendons, ligaments, and bone	scurvy



Nature and Properties of Enzymes

- All enzymes are proteins or protein in nature (except for a small group of catalytic RNAs)
 - RNA molecules with enzymatic activity are called ribozymes
 - But not all proteins are enzymes
- They are present at extremely low intercellular concentration (e.g. 10^{-7} molar)
- Small quantities are needed to catalyze reactions
- They are not altered or consumed during reaction
- They are reusable.
- They are not part of the reactants or products.

Nature and Properties of Enzymes Cont...

- Enzymes do not initiate chemical reactions or make reactions take place, they only stimulate and fasten the rate of **thermodynamically feasible** or favoured reactions to occur at a reasonable rate without thermal activation .
 - That is enzymes accelerate the approach of chemical reactions toward equilibrium without changing the equilibrium position.
- So any chemical reaction which proceeds in the presence of an enzyme will also proceed in the absence of the enzyme but at a much slower rate.
 - Enzyme accelerate reaction rates several fold compared to the rate of the same reaction in the absence of the enzyme.

Nature and Properties of Enzymes Cont...

- This is in accord with the Second Law of thermodynamics which tells whether a reaction can occur or not but say nothing about how *fast* it occurs.
- Enzyme accelerate reaction rates from 10^{17} to 10^{20} fold when compared to the uncatalyzed reaction.
- They accelerate the rate of both forward and backward reactions.
- They are selective, with a particular enzyme accelerating only a specific reaction.

Chemical catalysts versus enzymes

- Regardless of their chemical nature, all catalysts including enzymes, exhibit several common features:
- Both increase the rate of chemical reaction
- Both lower the **activation energy needed** to start a reaction.
- Both are not consumed or used up during the reaction i.e. they remain unchanged.
- Both are needed in small or minute quantities.
- Both do not alter equilibrium of the reactions they catalyzed
 - They simply decrease the amount of time required to achieve equilibrium.
 - They have no effect on the position of the equilibrium and therefore do not change the amount of free energy released or the direction in which the reaction will proceed.
- Both act by forming transient complexes with substrate molecules, ordering them in a manner that facilitates their interaction.
- Both are reusable

Unique properties of enzymes

- Enzymes exhibit several unique features that distinguish them from chemical catalysts routinely encountered in inorganic chemistry.
- These unique properties of enzymes relate to their:
- They have active site
- Catalytic power and efficiency
 - catalytic power is defined as the ratio of the enzyme-catalyzed rate of a reaction to the uncatalyzed rate.
- Catalytic power = $\frac{\text{Rate of enzyme-catalysed reaction}}{\text{Rate of uncatalysed reaction}}$
- Denaturation

Unique properties of enzymes cont...

- **Higher reaction rates**
 - The rates of enzyme catalysed reactions are typically several fold greater than those of the corresponding uncatalyzed reactions and several orders of magnitude than those of the corresponding chemically catalysed reactions.
- **Milder reaction conditions**
 - Enzyme catalysed reactions occurs under milder conditions such as; temperature, atmospheric pressure and near neutral pH. While chemical catalysts often require elevated temperature and pressure.
- **Greater reaction specificity**
 - Enzymes have greater degree of specificity with respect to the kind of reaction they catalyze as well as the identity of substrates acted upon than chemical catalysts.
- **They have capacity for regulation**
 - Ability to activate or inhibit enzymes can control which reactions occur and when.
 - The catalytic activities of many enzyme vary in response to regulations by substances other than their substrates and products.

Active site

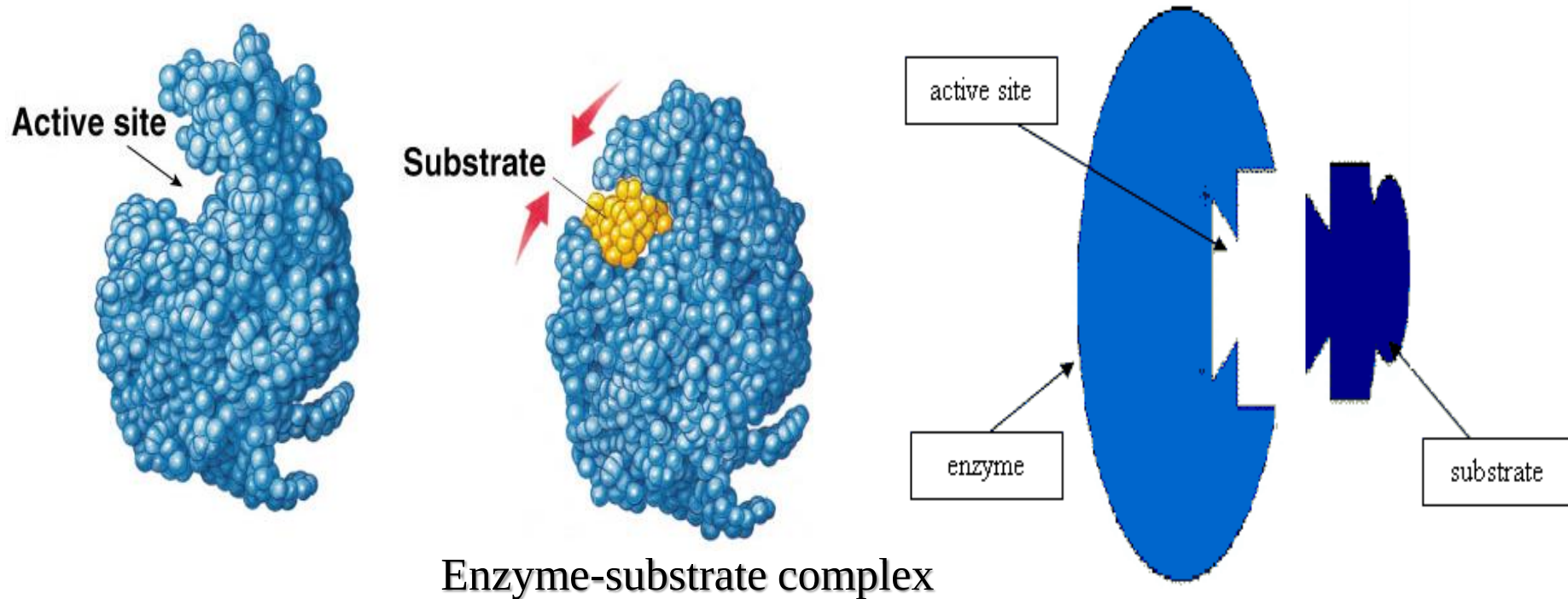
- The active site of an enzyme is the region on the surface of an enzyme that binds the substrate and convert it into products.
- Active site of an enzyme possesses a unique geometric shape and chemical properties that allow the enzyme to recognize a specific substrate and bind with it.
- The basic characteristics of the active sites include:
- Very small size compare to the entire enzyme molecule i.e. the active site of an enzyme occupies only a very small portion of the enzyme molecule.

Active site

- Its sequences of amino acid fold to form three dimensional structure which is found as a cleft or a crevice on the enzyme molecule.
 - Enzymes must be larger than their active sites so that the site chains responsible for binding and for catalyzing reactions can be juxtaposed appropriately into three dimensions such that the active site is always found in a **cleft** or **crevice**.
- The **active site** is also called the **binding site** or **catalytic site**. That is the substrate binds with the enzyme at the active site. It has two distinct functions binding substrate(s) and catalysis.

When the enzyme and substrate are connected, it is known as *enzyme-substrate complex*.

The enzyme-substrate complex is central to the action of enzymes.



The enzymatic reactions takes place by binding of the substrate with the active site of the enzyme molecule by several weak bonds.

Mechanisms of Enzyme Catalysis

- A number of mechanisms can be use to accelerate the conversion of substrates to the product of the reaction. These mechanisms include:
 - **Catalysis by proximity and optimal orientation**
 - Enzymes organize reactive groups between substrates and enzyme active into close proximity and proper orientation.
 - Substrates and enzyme active site must come together with the proper spatial relationship for a reaction to occur.
 - Proximity and optimal orientation effects suggest that molecules are maximally reactive when they are close enough and with optimal orientation.
 - This reduces their degrees of freedom and
 - Results in a large loss of entropy
 - Relatively enhanced the concentration of substrates since higher concentrations of substrate will increase their proximity to each other thereby promoting enhanced binding to enzyme resulting in increased catalysis

Mechanisms of Enzyme Catalysis Cont...

- **Acid-Base Catalysis**
- Biochemical processes are promoted by proton donors (acids) or proton acceptors (bases). That is acid-base catalysis gives and take protons.
- The active sites of some enzymes contain amino acid functional groups that can participate in the catalytic process as proton donors or proton acceptors.
- General acid-base catalysis involves partial proton transfer from a donor and partial proton abstraction from an acceptor that lowers the activation energy of the transition state.
- So, acid-base catalysts promote reactions by increasing the strength of the nucleophile (proton abstraction) and/or the stability of the leaving group (proton donation).

Mechanisms of Enzyme Catalysis Cont...

- **Catalysis by Strain** : Binding of Enzyme to substrates whose covalent bond are to be cleaved in an unfavorable configuration thereby exerting strain on the bonds ,stretching or distorting bonds.
- **Metal ion catalysis**
- **This** involves a metal ion bound to the enzyme interacts with substrate to facilitate binding, stabilizes negative charges and participates in oxidation reactions.

Mechanisms of Enzyme Catalysis Cont...

- **Covalent Catalysis**
- Covalent catalysis accelerates reaction rates through formation of transient covalent bond between the enzyme and the substrate which provides a reaction pathway with a lower activation energy.
- Covalent catalysis involves three stages: 1) nucleophilic reaction between enzyme and substrate 2) electrophilic withdrawal of electrons from substrate 3) elimination reaction (reverse of stage 1)
- On completion of reaction, enzyme returns to its original state. Cysteine, serine or histidine residues on enzyme participate in covalent catalysis

ENZYME NOMENCLATURE

- Nomenclature is a system of names or terms, or the rules for forming names.
- The principles of naming vary from discipline to discipline.
- The act of assigning names to enzymes is called enzyme nomenclature.
- The earliest enzymes discovered and studied were few and were given trivial or nondescriptive names as they were discovered

Enzyme nomenclature cont...

- Example:
 - Rennin- Curding of milk to start Cheese-making processes
 - Pepsin- hydrolyses protein at acidic pH.
 - Trypsin- hydrolyses protein at mild alkaline pH
 - Chymotrypsin
 - Catalase etc
- These names are historical and gives no idea of the source, function or reaction catalyzed by the enzyme.
- No direct relationship to the substrate or reaction type

Enzyme nomenclature cont...

- The nomenclature was later improved upon by adding the suffix -ase to the name of the substrate with which the enzyme functions i.e. react with.
 - In most cases, enzyme names end with –ase

- Example:

Substrate

Enzyme

- | | |
|---------------------|-------------------|
| • α -amylose | α -amylase |
| • Lactose | Lactase |
| • Maltose | Maltase |
| • Lipid(fat) | Lipase |
| • Urea | Urease |
| • Cellobiose | Cellobiase |
| • Fumarate | Fumarase |

Enzyme nomenclature cont...

- In the 1950's the increasing amounts of known enzymes were causing confusion. No official naming system for enzymes.
- The IUBMB created the International Commission on Enzymes in 1956 to deal with enzyme nomenclature
- Later replaced with the Nomenclature Committee of the International Union of Biochemistry and Molecular Biology (NC-IUBMB)

Enzyme nomenclature cont...

IUBMB has recommended system of nomenclature for enzymes & according to them each enzyme is assigned with two names:

- ✓ **Trivial name** (common name, recommended name).
- ✓ **Systemic name** (official name).
- ✓ **Systemic name**

Each enzyme is characterized by a code no.called Enzyme Code no. or **EC number** and contain four Figure (digit) separated by a dot.

e.g. **EC m. n. o. p**

First digit represents the class;

Second digit stands for subclass ;

Third digit stands for the sub-sub class or subgroup;

Fourth digit gives the serial number of the particular enzyme in the list.

e.g. **EC 2.7.1.1 for hexokinase.**

ENZYME CLASSIFICATION

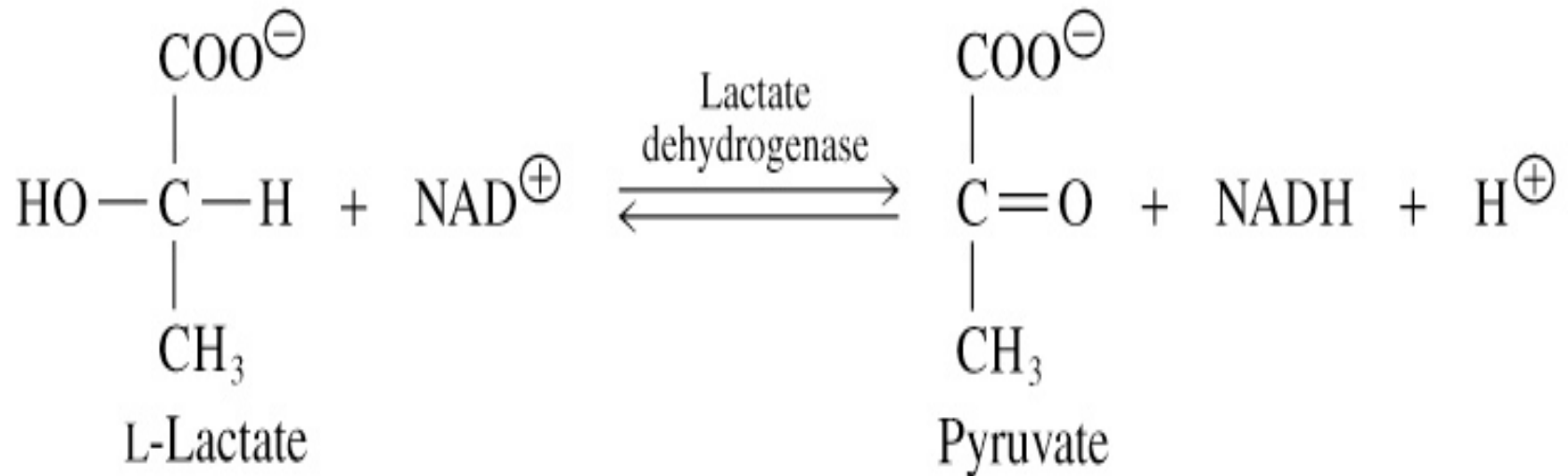
- The most accepted classification of enzymes is based on the types of reactions that they Catalyze.
- Enzymes are classified into six major classes (EC number Classification)
 - **EC 1. Oxidoreductases**
 - **EC 2. Transferases**
 - **EC 3. Hydrolases**
 - **EC 4. Lyases**
 - **EC 5. Isomerases**
 - **EC 6. Ligases**

EC 1. Oxidoreductases

- These groups of enzyme catalyse catalyze the transfer of electrons between species i.e., oxidation and reduction reaction.
- They act on many chemical groups to add or remove hydrogen atoms. They are subdivided into:
- Dehydrogenases:
 - which catalyse the removal of 2 atoms of hydrogen from substrates and their transference to a co-acceptor. (They remove 2H to form double bonds).
- Oxidases:
 - They catalyse the direct reduction of oxygen (oxidation).
- Oxygenases:
 - They catalyse the incorporation of oxygen into substratemolecule.

- Oxidative deaminates:
 - they catalyse the oxidation of amino compounds with elimination of a molecule of NH_3 ammonia.
- Examples:
 - Lactate dehydrogenase.
 - Glucose Oxidase.
 - Peroxidase.
 - Catalase.
 - Phenylalanine hydroxylase.

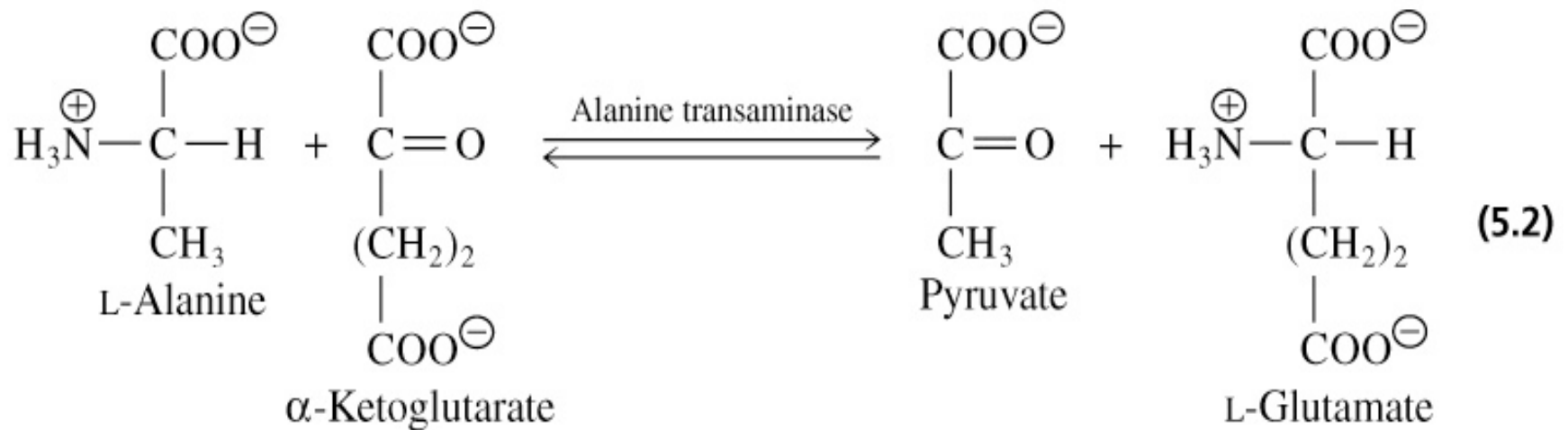
- Catalyze **oxidation-reduction** reactions



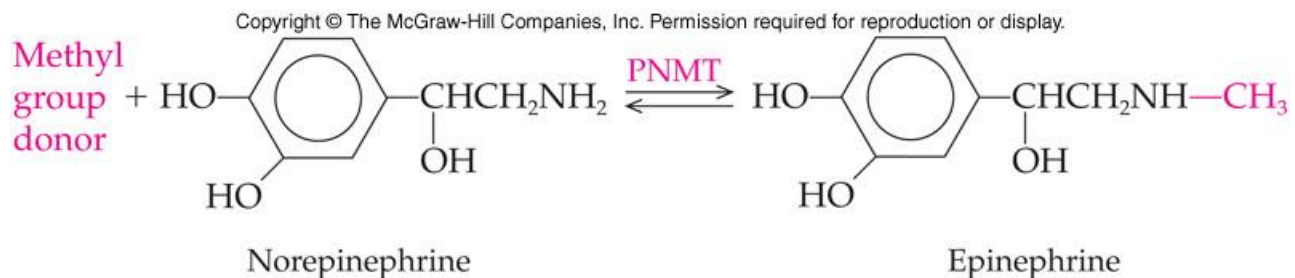
EC 2. Transferases

- These are enzymes that catalyze the transfer of a functional group from one molecule to another.
 - Transferases bring about the exchange of groups between donor and acceptor molecules.
$$AB + CD \rightarrow AC + BD$$
- They are subdivided into:
- Amino transferases:
 - These bring about the exchange of amino and keto group between an amino acid and a keto acid.
- Kinases:
 - These bring about the transfer of a phosphate radical using ATP as the donor or ADP acceptor.
- Phosphorylases:
 - which are phosphorylytic analog of the hydrolytic enzymes. They catalyse the splitting of the substrates with a molecule of phosphoric acid instead of water.

- Glycosyltransferases:
 - These bring about the transfer of a glycosyl group.
- (e)C1-Transferases:
 - These bring about the transfer of an acyl group.
- Examples:
 - Transaminases (ALT & AST).
 - Phosphotransferases (Kinases).
 - Transmethylases.
 - Transpeptidases.
 - Transacylases.



Transaminases catalyze transfer of an amino group
 Kinases transfer a phosphate group

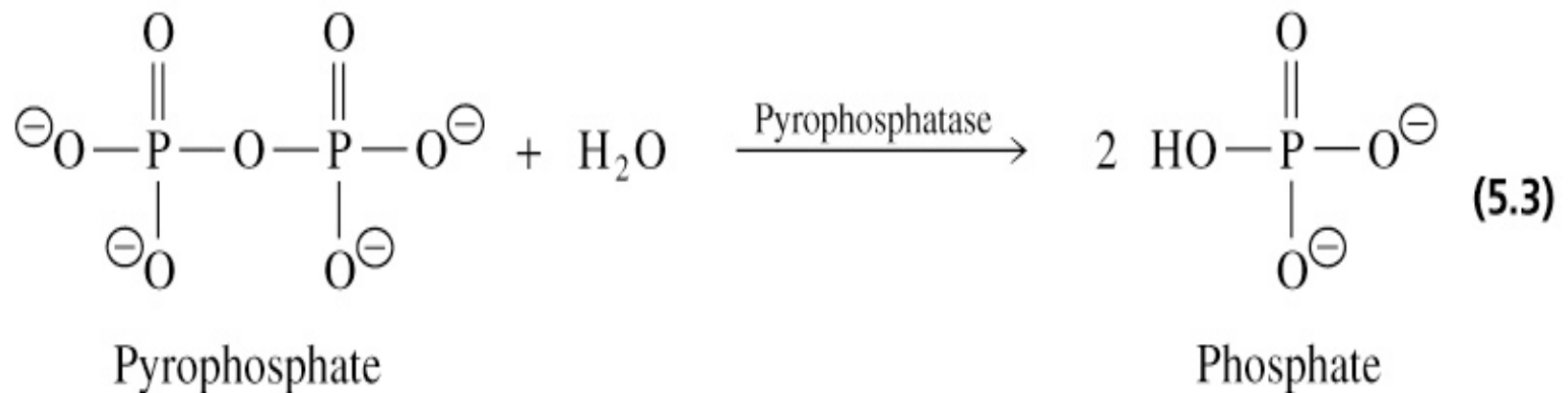


EC 3. Hydrolases

- These groups of enzymes employ water to cleave covalent bonds. They include the digestive enzymes. They catalyze the hydrolytic cleavage of C-C, C-O, C-N, P-O, and certain other bonds, including acid anhydride bonds
- They catalyse this type of reaction.
- $AB + H_2O \rightarrow AOH + HB$
- This group may be subdivided according to the type of substrates acted on.
- Peptidases
 - Enzymes (proteolytic enzymes or peptide hydrolase) which catalyse the hydrolysis of the peptide bonds.

- Amylases
 - These are enzymes (carbohydrases or glycoside hydrolases) which catalyse the hydrolysis of glycosidic bonds.
- Lipases
 - These are enzymes which catalyse the hydrolysis of lipids.
- Phosphatases
 - These are enzymes that catalyse the hydrolysis of phosphoric acids.
- Examples:
 - Protein hydrolyzing enzymes (Peptidases).
 - Carbohydrases (Amylase, Maltase, Lactase).
 - Lipid hydrolyzing enzymes (Lipase).
 - Deaminases.
 - Phosphatases.

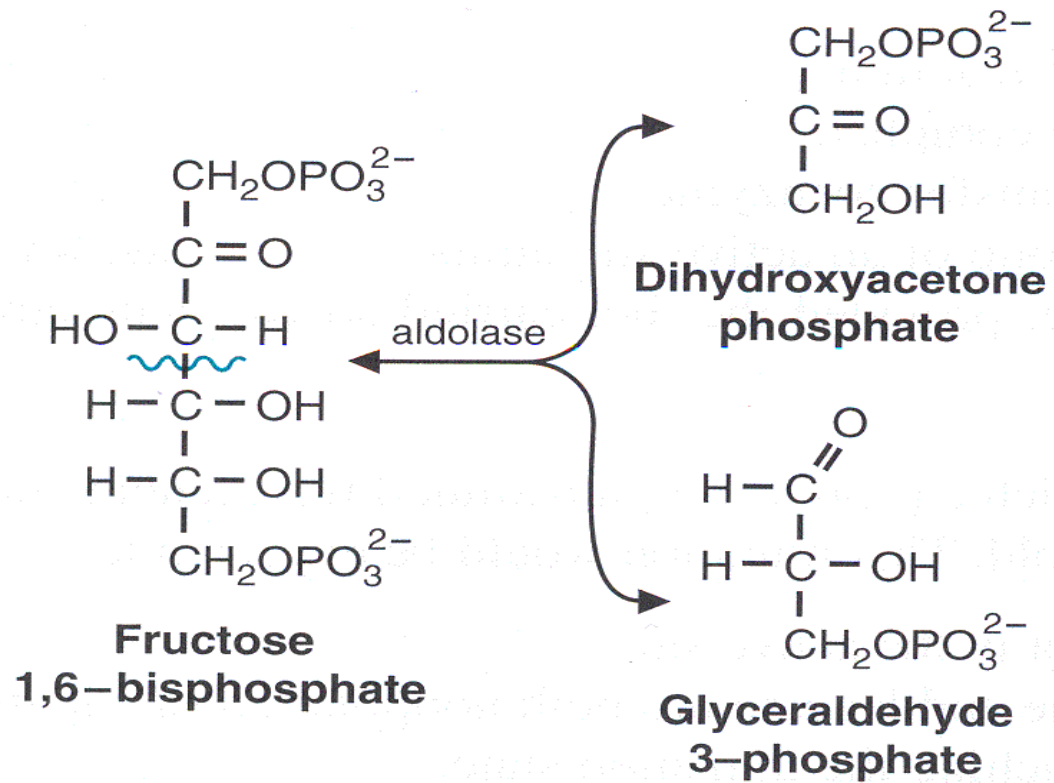
- Catalyze **hydrolytic reactions** where water is the acceptor of the transferred group



- **esterases**
- **peptidases**
- **glycosidases**

EC 4. Lyases

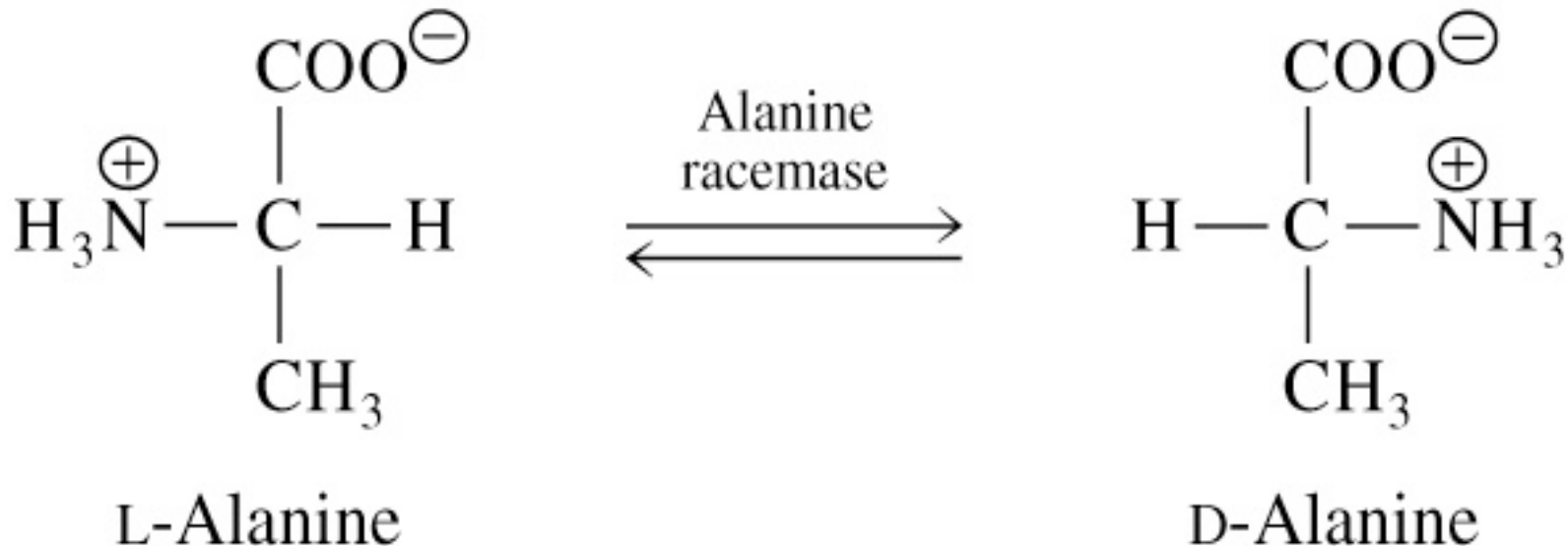
- These enzymes catalyze non-hydrolytic cleavage of various bonds
- **Lyases** catalyze removal of groups to form double bonds or the reverse break double bonds
 - Decarboxylases
 - Synthases
 - Add Water, Ammonia or Carbon dioxide across double bonds, or remove these elements to produce double bonds.
- Examples:
 - Fumarase.
 - Carbonic anhydrase.



EC 5. Isomerases

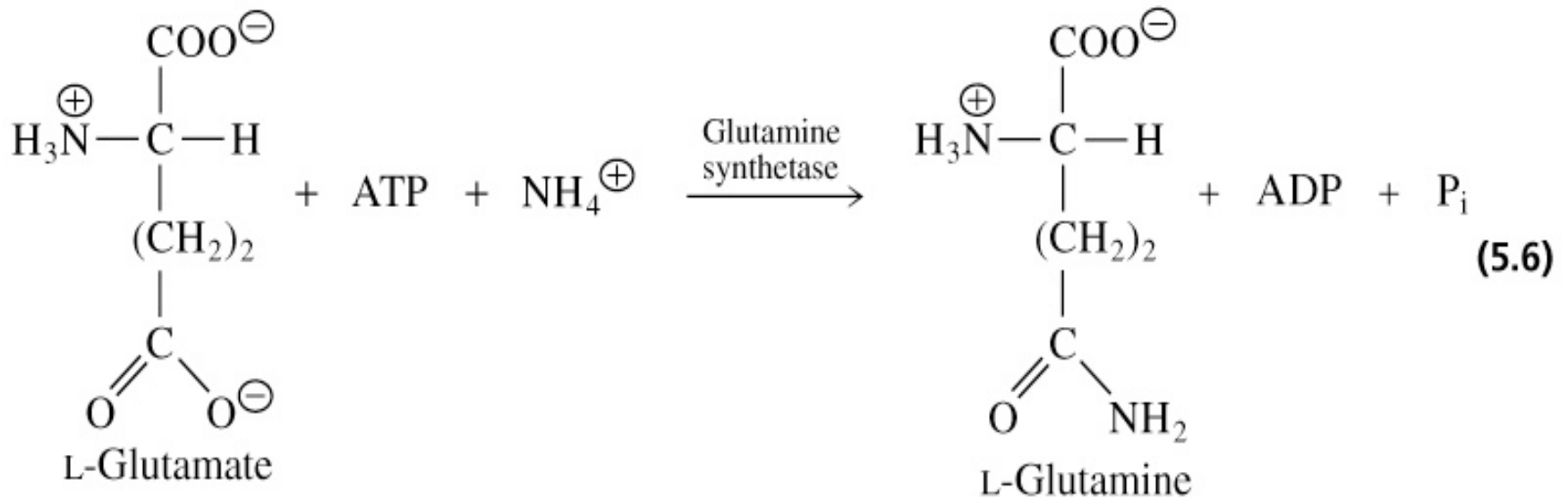
- They are enzymes which catalyse internal rearrangement within a molecule and therefore do not involve the addition or removal of group.
- They carry out many kinds of isomerization:
 - L to D isomerizations.
 - Mutase reactions (Shifts of chemical groups).
- Examples:
 - Isomerase.
 - Mutase.
 - Epimerases

- Catalyze **isomerization** reactions



6. Ligases (synthetases)

- Catalyze **ligation**, or joining of two substrates
- **Require chemical energy** (e.g. ATP)



Enzyme Specificity

- **Specificity** refers to the ability of an enzyme to discriminate between two competing substrates.
- Enzymes are highly specific in their action compared with chemical catalysts.
- Enzymes can distinguish between closely related
 - chemical species
 - D and L isomers
 - *cis and trans isomers*
 - Diastereomers (glucose and galactose)

Enzyme Specificity Cont...

- Enzymes have a high degree of specificity for their substrates and the type of reactions they catalyze.
- The specificity of enzymes is determined by its active site.
- The formation of the enzyme substrate complex can occur only if the substrate possesses groups which are in the correct three dimensional arrangement to interact with the binding groups of the active site.

Types of Enzyme Specificity

- Enzymes exhibit different types of specificity
- (a) Substrate specificity-this may be absolute, relative or broad:
 - **Absolute specificity:** Some enzymes act on only one substrate. Such enzymes are said to exhibit absolute specificity. For example, succinate dehydrogenase, a key enzyme of TCA cycle catalyzes only the oxidation of succinate to fumarate.
 - **Absolute group specificity:** Some other enzymes act on a very small group of substrates having the same functional group but at different rates. Such enzymes are said to exhibit absolute group specificity. For example, alcohol dehydrogenase oxidizes both ethanol and methanol which have common hydroxyl group. Similarly, hexokinase not only phosphorylates glucose but also fructose and mannose.
 - **Relative group specificity:** Some other enzymes exhibit relative group specificity where a given enzyme can act upon more than one group of substrates. For example, trypsin catalyzes the hydrolysis of both ester and amide bonds.

Types of Enzyme Specificity

- **Bond Specificity:** Enzymes that are specific for a bond or linkage such as ester, peptide or glycosidic belong to this group. Examples:
 - Esterases- acts on ester bonds
 - Peptidases-acts on peptide bonds
 - Glycosidases- acts on glycosidic bonds
- **(b) Stereospecificity:**
 - The enzymes act only on one isomer and therefore exhibit stereoisomerism. e.g., L-amino acid oxidase act on L-amino acids; hexokinase on D-hexose (Note: isomerases do not exhibit stereospecificity).
- **(c) Reaction specificity:**
 - The same substrate can undergo different types of reactions, each catalysed by a separate enzyme e.g., amino acids undergoing transamination, decarboxylation etc.

Models of Enzyme Specificity

- Enzymes are so specific because the active site of each enzyme has the proper
 - shape,
 - size and
 - chargeto bind certain substrates only and to catalyze the conversion of these substrates to specific products.
- There are two models that explain enzyme specificity:
 - The Lock and Key model by Emil Fischer
 - The induced-fit model by Daniel Koshland

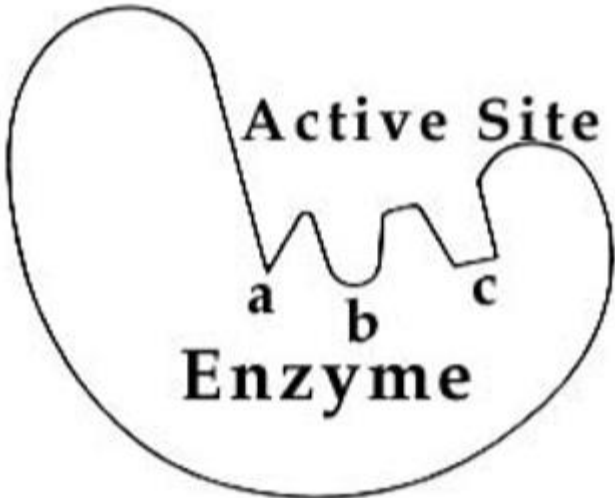
Lock and Key Model

- The lock and key model of enzyme specificity proposed by Emil Fischer in 1894 holds that the active site of an unbound enzyme is complementary in shape to that of the substrate.
- Enzymes exhibit remarkable specificity because of the precise fit between their binding sites and their substrates.
- Each enzyme binds to a single type of substrate because the active site of the enzyme and substrate have rigid complementary structures.
- The substrate overall shape and charge distribution allow it to enter and interact with the enzyme's active like the lock and key.
- In the **lock-and-key model**, the enzyme is assumed to be the lock and the substrate the key.

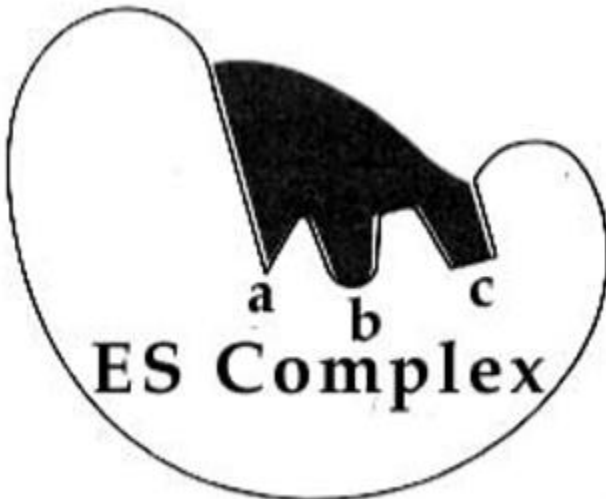


Substrate

+



Active Site
Enzyme



ES Complex

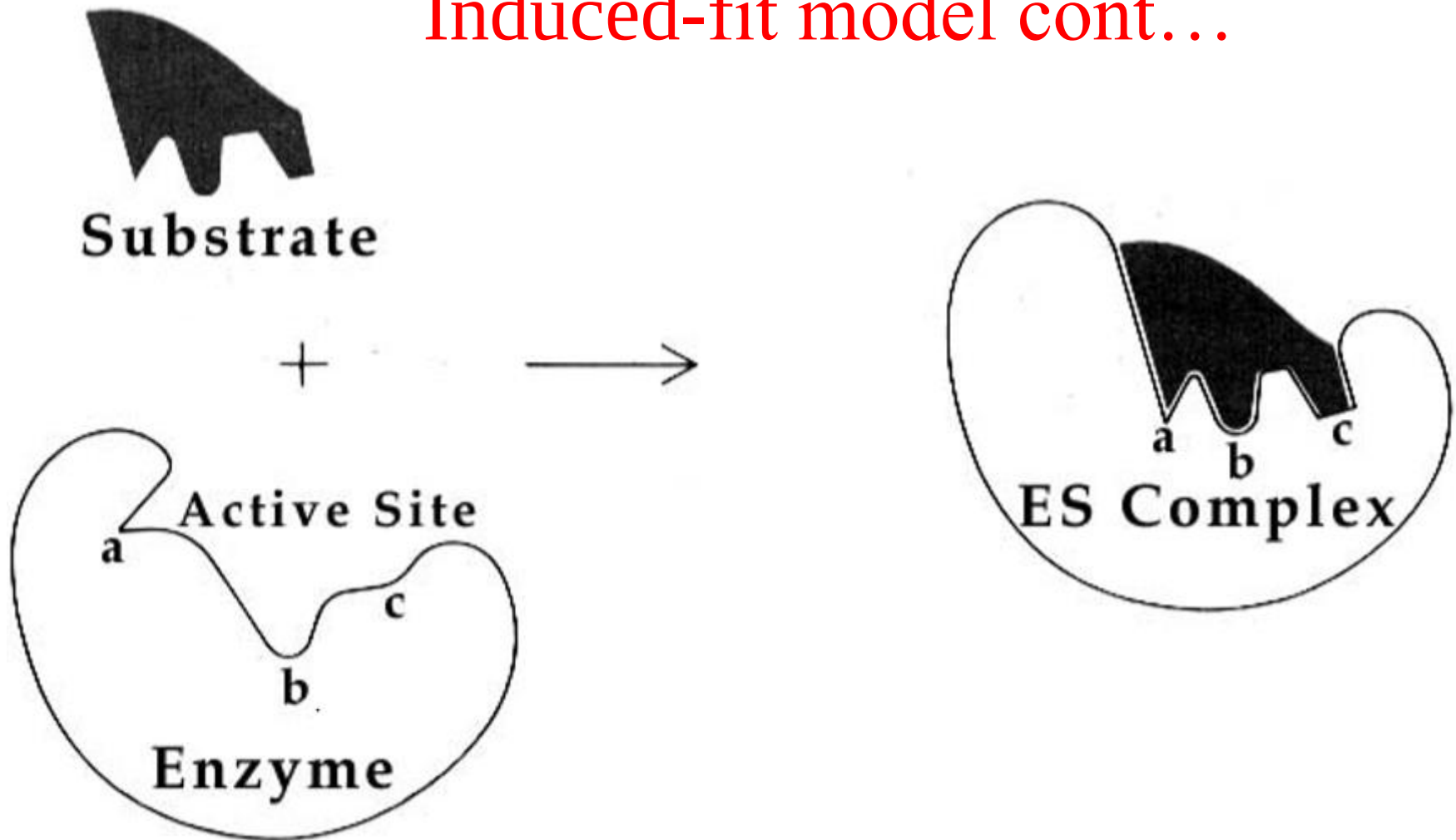
Lock and key model cont...

- The lock and key analogy has been widely used to explain enzyme specificity
- This is inadequate because the formation of enzyme substrate complex involves more than a steric complementarity between the enzyme and the substrate.
- It involves chemical interaction between the enzyme and the substrate in a solution.
- Hence, there may not be that rigid shape like the lock and key.
- This model fails to take into account proteins conformational changes to accommodate a substrate molecule

Induced-fit Model

- The induced fit model proposed by Daniel Koshland in 1968 holds that the active site of an enzyme has a shape complementary to that of the substrate only after the substrate is bound to the enzyme.
- The enzyme changes shape upon binding substrate i.e. substrate binding causes a conformational change in the enzyme which make the substrate to fit precisely into the active site.
- In the induced fit model, substrate does not fit precisely into a rigid active site.
- Instead noncovalent interactions between the enzyme and the substrate changes the structure of the active site conforming it to the shape of the substrate.

Induced-fit model cont...



Induced-fit model cont...

- The induced-fit model of enzyme action assumes that the enzyme active site:
 - is a flexible pocket
 - changes conformation to accommodate the substrate molecule.
- The induced-fit model replaced the lock and key model because it accounts for the flexible nature of proteins of which enzymes are.

ENZYME CATALYSIS

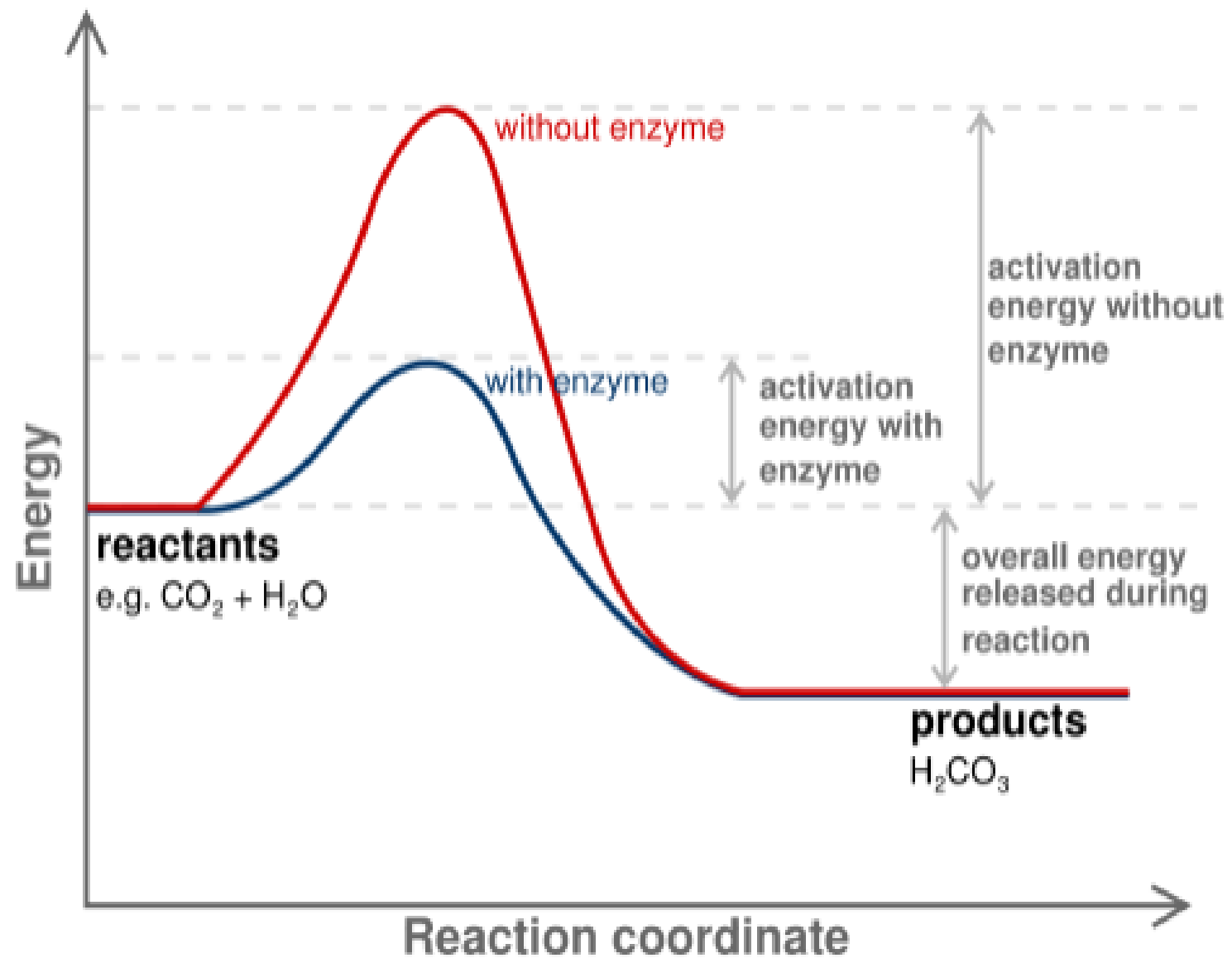
- The catalytic action of an enzyme is referred to as its activity, is measured by determining the increase in the reaction rate under precisely defined conditions.
- In other words, the process of increasing the rate of an enzyme catalyzed reaction is called enzyme catalysis.
- The catalytic ability of an enzyme is determined by the active site because it is the region on the enzyme responsible for binding substrate (and the prosthetic group if any).
- Every chemical reaction is characterized by an equilibrium constant which is a reflection of the difference in energy between reactants and products

Enzyme Catalysis Cont...

- Generally, enzymes increase the rate or velocity, V , of many physiological reactions in biological systems by lowering the activation energy of the reactions they catalyze.
- Activation energy is the minimum amount of energy that must be imputed into the reaction to get it started so that reactants are converted to products.
- In other words, activation energy is the minimum energy barrier that must be met for a chemical reaction to occur.
- It is believed that enzymes bind to their substrates and lower the energy required for activation of the reaction, the reaction occurs and the enzyme is then released unchanged to be used again.

Enzyme Catalysis Cont...

- The initial interaction of the enzyme with the substrate leads to the formation of ***enzyme–substrate complex (ES)***.
- Enzymes lower the activation energy by forming ES complex.
- ES is formed through the interactions between the enzyme and substrate. Each interaction releases a small amount of energy to stabilize the complex.
- These interactions combine to lower the activation energy of the reaction.
- The uncatalyzed reaction has a large activation energy, E_a , than the catalyzed reaction
- In the catalyzed reaction, the activation energy has been lowered significantly leading to increased rate of the reaction. This provides a lower energy route for conversion of substrate to product.
- Enzymes lower the activation energy, but do not change the free energy required for the reaction to occur (alters the rate, but not thermodynamics).



Enzyme kinetics

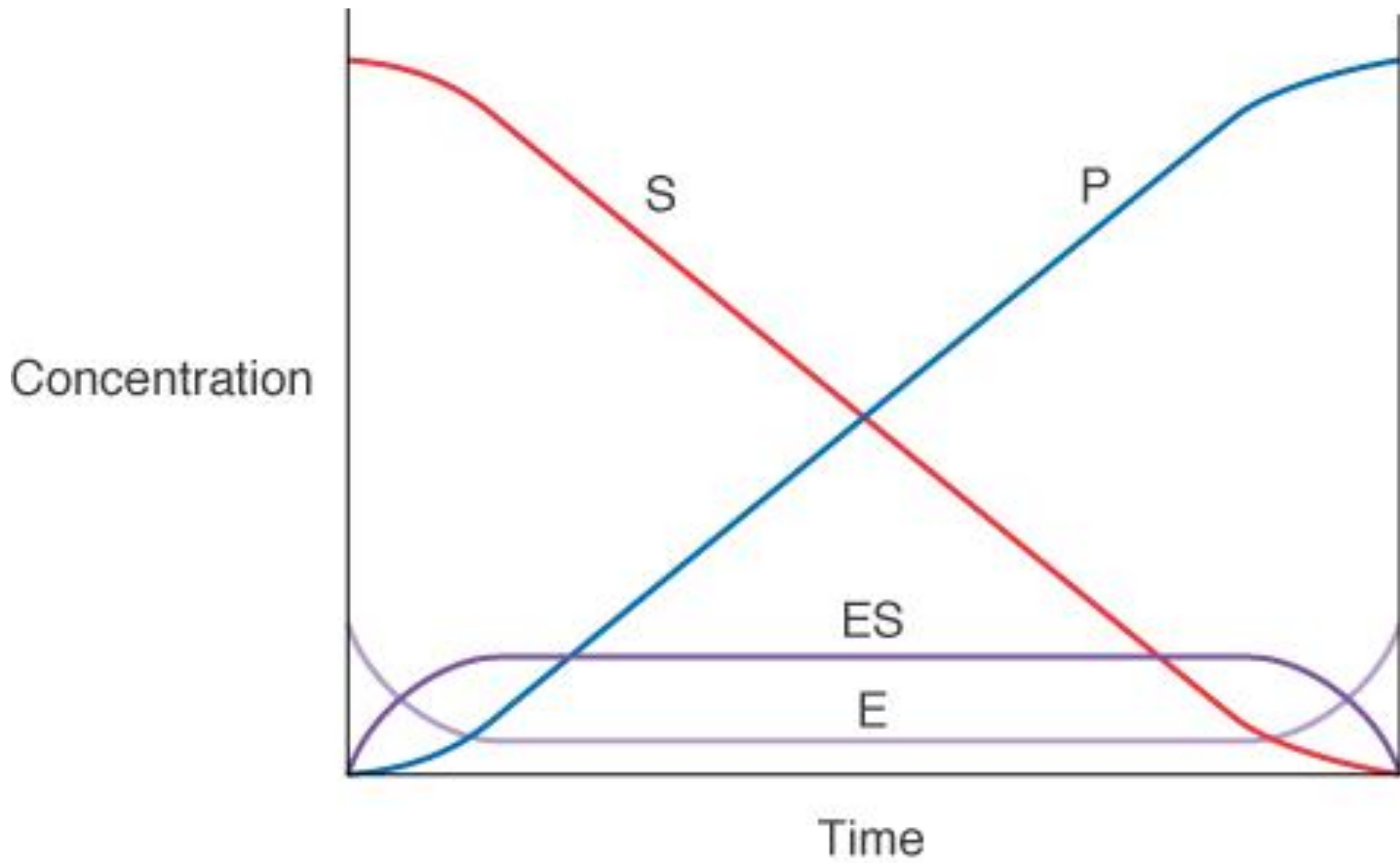
- Enzyme kinetics is the study of the velocity of enzyme catalyzed reactions and factors that affect the rate of enzyme catalyzed reaction.
- Enzyme kinetics provides information about the basic mechanisms of enzyme catalyzed reaction and other parameters that characterize the properties of enzymes.
- The velocity of all enzymes is dependent on the concentration of the substrate.
- The velocity of an enzyme catalyzed reaction is greatest when all the sites at which catalytic activity can take place on the enzyme molecule (active sites) are filled with substrate i.e when substrate concentration is very high.
- Enzyme kinetics is a branch of chemical kinetics applies to enzyme catalyzed reaction where a rate expression relates the rate of reaction to the concentration of its reactants.
- Considering a reaction type

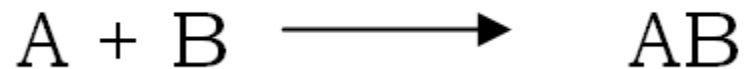
Over the course of reaction, the concentration of the substrate falls as the concentration of the product rises.

The progress of this reaction or any other reaction can be expressed as **velocity** (v); *i.e.*, either the rate of disappearance of the substrate (S) or the rate of appearance of the product (P).

$$v = -\frac{d[S]}{dt} = \frac{d[P]}{dt}$$

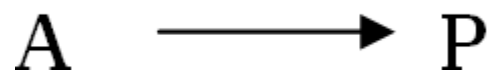
Considering a reaction type





The rate expression is = $K[A][B]$

The rate or velocity(v) at which a reactant is consumed or a reaction product appears can be mathematically described for a reaction type

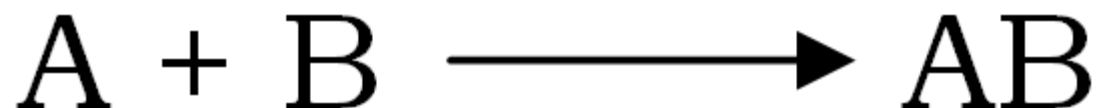


$$V = \frac{d[P]}{dt} = -\frac{d[A]}{dt} = K[A]$$

This implies that the rate of chemical change varies directly with the active concentrations of the reactants.

The value of the rate constant for enzyme catalyzed reaction is larger than that for the same reaction which is not enzyme-catalyzed.

The order of a chemical reaction is the sum of the powers to which the reactant concentrations are raised in the rate expression. For a reaction type



the order of the reaction is first order with respect to each reactant A and B but second order with respect to the overall reaction

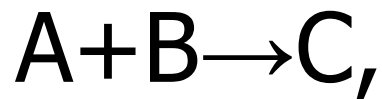
The order of a reaction is given by the exponents of the reactant concentrations in the rate law.

- $v = k[A]^n$
- $n=0$, zero order (no dependence on $[A]$)
- $n=1$, first order
- $n=2$, second order
- $v = k[A]^2[B]$
- second order in A, first order in B, third order overall

- The rate or velocity of a given chemical reaction is the measure of the change in concentration of the reactants or the change in concentration of the products per unit time.
- The speed of a chemical reaction may be defined as the change in concentration of a substance divided by the time interval during which this change is observed:

$$\text{Rate or velocity} = \frac{\Delta \text{concentration}}{\Delta \text{time}}$$

For a reaction type



the rate can be expressed in terms of the change in concentration of any of its components

$$\text{Rate or velocity} = -\frac{\Delta[A]}{\Delta t}$$

$$\text{Rate or velocity} = -\frac{\Delta[B]}{\Delta t}$$

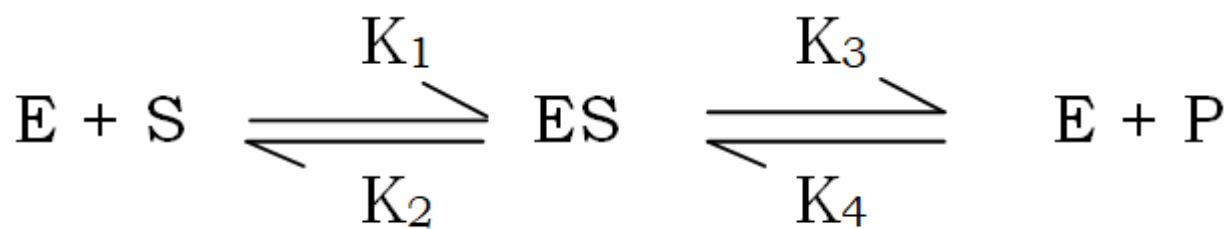
$$\text{Rate or velocity} = \frac{\Delta[C]}{\Delta t}$$

in which $\Delta[A]$ is the difference between the concentration of A over the time interval $t_2 - t_1$:

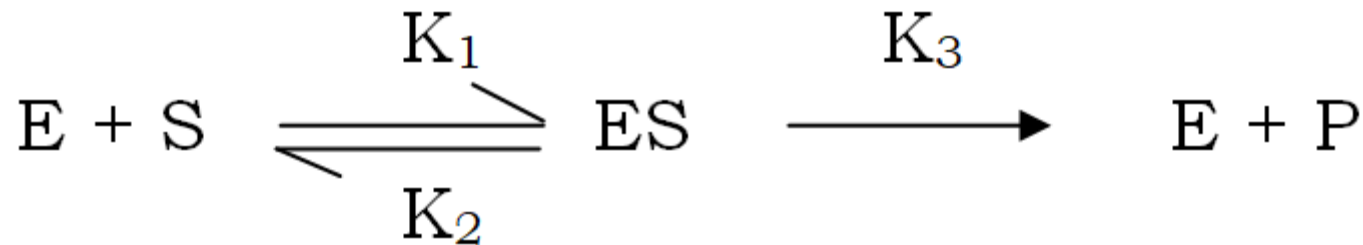
$$\Delta[A] = [A]_{\text{Final}} - [A]_{\text{Initial}}$$

MICHAELIS-MENTEN KINETICS

- In 1913 Leonor Michaelis and Maud Menten proposed a theory to explain the relationship between the velocity of enzyme catalyzed reaction and $[S]$.
- According to Michaelis-Menten model of enzyme action, the substrate, S binds reversibility to an enzyme, E to form an enzyme substrate complex, ES which then reacts irreversibly to generate a product, P and to regenerate the free enzyme, E .
- Michaelis-Menten model is very useful in the quantitative analysis of almost all aspects of enzyme kinetics.
- The Michaelis-Menten reaction scheme:



But $K_4 = 0$



E = free enzyme

S = substrate

ES = enzyme-substrate complex

P = product

K_1 , K_2 & K_3 = rate constants for the formation and dissociation of ES

K_1 = the forward rate constant for substrate binding

K_2 = the reverse rate constant for substrate binding

K_3 = catalytic rate constant

Michaelis-Menten kinetics cont...

- The Michaelis-Menten model of enzyme action is based on the following assumptions:
 1. $K_2 \gg K_3$, therefore, the first step of the reaction achieves equilibrium
 2. [ES] maintains a steady state such that the rate of formation of ES is equal to its rate of breakdown over most period of the reaction
 3. The reaction of ES to P + E is not reversible
- The reaction $E + S \leftrightarrow ES$ reaches equilibrium very quickly and maintains it throughout the overall reaction.
- This is the rapid equilibrium assumption upon which Michaelis-Menten equation is based.
- At equilibrium the rate of formation of ES equal to breakdown

The rate of formation of ES = $k_1 [E] [S]$

The rate of breakdown of ES = $k_2 [ES]$

at equilibrium, the rate of formation of ES equal to the rate of breakdown of ES, therefore

$$k_1 [E] [S] = k_2 [ES]$$

$$[E] = \frac{k_2 [ES]}{k_1 [S]}$$

$$[E] = [ES] \frac{k_2}{k_1 [S]}$$

The total amount of enzyme, $[E]_t$, in the system at any give time is conserved and can be given as

$$[E]_t = [E] + [ES]$$

Substituting the value for $[E]$ into the enzyme conservation equation

$$[E]_t = [ES] \frac{k_2}{k_1[S]} + [ES]$$

Factorizing $[ES]$

$$[E]_t = [ES] \left(1 + \frac{k_2}{k_1[S]} \right)$$

Rearrange and express in terms of ES

$$[ES] = \frac{[E]_t}{1 + \frac{k_2}{k_1[S]}}$$

Multiplying both the numerator and denominator by [S]

$$[ES] = \frac{[E]_t[S]}{[S] + \frac{k_2}{k_1}}$$

the over all rate of the reaction, v , is given as

$$v = k_3 [ES]$$

$$[ES] = \frac{v}{k_3}$$

Substituting for [ES]

$$V = \frac{K_3[E]_t[S]}{\frac{k_2}{k_1} + [S]}$$

At high substrate concentration when all the enzyme is bound to substrate then $[E]_t = [ES]$ and under these conditions the velocity of the reaction is maximum, V_{max}

$$k_3[E]_t = V_{max}$$

$$\frac{k_2}{k_1} = K_M \text{ (Michaelis-Menten constant).}$$

$$V = \frac{V_{\text{Max}}[S]}{K_M + [S]}$$

This is Michaelis-Menten equation under equilibrium assumption

A more general model of enzyme kinetics describing the relationship between velocity and substrate concentration was provided by George Briggs and G.E.S. Haldane which is based on steady state

The steady state is based on the assumption that the intermediate enzyme-substrate complex in the reaction is consumed as quickly as it is generated such that the overall reaction is in a steady state i.e. the rate of formation equals to the rate of breakdown

$$\frac{d[ES]}{dt} = 0$$

$$\frac{d[ES]}{dt} = K_1[E][S] - K_2[ES] - K_3[ES] = 0$$

From enzyme conservative equation

$$[E]_t = [E] + [ES]$$

$$[E] = [E]_t - [ES]$$

Substituting and rearranging

$$K_1([E]_t - [ES])[S] - K_2[ES] - K_3[ES] = 0$$

$$K_1[E]_t[S] - K_1[S][ES] - K_2[ES] - K_3[ES] = 0$$

$$K_1[E]_t[S] = K_1[S][ES] + K_2[ES] + K_3[ES]$$

$$K_1[E]_t[S] = [ES] (K_1[S] + K_2 + K_3)$$

$$[ES] = \frac{K_1[E]_t[S]}{K_1[S] + K_2 + K_3}$$

Dividing both the numerator and denominator K_1

$$[ES] = \frac{[E]_t[S]}{[S] + \frac{K_2 + K_3}{K_1}}$$

the over all rate of the reaction, v , is given as

$$v = k_3 [ES], \quad [ES] = \frac{v}{k_3}$$

Substituting for $[ES]$

$$V = \frac{k_3 [E]_t[S]}{[S] + \frac{K_2 + K_3}{K_1}}$$

At high substrate concentration when all the enzyme is bound to substrate then $[E]_t = [ES]$ and under these conditions the velocity of the reaction is maximum, V_{max}

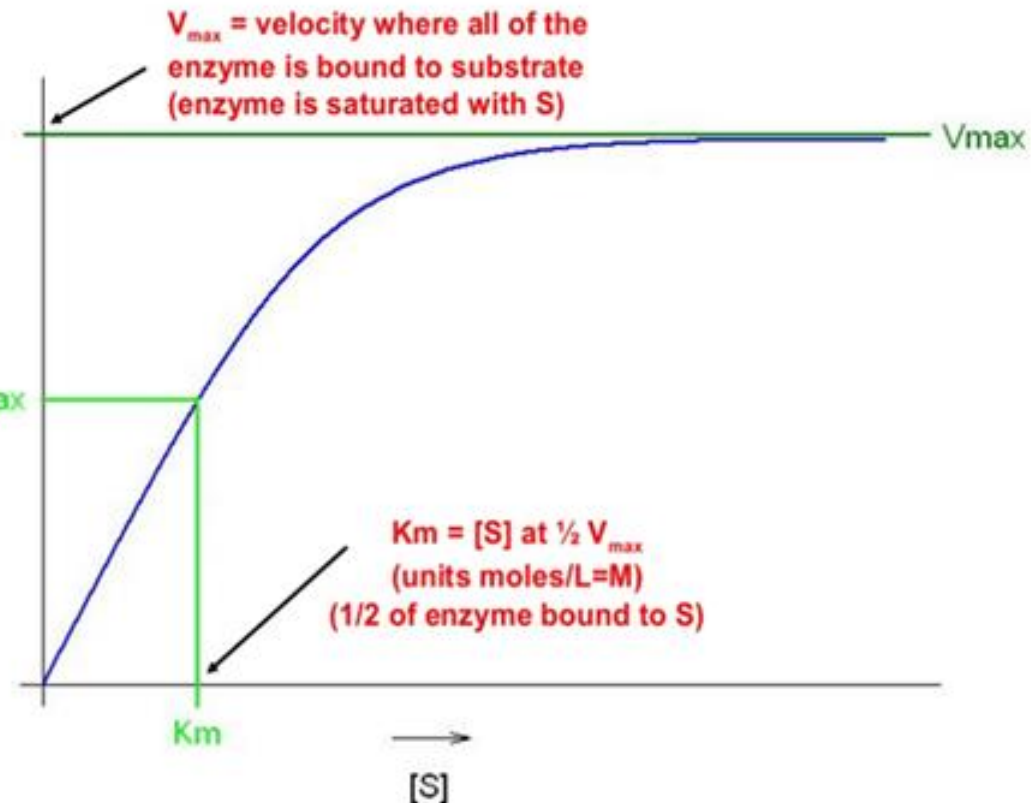
$$k_3[E]_t = V_{max}$$

$$\frac{K_2 + K_3}{K_1} = K_M \quad (\text{Michaelis-Menten constant}).$$

$$V = \frac{V_{\text{Max}} [S]}{[S] + K_M}$$

This is Michaelis-Menten equation under steady state assumption

A plot of Michaelis-Menten equation represents the initial velocity, V as a function of substrate concentration, S with the kinetic parameters; Michaelis-menten constant, K_m and the maximal velocity, V_{max}



$$V_0 = \frac{V_{\text{max}} [S]}{K_m + [S]}$$

The two equations i.e. Michaelis-Menten and Briggs-Haldane are identical except that Michaelis-Menten equation has its K_M

$$K_M = \frac{K_2}{K_1} = K_d \text{ (Enzyme dissociation)}$$

While in the steady state assumption, Briggs Haldane equation has

$$K_M = \frac{K_2 + K_3}{K_1}$$

When $K_2 \gg K_3$, then the Briggs-Haldane equation becomes equal to Michaelis-menten equation.

Important Conclusions of Michaels - Menten Kinetics

- when $[S] \gg K_M$, the equation reduces to

$$V = \frac{V_{\max} [S]}{K_M + [S]} = \frac{V_{\max} [S]}{[S]} = V_{\max}$$

- when $[S] \ll K_M$, the equation reduces to

$$V = \frac{V_{\max} [S]}{K_M + [S]} = \frac{V_{\max} [S]}{K_M} = \frac{V_{\max}}{K_M} [S]$$

- when $[S] = K_M$, the equation reduces to

$$V = \frac{V_{\max} [S]}{K_M + [S]} = \frac{V_{\max} [S]}{[S] + [S]} = \frac{V_{\max}}{2}$$

Enzyme activity

- Enzyme activity is the ability of an enzyme to catalyze a specific reaction.
- It is a measure of the rate at which the reaction proceeds from substrate to products.
- Enzymes are measured in terms of activity rather than concentration because of their chemical similarities, and their extremely low concentrations in biological fluids.

- Enzyme activity is normally expressed in international units per milliliter of enzyme-containing solution, or where this value is inconveniently small: as milli-units per milliliter.
- The international unit of enzyme activity is defined as the amount of enzyme which will catalyze the transformation of one micro-mole of substrate per minute under certain specified conditions such as pH and temperature.
- Enzyme activity depends on a number of factors such as temperature, pH, substrate, coenzymes and inhibitors.

Enzyme activity is measured as an initial reaction velocity. Reaction rates are initial rates determined when the substrate is in vast excess and is not changing much.

At the start of a reaction, $[S]$ is in large excess of $[P]$, thus the initial velocity of the reaction will be dependent on substrate concentration

That is the velocity before much substrate has been depleted and before much product has been generated.

It is easier to measure the appearance of a small amount of product from a baseline of zero products

Than to measure the disappearance of a small amount of substrate against a background of high concentration of substrate.

KINETIC PARAMETERS

- Michaelis-Menten Constant, K_M
- Maximum velocity, $V_{\max}$
- Catalytic Constant, K_{cat} or turnover number
- Catalytic efficiency, K_{cat}/K_m

K_m (Michaelis Constant)

- K_m is the substrate concentration at which the initial rate is one-half of the maximum rate ($\frac{1}{2} V_{\max}$)
- It is the $[S]$ required to saturate half of all of the active sites of the enzymes
- The unit of K_m is moles per litre
- K_m is determined by measuring the reaction velocity at different substrate concentration
- It is independent of enzyme concentration.
- It is a constant for a given enzyme at a particular temp and pressure.

- The K_m value of a substrate depends on its affinity with the enzyme since it is an estimate of equilibrium constant for substrate binding to enzyme.
 - High K_m means low affinity with enzyme (more substrate needed to saturate the enzyme)
 - Low K_m means high affinity with enzyme (less substrate needed to saturate the enzyme)
- The lower the K_m the better the enzyme recognizes substrate and bind it at low concentration.
- Small K_m = tight binding, large K_m =weak binding
- It is a measure of substrate concentration required for effective catalysis.

- K_m varies considerably from enzyme to enzyme
- It also varies with different substrates for the same enzyme.
- When the substrate concentration is equal to the K_m value, half of the enzyme's active sites are occupied by substrate molecules.
- K_m also depends on temperature, nature of substrate, pH of the reaction medium, ionic strength, and any other reaction condition.
- Hence, it is important to characterize enzyme-substrate reactions under specific or well defined conditions.
- K_m is associated with *binding* of S to E, not a property of turnover

Maximum velocity, $V_{\max}$

- An essential feature of enzyme-catalyzed reactions is saturation: at increasing concentrations of substrates the rate increases and approaches a limit where there is no dependence of rate on concentration
- $V_{\max}$ is the maximum rate of reaction attainable with an excess of substrate in the presence of a given concentration of enzyme.
- That is $V_{\max}$ is the limiting value of the initial rate or velocity, V at saturating substrate concentration.

Maximum velocity, $V_{\max}$

- The higher the $V_{\max}$, the more the substrates an enzyme can process per min.
- $V_{\max}$ is reached when all enzyme sites are saturated with the substrate.
- This will happen when substrate concentration $[S]$ is greater than K_m so that $[S]/([S]+K_m)$ approaches 1.
- $V_{\max}$ is a constant with units S^{-1}
- $V_{\max}$ is the theoretical maximal rate of the reaction; in reality, it is never achieved.

Catalytic Constant, K_{cat}

- The k_{cat} is a direct measure of the catalytic production of product under saturating substrate conditions.
- K_{cat} or the turnover number, is the maximum number of substrate molecules converted to product per enzyme molecule per unit of time under conditions where enzyme is completely saturated with substrate.
- K_{cat} is the number of reaction processes (turnover) that each active site catalyzes per unit time.
- That is K_{cat} is the maximum velocity divided by the total enzyme concentration.

$$K_{cat} = \frac{V_{Max}}{[E]_T}$$

- For the simple kinetic scheme used to derive Michaelis-menten equation, $K_{cat} = K_3$.

Catalytic efficiency

- Catalytic efficiency, calculated as K_{cat} / K_m indicates how often the enzyme catalyzes a reaction upon encountering its substance.
- The value of K_{cat}/k_m is indicative of an enzyme's catalytic efficiency.
- It takes into account enzyme's affinity for the substrate (K_m) as well as the rate that the enzyme converts ES to P which (K_{cat}).
- The unit of K_{cat}/k_m is $\text{M}^{-1}\text{sec}^{-1}$
- It tells how fast an enzyme catalyzed reaction can proceed

Catalytic efficiency

- Catalytic efficiency is used to compare enzymes.
- This measure of efficiency is helpful in determining whether the rate is limited by the:
 - creation of product or
 - the amount of substrate in the reaction mixture.
- Generally, the upper limit of k_{cat}/K_m is determined by the rate of diffusion of substrate to the active site of enzyme.
- Under certain conditions, an enzyme may act on two related substrates.
 - Here, one has to consider the relative rates of reaction with each substrate, because each substrate has its own K_m value.
- However, if K_m is the sole determinant of enzyme specificity, then, as the ratio $[S]/K_m$ increases above 1,
- k_{cat} becomes the best parameter to determine which substrate is better.

Linear forms of Michaelis-Menten equation

- The Kinetic parameters for an enzymatic reactions can be determined graphically from experimental data by modifying the Michaelis-Menten equation into a number of linear relationships:
- Lineweaver-Burk method
- Eadie-Hofstee method
- Hanes-woolf method
- Eisenthal and cornish-Bowden method

Lineweaver-Burk plot

- The Lineweaver - Burk equation is obtained by taking the reciprocal of both sides of the Michaelis – Menten equation.
- Lineweaver - Burk plot is also called the double-reciprocal plot.
- It is plotted to calculate the K_m and V_{max} values and to determine the mechanism of action of enzyme inhibitors.
- Lineweaver-Burk plots are the most widely used for linearizing kinetic data

Lineweaver-Burk plot

- They give the most precise estimates of K_m and V_{max} .
- However, this method places undue weight on the points obtained at lower substrate concentrations, i.e., the highest values of $1/[S]$ and $1/V$).
- The double reciprocal plot distorts the error structure of the data, and it is therefore unreliable for the determination of enzyme kinetic parameters

The Michaelis Menton equation:

$$v = \frac{V_{\max}[S]}{K_m + [S]}$$

Take the reciprocal of both sides:

$$\frac{1}{v} = \frac{K_m + [S]}{V_{\max}[S]}$$

Split terms:

$$\frac{1}{v} = \frac{K_m}{V_{\max}[S]} + \frac{[S]}{V_{\max}[S]}$$

$$\frac{1}{v} = \frac{K_m}{V_{\max}[S]} + \frac{\cancel{[S]}}{V_{\max}\cancel{[S]}}$$

Cancelling to obtain the final transformed equation:

$$\boxed{\frac{1}{v} = \frac{K_m}{V_{\max}[S]} + \frac{1}{V_{\max}}}$$

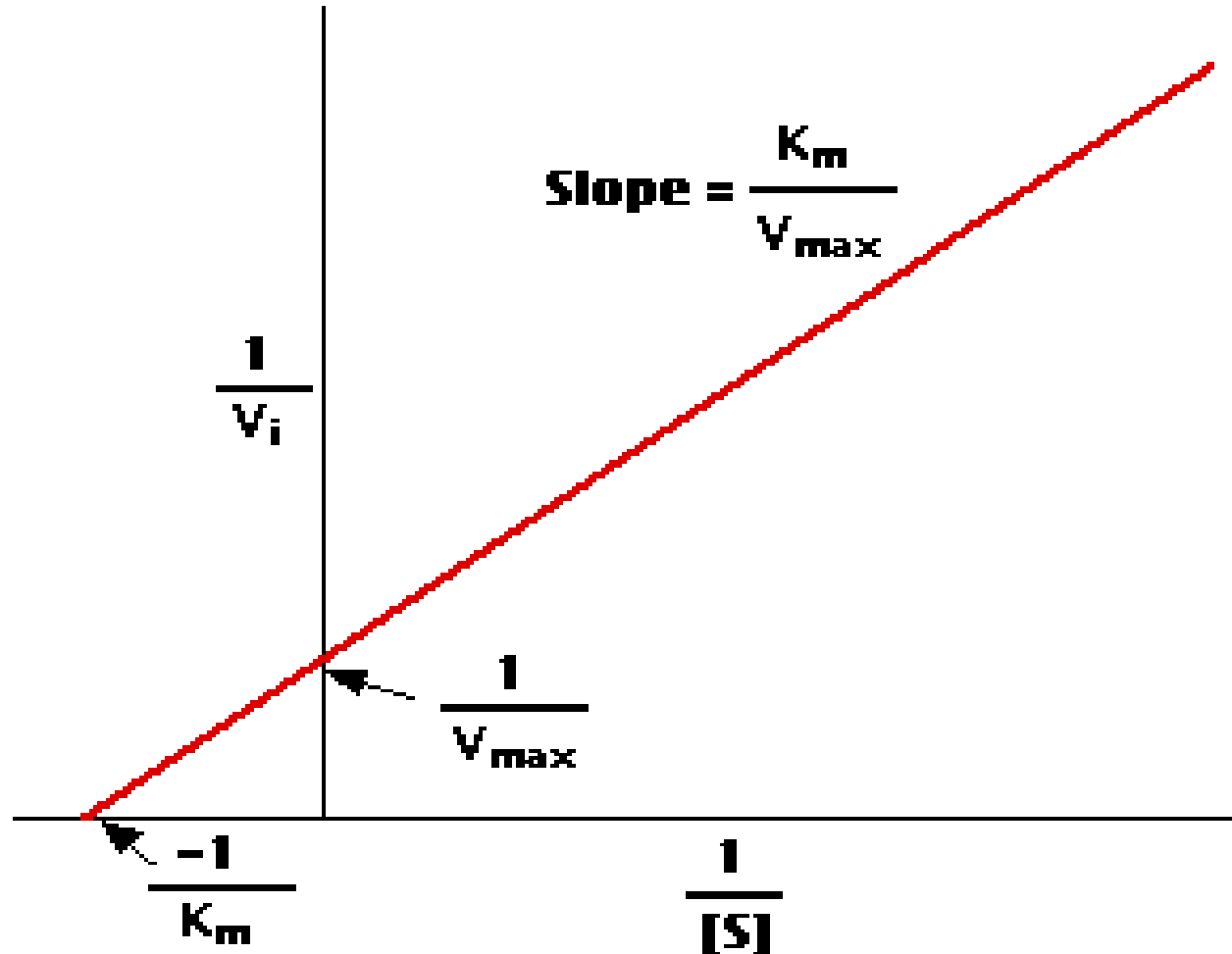
Lineweaver-Burk plot
equation

$$\frac{1}{v_o} = \left(\frac{K_m}{v_{\max}} \right) \frac{1}{(S)} + \frac{1}{v_{\max}}$$

$$y = m x + b$$

Lineweaver-Burk plot

A plot of $1/V_o$ against $1/[S]$ gives



The value of K_m may be obtained more accurately

Eadie-Hofstee

- Another method for graphically representing enzyme kinetics is the Eadie-Hofstee plot
- In this method, reaction rate is plotted as a function of the ratio between reaction rate and substrate concentration.
- The disadvantage of this method is that neither X nor Y axes represent independent variables
- Both the X and Y axes are dependent on reaction rate.
 - Any experimental or instrument error will affect both axes to a larger extent.
 - As such the Eadie-Hofstee plot can be subjected to large errors since both coordinates contain V_o , but there is less bias on the point at low $[S]$.

The Michaelis Menton equation: $v = \frac{V_{\max}[S]}{K_m + [S]}$

Take the reciprocal of both sides:

$$\frac{1}{v_o} = \left(\frac{K_M}{v_{\max}} \right) \frac{1}{(S)} + \frac{1}{v_{\max}}$$

multiply both sides by $v_o v_{\max}$

$$\frac{v_o v_{\max}}{v_o} = \left(\frac{K_M}{v_{\max}} \right) \frac{v_o v_{\max}}{(S)} + \frac{v_o v_{\max}}{v_{\max}}$$

cancel terms

$$\frac{\cancel{v_o} v_{\max}}{\cancel{v_o}} = \left(\frac{K_M}{\cancel{v_{\max}}} \right) \frac{v_o \cancel{v_{\max}}}{(S)} + \frac{v_o \cancel{v_{\max}}}{\cancel{v_{\max}}}$$

$$v_{\max} = K_M \left(\frac{v_o}{(S)} \right) + v_o$$

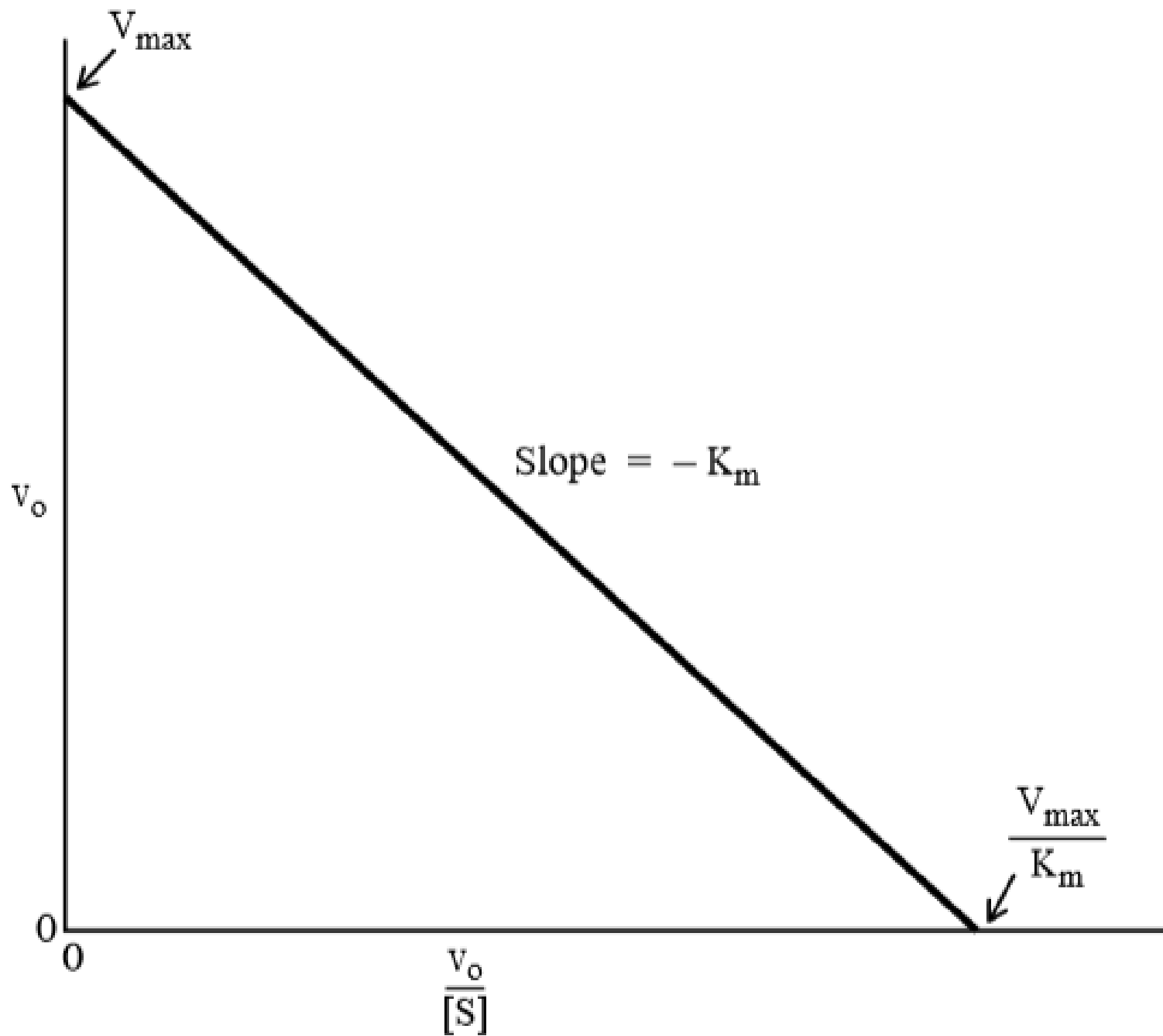
solve for v_o

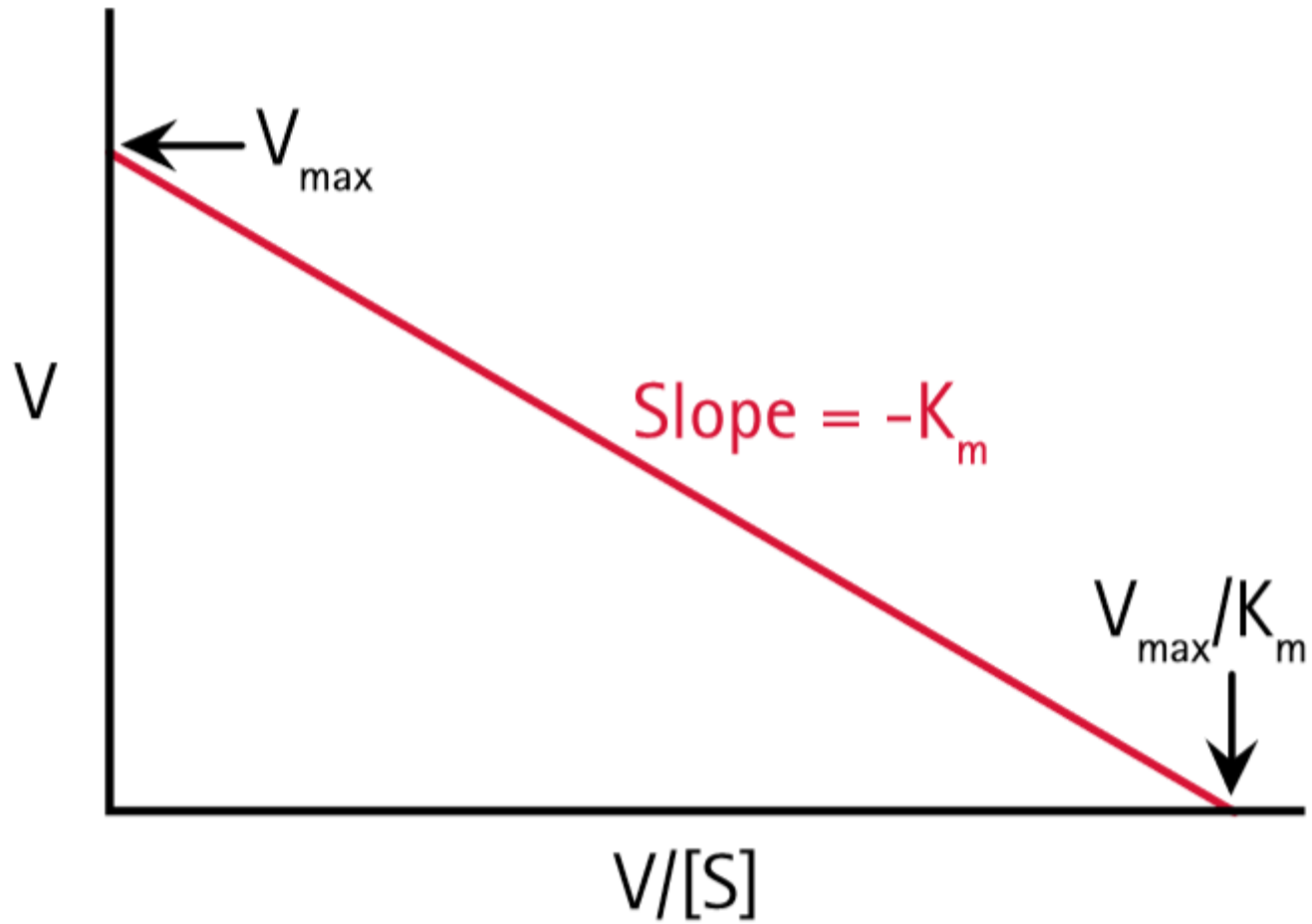
$$v_o = -K_M \left(\frac{v_o}{(S)} \right) + v_{\max}$$

— This is Eadie-Hofstee equation

$$y = m x + b$$

A plot of V against $V/[S]$ gives





Hanes-Woolf Plot

- Hanes–Woolf equation is obtained by taking the reciprocal and multiplying both sides of Michaelis-Menten equation by $[S]$.
- It is a graphical representation of enzyme kinetics in which the ratio of the initial substrate concentration $[S]$ to the reaction velocity v is plotted against $[S]$.
- Both axes are dependent on substrate concentration

The Michaelis Menton equation: $v = \frac{V_{\max}[S]}{K_m + [S]}$

Take the reciprocal of both sides:

$$\frac{1}{v_o} = \left(\frac{K_M}{v_{\max}} \right) \frac{1}{(S)} + \frac{1}{v_{\max}}$$

multiply both sides by (S)

$$\frac{1 (S)}{v_o} = \left(\frac{K_M}{v_{\max}} \right) \frac{1 (S)}{(S)} + \frac{1 (S)}{v_{\max}}$$

regroup and cancel

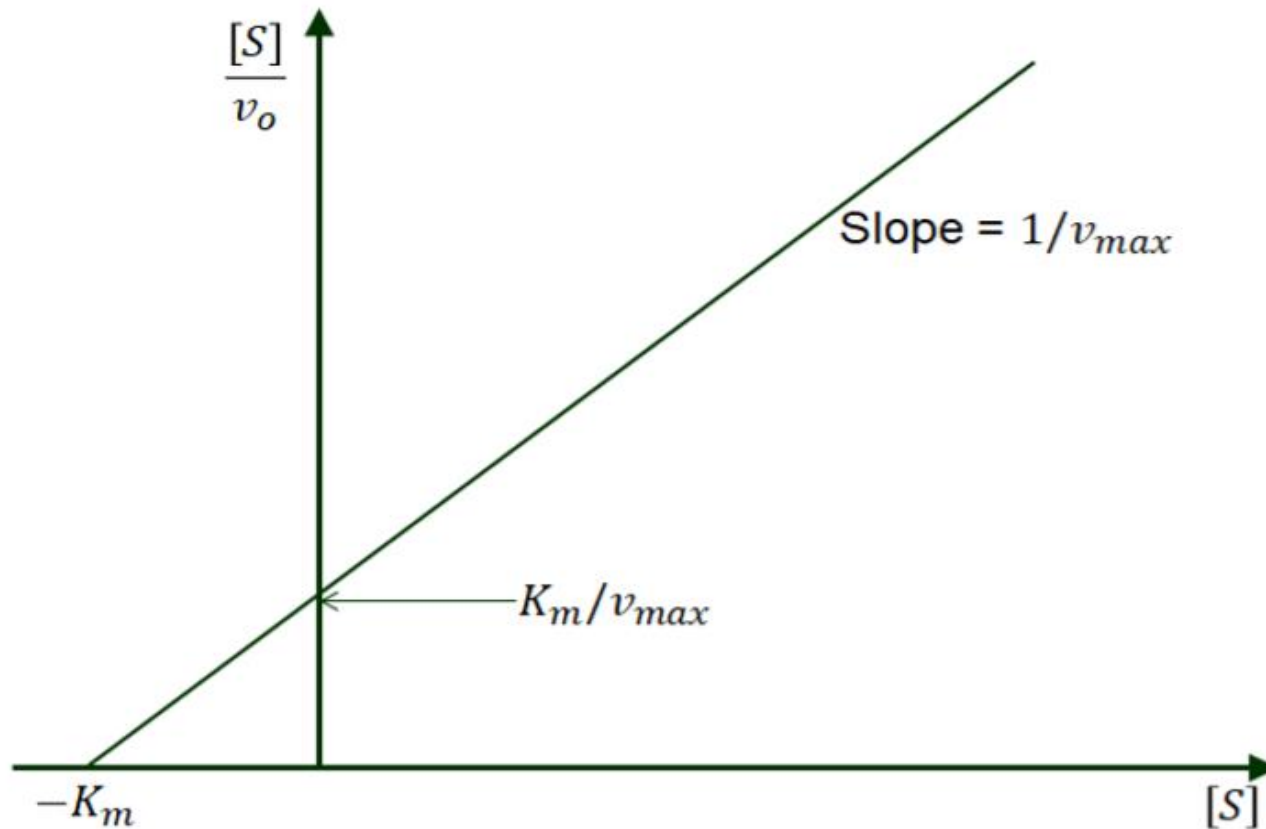
$$\frac{(S)}{v_o} = \left(\frac{K_M}{v_{\max}} \right) \frac{\cancel{(S)}}{\cancel{(S)}} + \frac{1}{v_{\max}} (S)$$

$$\frac{[S]}{v_o} = \frac{1}{v_{\max}} [S] + \left(\frac{K_m}{v_{\max}} \right)$$

Hanes Woolf equation

$$y = mx + b$$

A plot of $[S]/V_o$ against $[S]$ gives



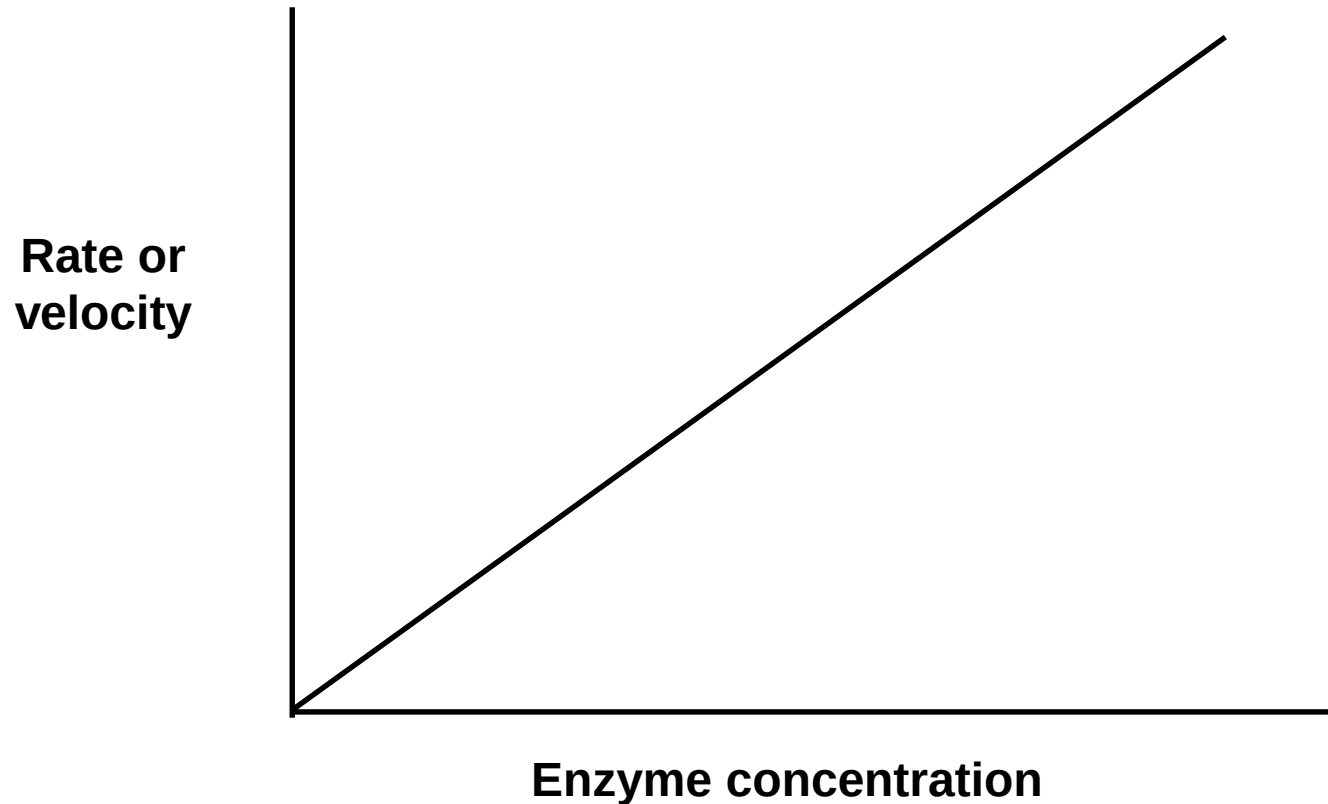
Factors affecting the rate of enzyme-catalyzed reactions

- Several factors are known to influence the rate of enzyme-catalyzed reaction, notable among them are:
- Enzyme concentration
- Substrate concentration
- Temperature
- pH and
- Activators and inhibitor

Effect of enzyme concentration

- The rate of enzyme-catalyzed reactions depends directly on the enzyme concentration.
- That is at a fixed substrate concentration, an increase in enzyme concentration increases the rate of an enzyme catalyzed reaction until the substrate concentration becomes the limiting factor.
- As the concentration of the enzyme is increased, the velocity of the reaction proportionately increases.
- When the substrate concentration is not limiting, the rate of an enzyme catalyzed reaction is directly proportional to the enzyme concentration.
- The rate slows down only when $[s]$ starts becoming a limiting factor. The rate remains unchanged at high $[E]$ when $[s]$ is the limiting factor.

Enzyme concentration



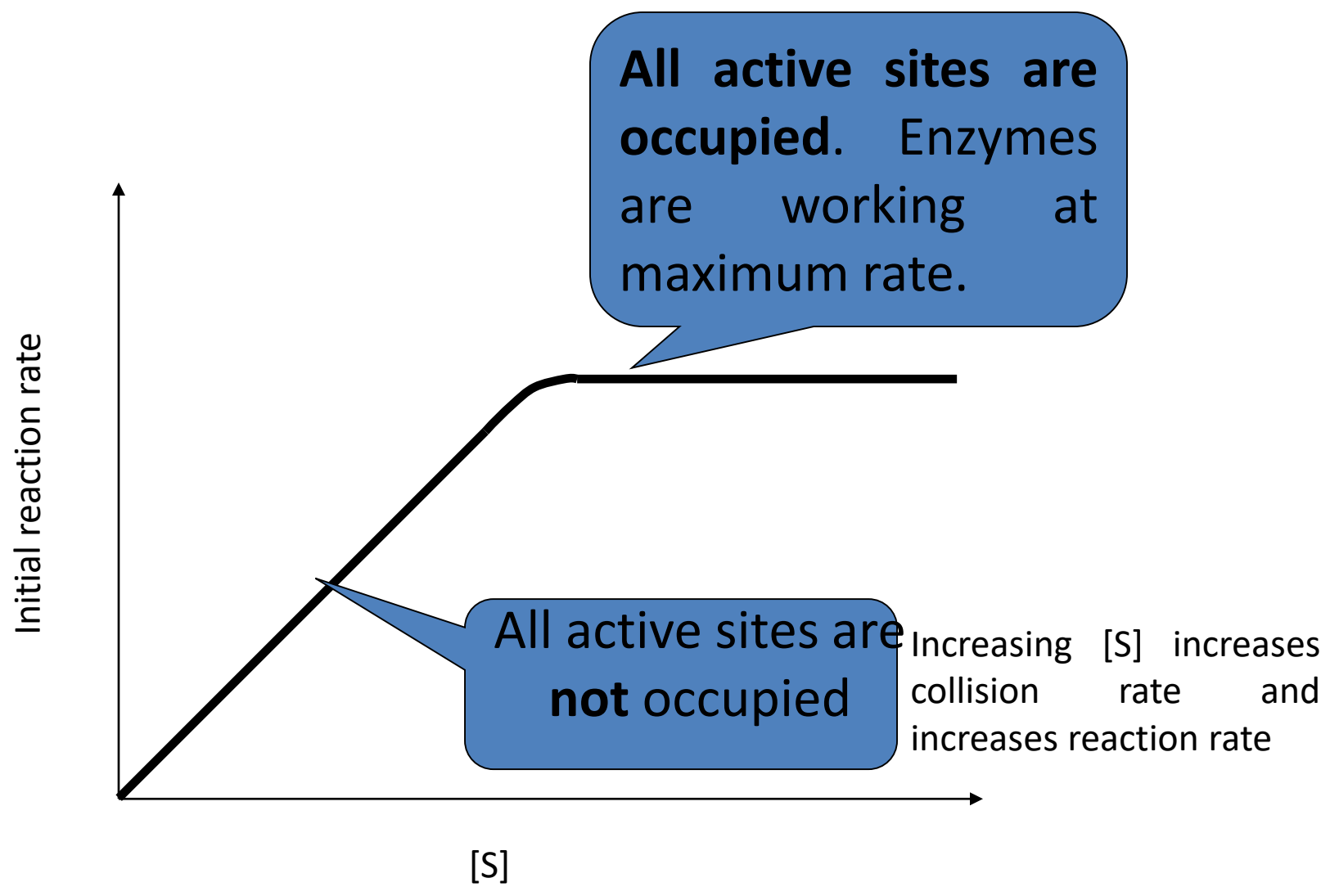
The increase in velocity is proportional to the enzyme concentration.

Effect of substrate concentration

- In the presence of a fixed enzyme concentration, the rate of enzyme catalyzed reaction increases as the substrate concentration increases **up to a point at which the substrate eventually saturates the enzyme to give maximum activity (V_{max})**, after which further increase in the substrate concentration produces no significant change in the reaction rate.
- At V_{max} , the enzyme is completely saturated with the substrate i.e. all the active sites of the enzymes are occupied by the substrate.
- The typical shape of the curve obtained between the velocity and substrate concentration is called a **rectangular hyperbola**.

Enzymes follow zero order kinetics when substrate concentrations are high.

Zero order means there is no increase in the rate of the reaction when more substrate is added.



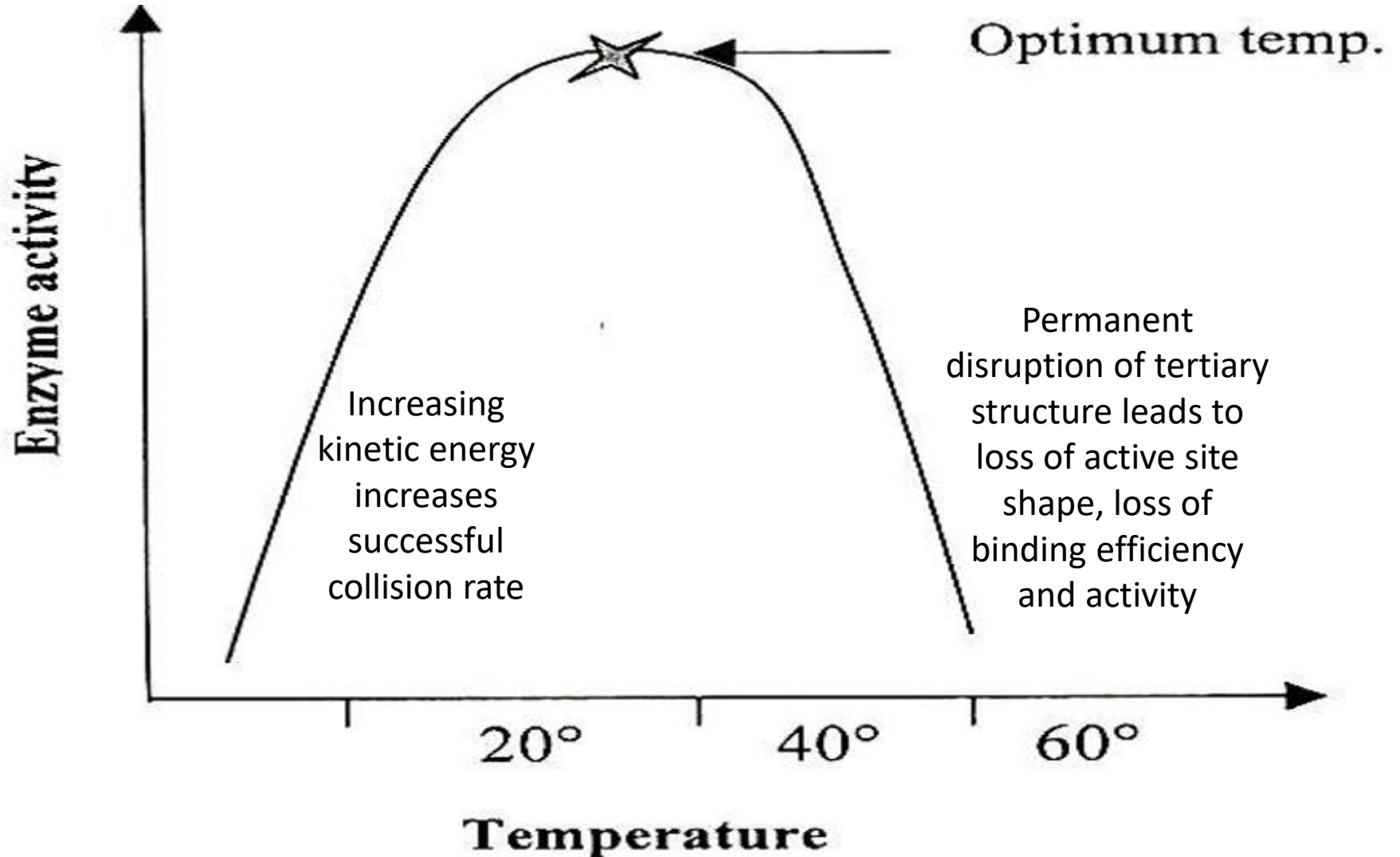
Effects of temperature

- The rate of enzyme catalyzed reaction increases gradually with increase or rise in temperature until it reaches a certain temperature, called optimum temperature.
- At very low temperature say 0°C , the rate of reaction may be almost zero but increase in temperature imparts more kinetic energy to the enzyme and substrate molecules resulting in more productive collisions per unit time which leads to an overall increase in the rate of reaction.
- Generally, for every 10°C rise in temperature, the rate is double, but this occurs only up to a specific temperature known as the optimum temperature.

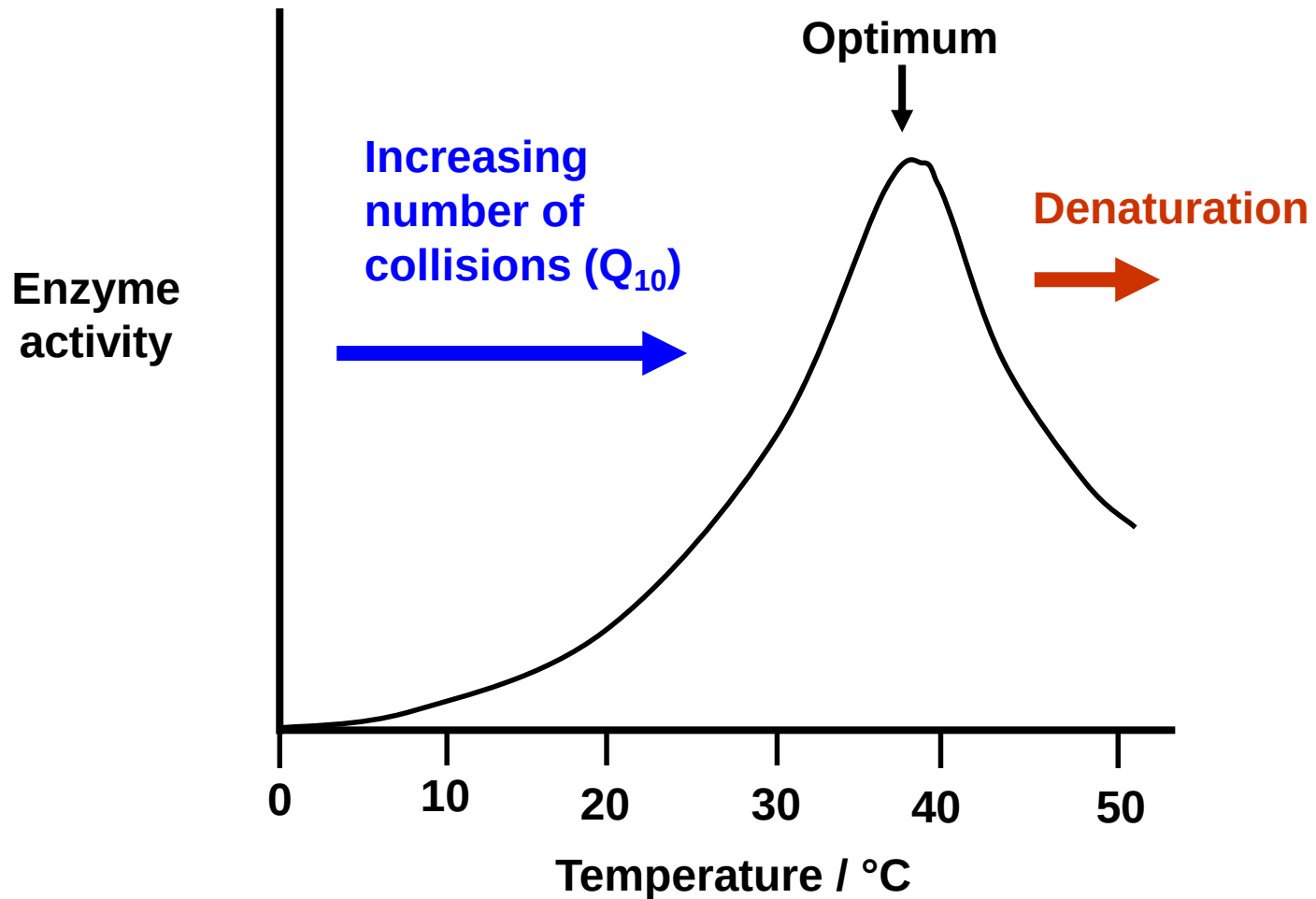
Effects of temperature cont...

- This increase in temperature imparts substrate molecules with sufficient energy to overcome the activation energy and hence increases the rate of the reaction.
- However, increasing the temperature usually beyond the optimum temperature leads to a sharp decrease in the rate of reaction due to denaturation of the enzyme.
- The optimum temperature for an enzyme assay is the temperature at which the enzyme exhibits its maximum activity.
- The optimum temperature is 37-40 °C in humans

Effects of temperature



The effect of temperature



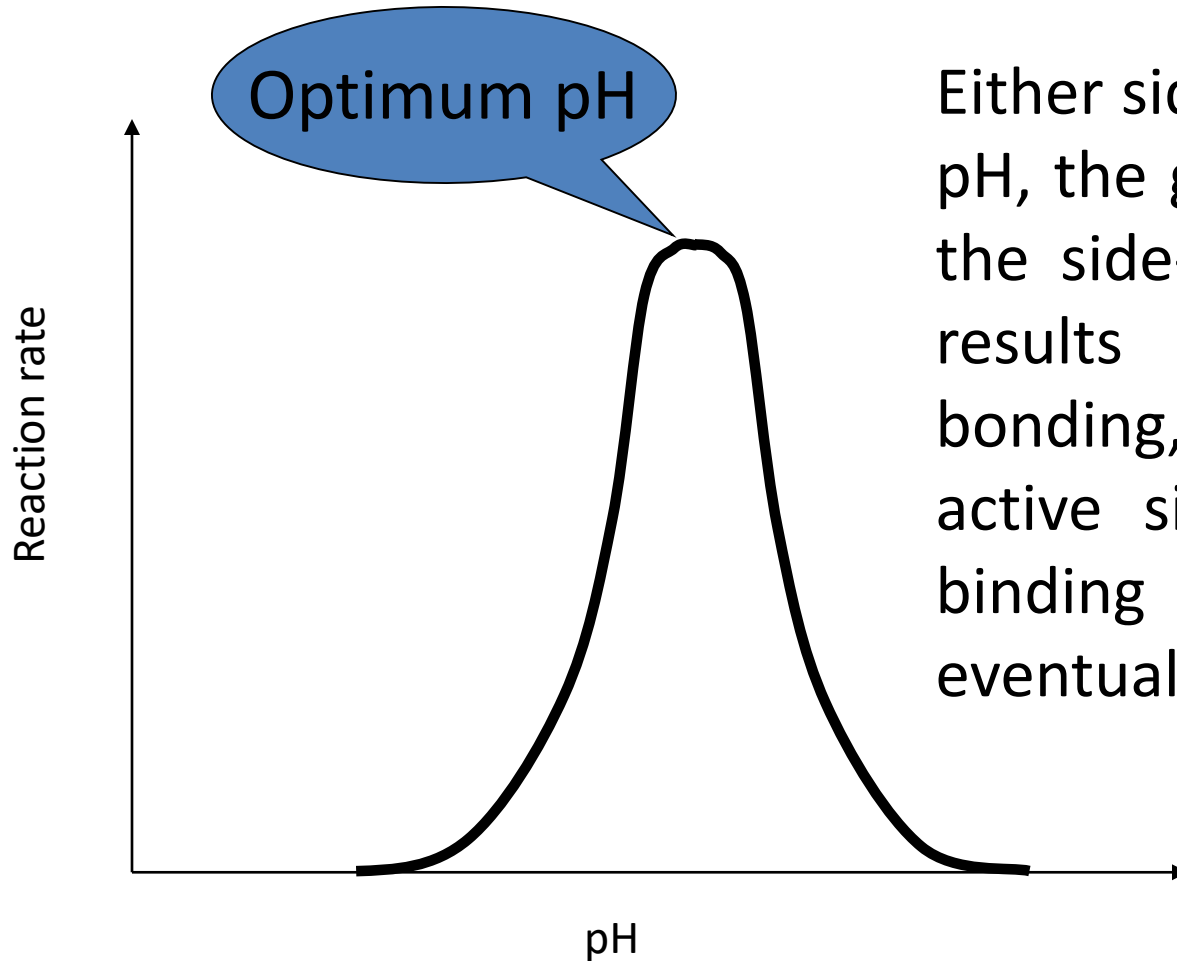
Effect of pH

- The rate of an enzyme catalyzed reaction is strongly influenced by the pH of the reaction solution both in *in vivo* and *in vitro*.
- This is because the active sites on enzymes are composed of ionizable groups that must be in the proper ionic form in order to maintain the conformation of the active site, bind the substrate or catalyze the reaction.
- Furthermore, substrates mainly contain ionizable groups that may be required to be in certain ionic form for it to bind to the enzyme or undergo catalysis.
- Changes in pH will modify the degree of ionization of some or all of these groups and this will affect the ionization of the enzyme molecule as a whole.

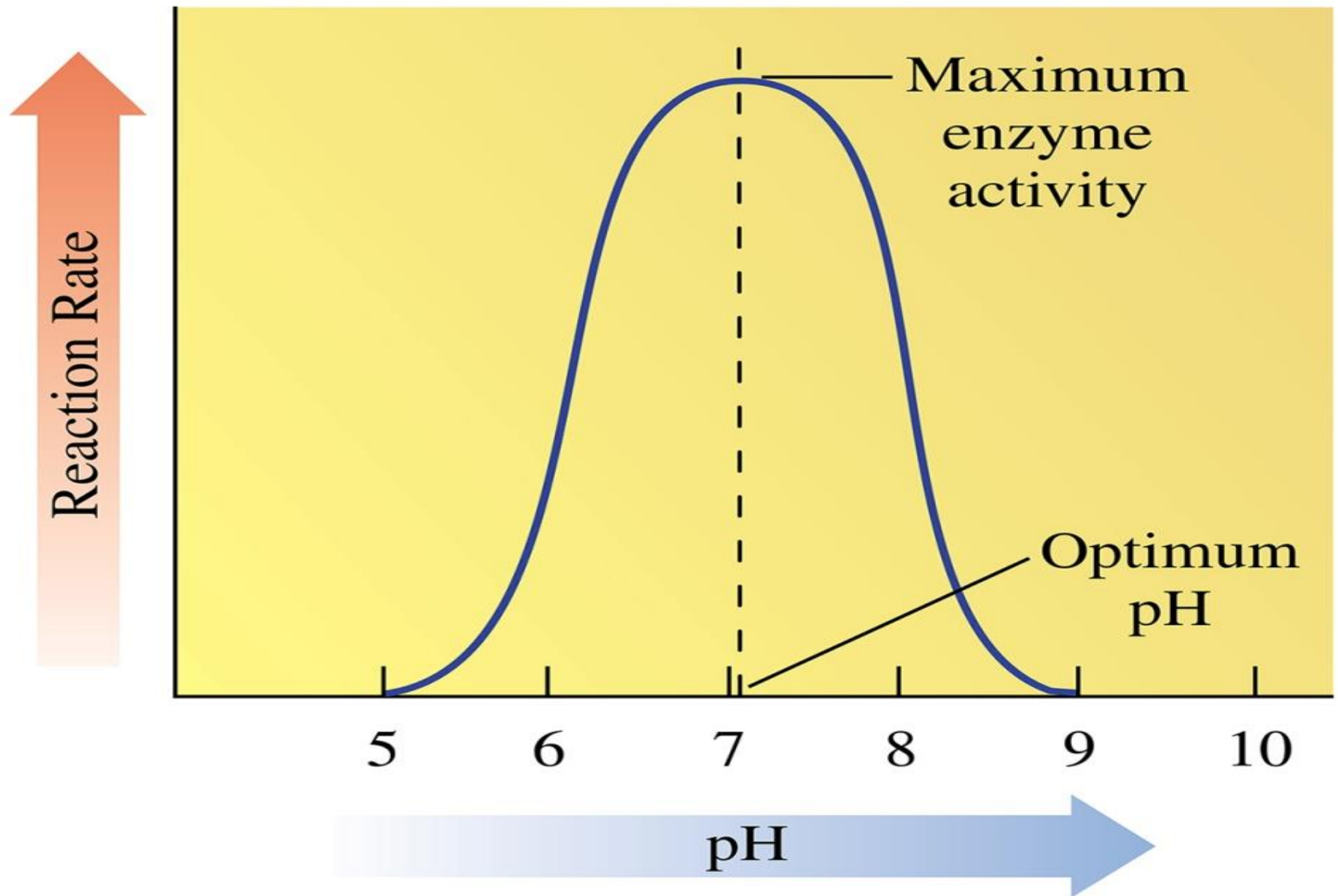
Effect of pH

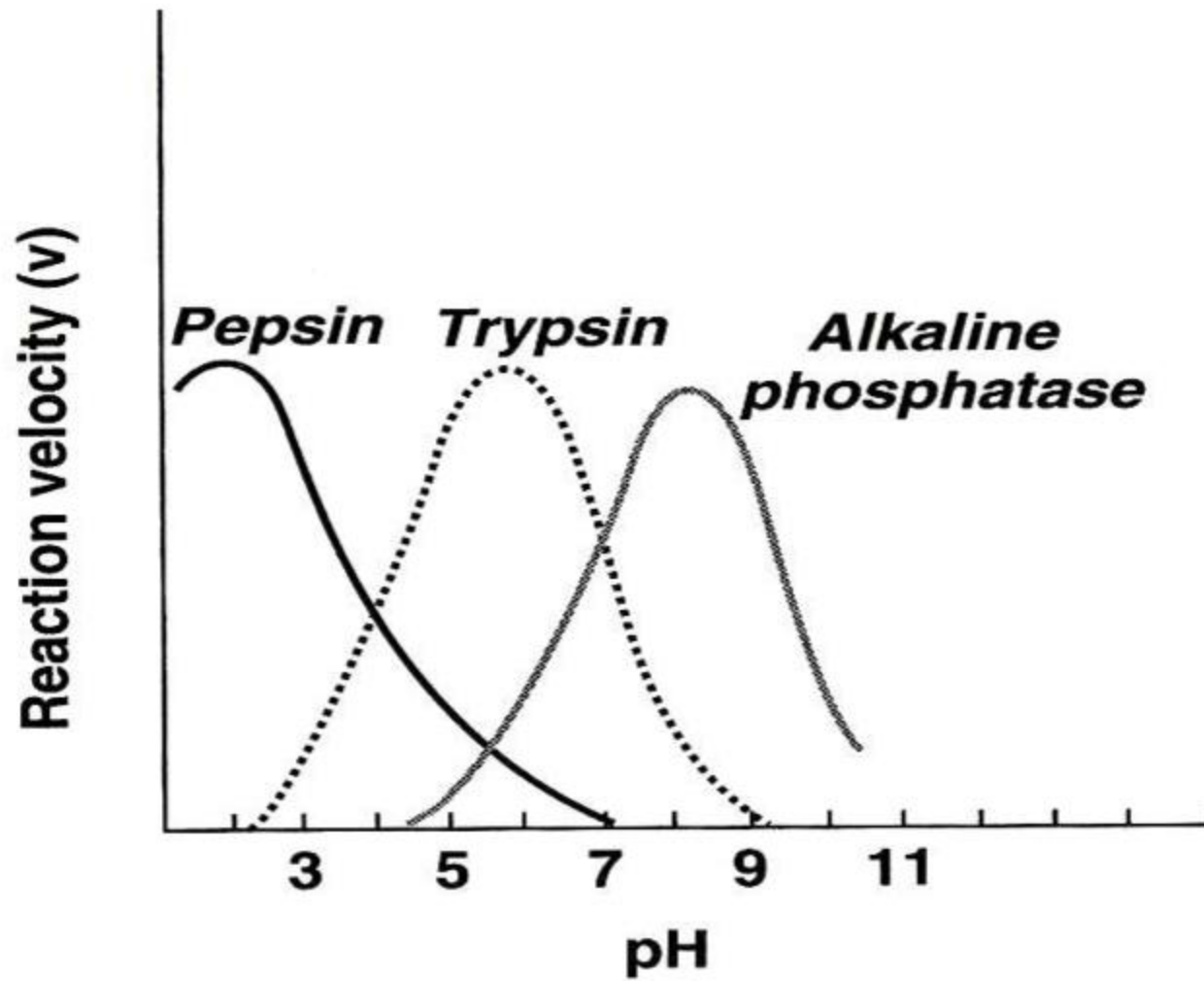
- Generally, pH can affect enzyme activity by changing the structure or the charge on the amino acids which function in substrate binding or catalysis.
- The optimum pH value varies from enzyme to enzyme.
- The optimum pH of an enzyme is the pH at which the enzyme exhibit its highest activity.
- Change of pH above or below optimum pH decrease rate of enzyme action due to:

Enzymes and pH



Either side of the optimum pH, the gradual ionising of the side-chains (R-groups) results in loss of H-bonding, 3^o structure, active site shape loss of binding efficiency and eventually enzyme activity





ENZYME INHIBITION

- Enzyme inhibitors are compounds that combine with enzymes and decrease or reduce the rate of enzyme activity by influencing either the binding of substrates or the turnover number.
- If a chemical binds to an enzyme and prevent it from carrying out its catalytic activity, the enzyme is said to be inhibited, and the chemical is called an inhibitor.
- Some enzyme inhibitors are normal body metabolites that inhibit a particular enzyme as part of the normal metabolic control of the pathway.
- Other inhibitors may be foreign substances, such as drugs or toxins, where their effects on enzyme activity could be either therapeutic or, at the other extreme, lethal.

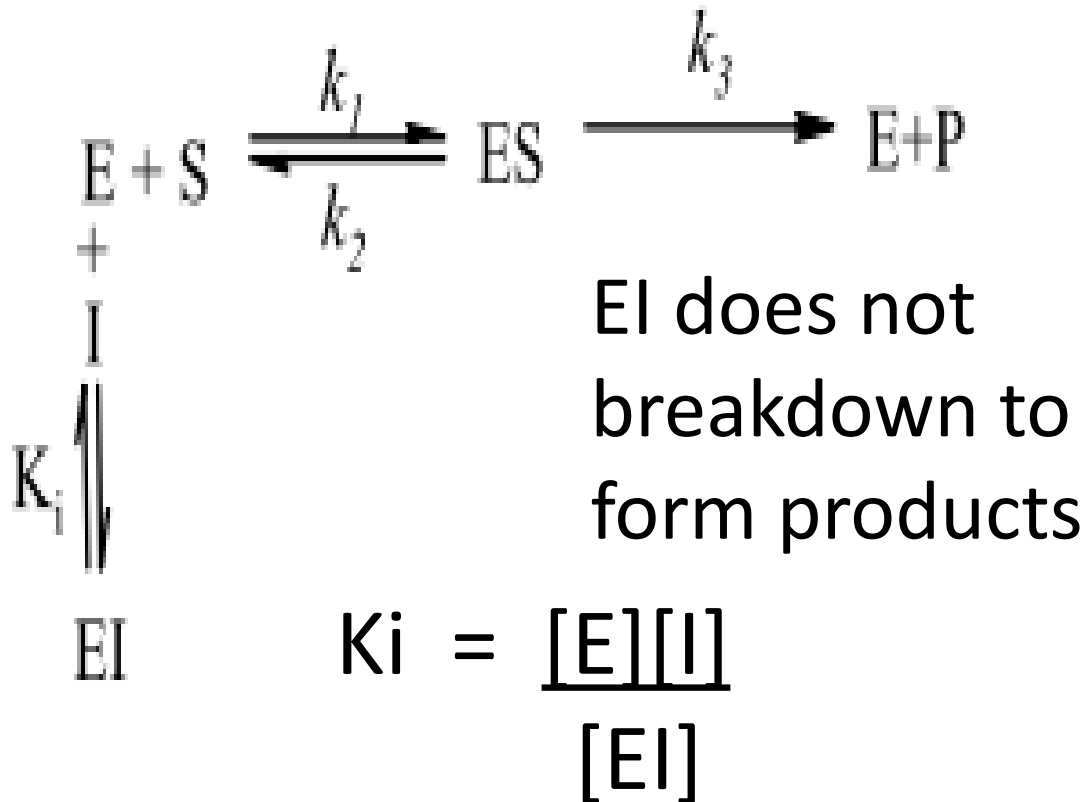
Types of Enzyme Inhibitor

- The action of an inhibitor on an enzyme can be described as either reversible or irreversible.
- That is enzyme inhibitors are divided into two:
- Reversible Inhibitors
- Irreversible Inhibitors
- REVERSIBLE INHIBITORS
- A reversible inhibitor binds to enzymes reversibly through weak non-covalent interactions in such a way that slows down or prevents enzyme turnover.
- An equilibrium exists between the free inhibitor and the enzyme-inhibitor complex (EI) and is defined by the equilibrium constant K_i , $E + I \rightleftharpoons EI$
- The activity of the enzyme is fully restored on removal of the inhibitor by dialysis.

- There are three types of reversible inhibitors:
 - Competitive Inhibitor
 - Noncompetitive Inhibitor
 - Uncompetitive Inhibitor
- **COMPETITIVE INHIBITOR**
- In competitive inhibition, the inhibitor competes with the substrate for binding to the active site of the enzyme.
- That is the inhibitor binds only to the free enzyme at the active site.

- A competitive inhibitor is a structural analog of the substrate i.e. it has a structure similar to that of the substrate
 - This allows it to form interactions with the enzyme active site but the enzyme–inhibitor complex formed does not breakdown to form products
- The effect of the inhibitor can be reversed by increasing the substrate concentration.
- The degree of inhibition depends on the concentrations of the S & I and the relative affinity of the enzyme for the S & I.
- K_i is the dissociation constant for the inhibitor binding with enzyme.

Using the steady state approach inhibitor combine with the enzyme according to the reaction scheme shown below:



$$[\text{EI}] = \frac{[\text{E}][\text{I}]}{K_i}$$

From the uninhibited reaction, the rate of formation of ES is equal to the rate of breakdown of ES.

$$K_1[E][S] = K_2[ES] + K_3[ES]$$

$$[E] = \frac{[ES](K_2 + K_3)}{K_1[S]}$$

But enzyme concentration is conserved

$$[E_T] = [E] + [ES] + [EI]$$

Substituting for EI in the enzyme conservation equation

$$[E_T] = [E] + [ES] + \frac{[E][I]}{K_i}$$

Factorizing E

$$[E_T] = [ES] + [E] \left(1 + \frac{[I]}{K_i}\right)$$

Substituting for E from the uninhibited reaction and rearranging

$$[E_T] = [ES] + [ES] \left[\frac{K_2 + K_3}{K_1[S]} \right] \left(1 + \frac{[I]}{K_i} \right)$$

$$[E_T] = [ES] \left[1 + \frac{K_2 + K_3}{K_1[S]} \right] \left(1 + \frac{[I]}{K_i} \right)$$

Isolating ES

$$[ES] = \frac{[E_T]}{1 + \frac{K_2 + K_3}{K_1[S]} \left(1 + \frac{[I]}{K_i} \right)}$$

Multiplying both the numerator and denominator by [S]

$$[ES] = \frac{[E_T][S]}{[S] + \frac{K_2 + K_3}{K_1} \left(1 + \frac{[I]}{K_i} \right)}$$

But $V_0 = K_3[ES]$ $[ES] = \frac{V}{K_3}$

$$V = \frac{V_{\text{Max}}[S]}{[S] + K_M \left(1 + \frac{[I]}{K_i}\right)}$$

Substituting for ES

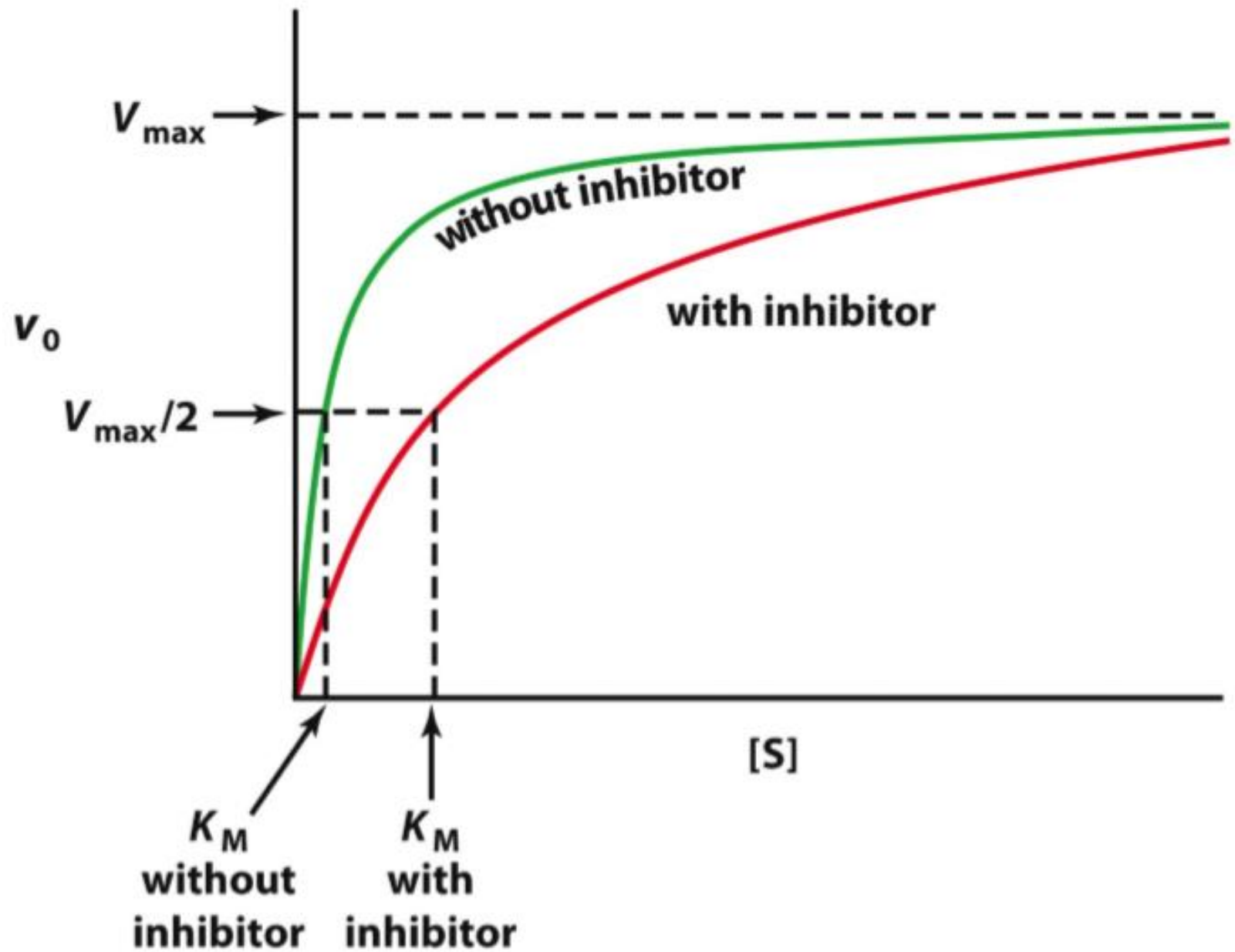
$$V = \frac{K_3[E_T][S]}{[S] + \frac{K_2 + K_3}{K_1} \left(1 + \frac{[I]}{K_i}\right)}$$

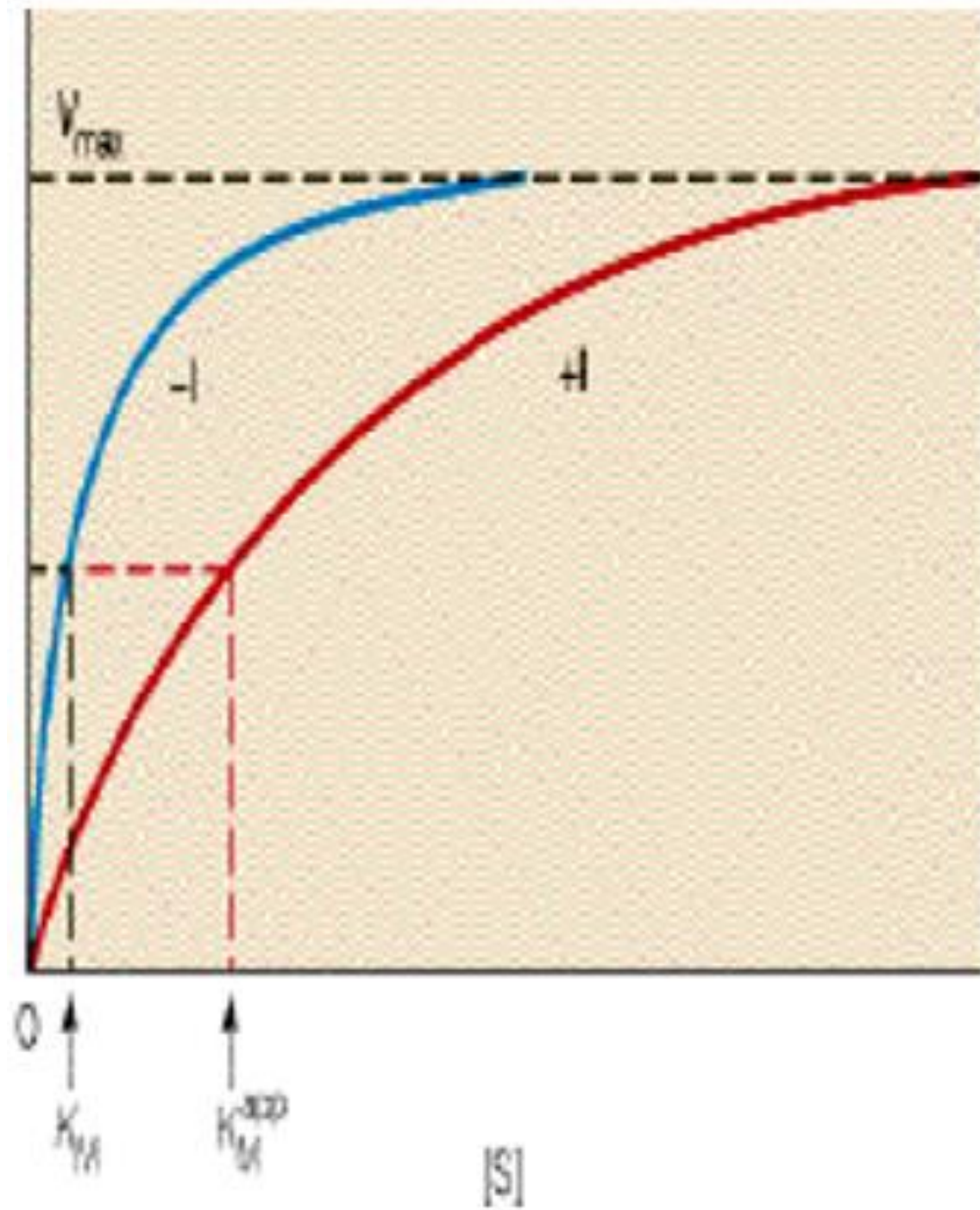
But $K_M = \frac{K_2 + K_3}{K_1}$ and

$$K_3E_T = V_{\text{max}}$$

$$v = \frac{V_{\text{max}} [S]}{[S] + K_m \left(1 + \frac{[I]}{K_i}\right)}$$

Michaelis-Menten
equation in the
presence of a
competitive inhibitor

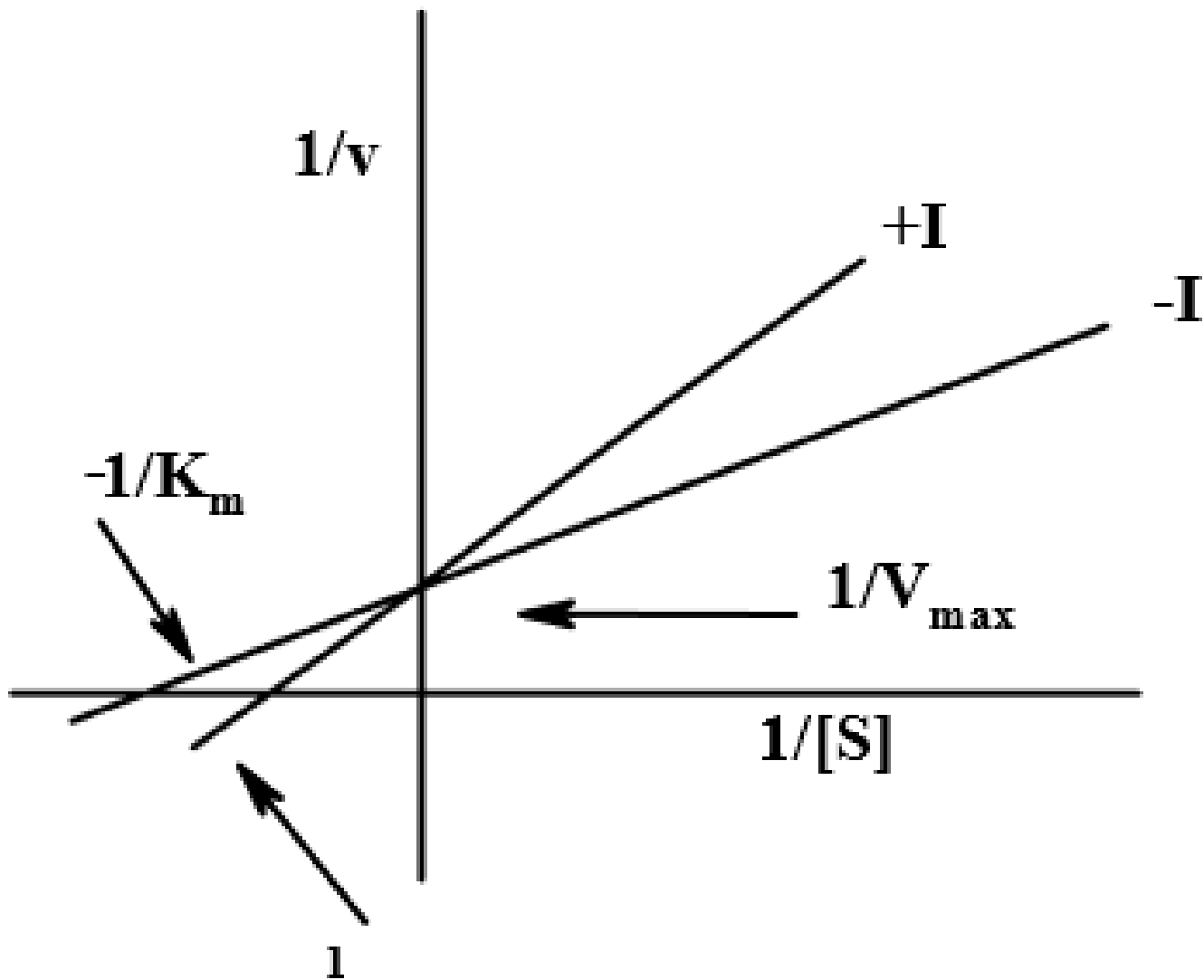




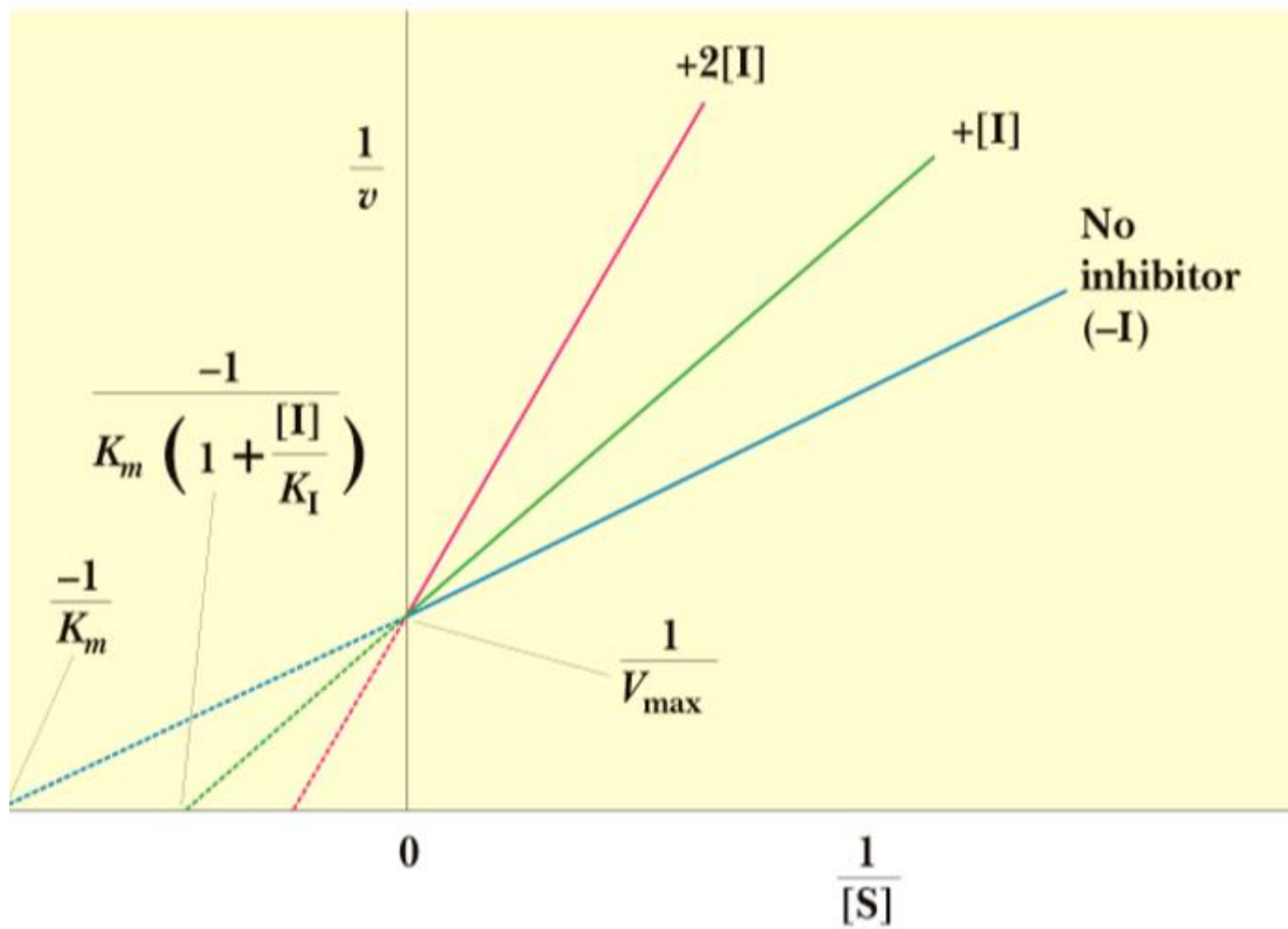
Lineweaver-Burk equation in the presence of a competitive inhibitor becomes

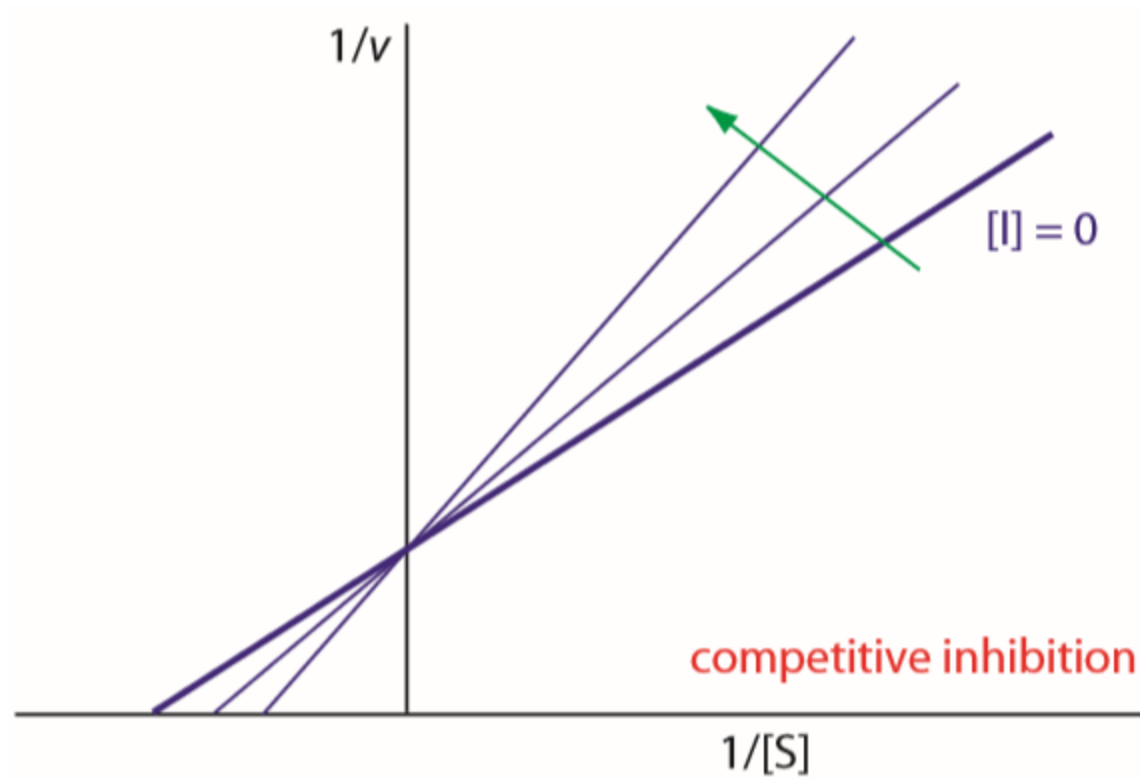
$$\frac{1}{v} = \frac{K_m (1 + [I]/K_i)}{V_{\max}} \cdot \frac{1}{[S]} + \frac{1}{V_{\max}}$$

Lineweaver-Burk plots in absence and presence of a competitive inhibitor becomes



$$-\frac{1}{K_m \left(1 + \frac{[I]}{K_m}\right)}$$





Comparing this with Michaelis-Menten equation, the K_m in the presence of a competitive inhibitor is increased by a factor

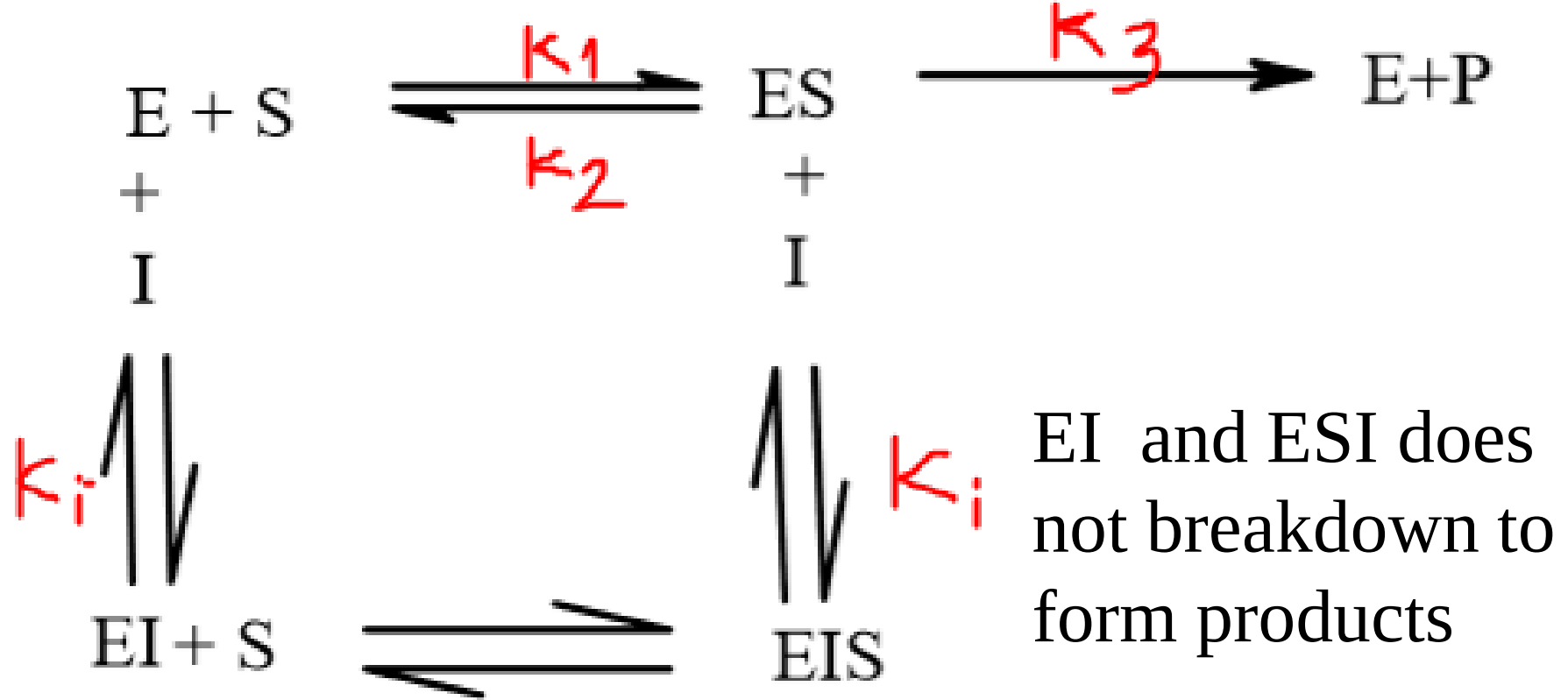
$$\left(1 + \frac{[I]}{K_i}\right)$$

This suggesting that v is less in presence of the inhibitor.

The V_{max} remains unchanged i.e. V_{max} is the same both in the absence and presence of a competitive inhibitor.

NONCOMPETITIVE INHIBITOR

- A Non-competitive inhibitor is not structurally related to the substrate
- Does not compete with the substrate for binding to the enzyme active site
- Instead it binds to a site that is distinct from the enzyme active site i.e. at the **allosteric site**.
- Non-competitive inhibitor binds to both free enzyme and enzyme-substrate complex (ES).
- The effect of a noncompetitive inhibitor cannot be overcome by increasing the substrate concentration.



$$K_i = \frac{[\text{E}][\text{I}]}{[\text{EI}]}$$

$$[\text{EI}] = \frac{[\text{E}][\text{I}]}{K_i}$$

Similarly

$$K_i = \frac{[\text{ES}][\text{I}]}{[\text{ESI}]}$$

$$[\text{ESI}] = \frac{[\text{ES}][\text{I}]}{K_i}$$

The total enzyme concentration is conserved

$$[E_T] = [E] + [ES] + [EI] + [ESI]$$

Substituting for EI and ESI in the enzyme conservation equation

$$[E_T] = [E] + \frac{[E][I]}{K_i} + [ES] + \frac{[ES][I]}{K_i}$$

Factorizing E

$$[E_T] = [E] \left(1 + \frac{[I]}{K_i} \right) + [ES] + \frac{[ES][I]}{K_i}$$

From the uninhibited reaction, the rate of formation of ES is equal to the rate of breakdown of ES.

$$[E] = [ES] \frac{(K_2 + K_3)}{K_1[S]}$$

Substituting for E

$$[E_T] = [ES] \frac{(K_2 + K_3)}{K_1[S]} \left(1 + \frac{[I]}{K_i}\right) + [ES] + \frac{[ES][I]}{K_i}$$

Factorizing ES

$$[E_T] = [ES] \left[\frac{(K_2 + K_3)}{K_1[S]} \left(1 + \frac{[I]}{K_i}\right) + \left(1 + \frac{[I]}{K_i}\right) \right]$$

Isolating ES

$$[ES] = \frac{[E_T]}{\left[\frac{(K_2 + K_3)}{K_1[S]} \left(1 + \frac{[I]}{K_i}\right) + \left(1 + \frac{[I]}{K_i}\right) \right]}$$

Multiplying both the numerator and denominator by [S]

$$[ES] = \frac{[E_T][S]}{\frac{(K_2 + K_3)}{K_1} (1 + \frac{[I]}{K_i}) + [S] (1 + \frac{[I]}{K_i})}$$

But $V_0 = K_3[ES]$ $[ES] = \frac{V}{K_3}$

Substituting for ES

$$V = \frac{K_3[E_T][S]}{\frac{(K_2 + K_3)}{K_1} (1 + \frac{[I]}{K_i}) + [S] (1 + \frac{[I]}{K_i})}$$

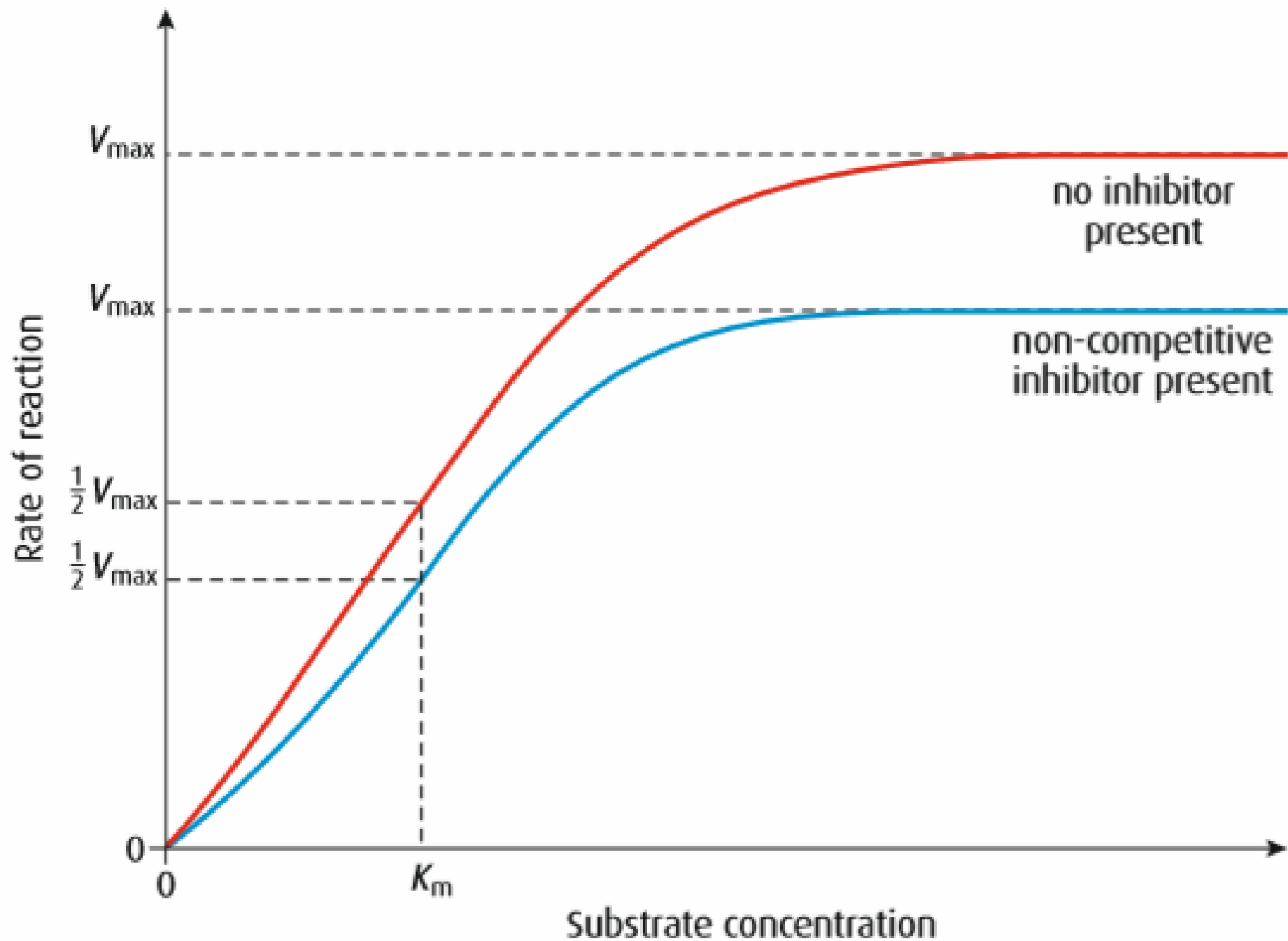
But $K_M = \frac{K_2 + K_3}{K_1}$ and $K_3E_T = V_{\max}$

$$V = \frac{V_{\max} [S]}{K_M (1 + \frac{[I]}{K_i}) + [S] (1 + \frac{[I]}{K_i})}$$

Dividing both the numerator and denominator by $(1 + \frac{[I]}{K_i})$

$$v = \frac{V_{\max} [S] / (1 + [I] / K_i)}{K_m + [S]}$$

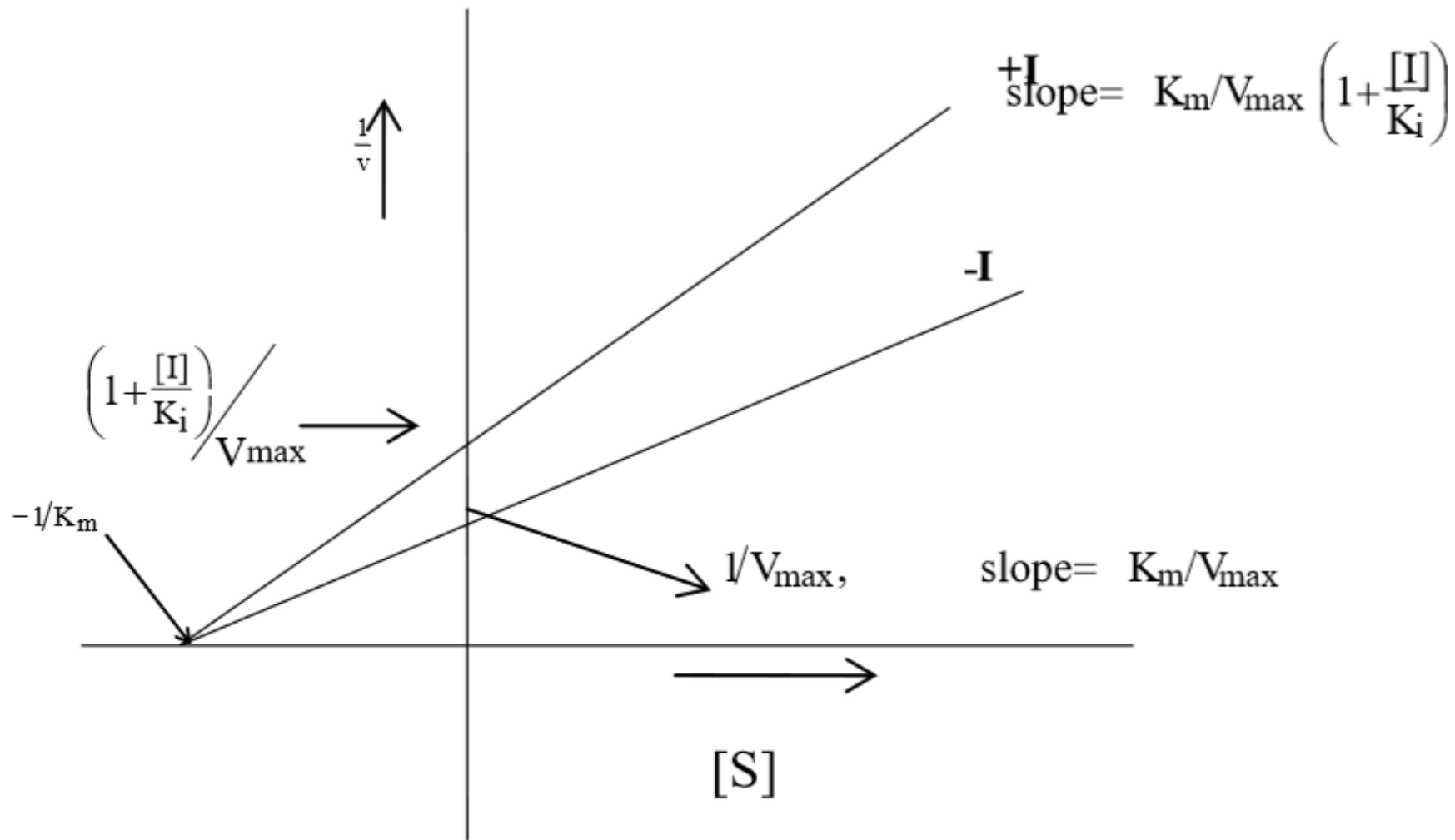
Michaelis-Menten
equation in the
presence of a
noncompetitive
inhibitor



Lineweaver-Burk equation in the presence of a non-competitive inhibitor becomes

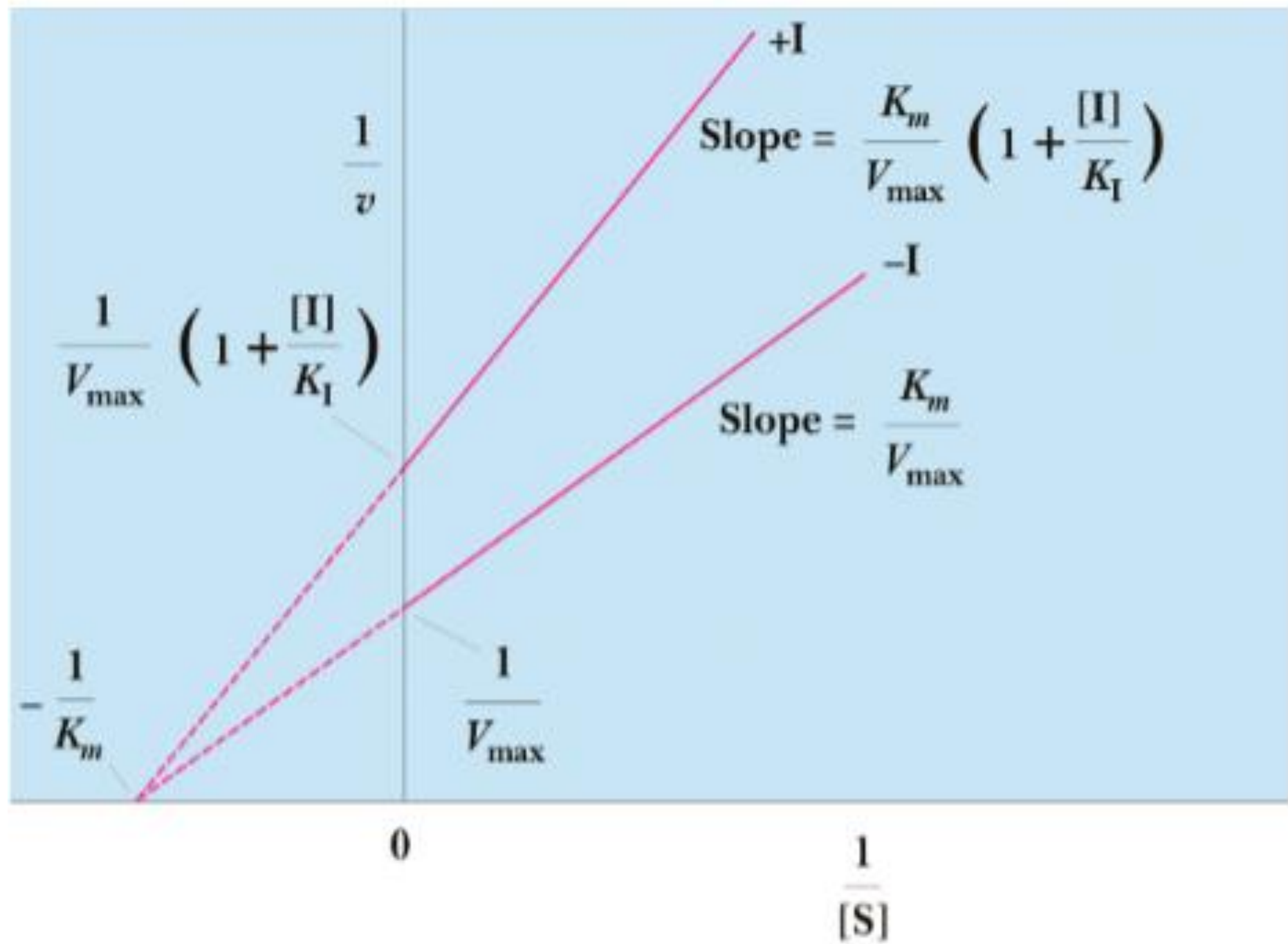
$$\frac{1}{v} = \frac{K_m}{V_{\max}} \left(1 + \frac{[I]}{K_i} \right) \frac{1}{[S]} + \frac{1}{V_{\max}} \left(1 + \frac{[I]}{K_i} \right)$$

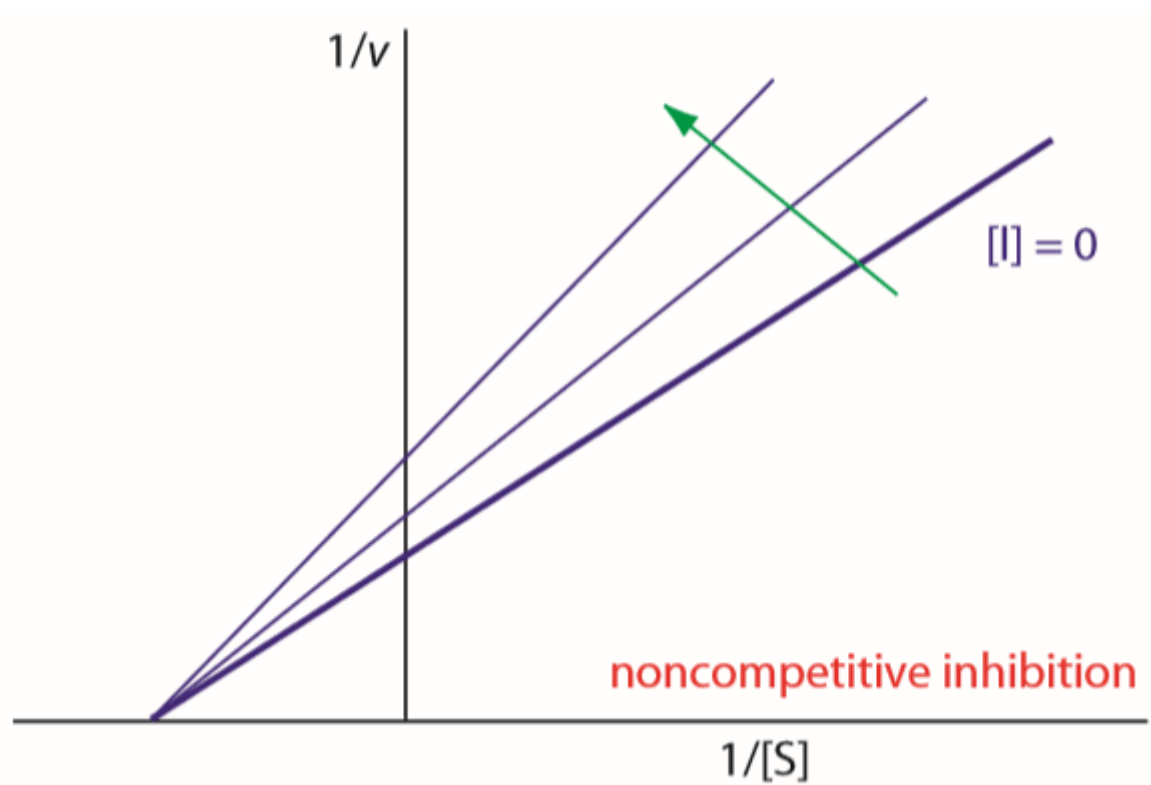
Lineweaver-Burk plots in absence and presence of non-competitive inhibitor becomes



Comparing this with Michaelis-Menten equation, the K_m in the presence of a noncompetitive inhibitor remains unchanged

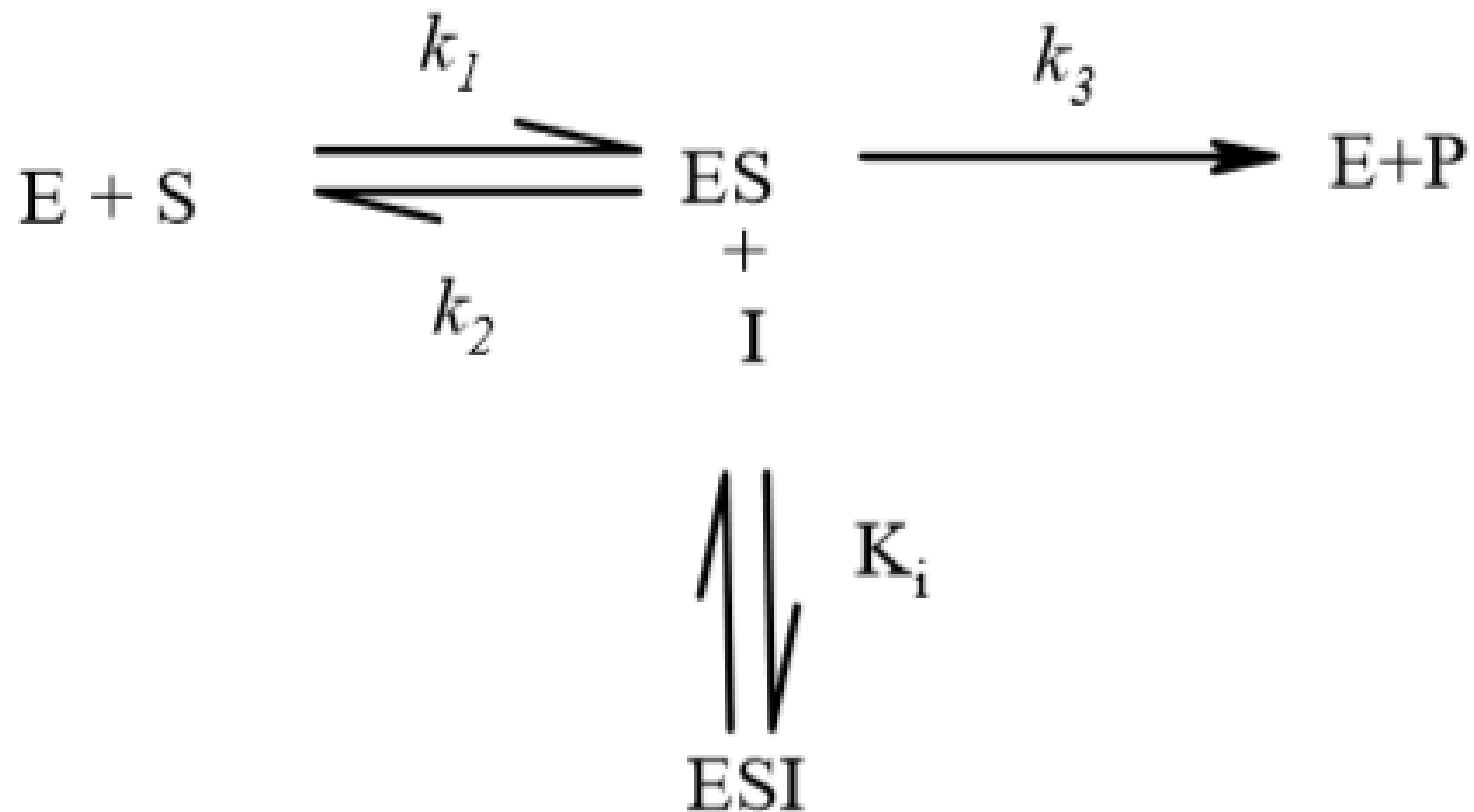
The V_{max} decreased by a factor





Uncompetitive inhibitor

- An uncompetitive inhibitor is not structurally related to the substrate and binds only to the enzyme-substrate complex (ES) at a site that is distinct from enzyme active site to form ESI.
- ESI is a dead end complex, it cannot form product
- That is an uncompetitive inhibitor does not bind to the free enzyme.
- The net effect of an uncompetitive inhibitor is that both K_M and maximal rate of enzyme reaction decreases.



ESI does not breakdown to form products

$$K_i = \frac{[\text{ES}][\text{I}]}{[\text{ESI}]}$$

$$[\text{ESI}] = \frac{[\text{ES}][\text{I}]}{K_i}$$

The total enzyme concentration is conserved

$$[E_T] = [E] + [ES] + [ESI]$$

Substituting for ESI in the enzyme conservation equation

$$[E_T] = [E] + [ES] + \frac{[ES][I]}{K_i}$$

Factorizing ES

$$[E_T] = [E] + [ES] \left(1 + \frac{[I]}{K_i}\right)$$

From the uninhibited reaction, the rate of formation of ES is equal to the rate of breakdown of ES.

$$[E] = [ES] \frac{(K_2 + K_3)}{K_1[S]}$$

Substituting for E

$$[E_T] = [ES] \frac{(K_2 + K_3)}{K_1[S]} + [ES] \left(1 + \frac{[I]}{K_i}\right)$$

Factorizing ES

$$[E_T] = [ES] \left[\frac{(K_2 + K_3)}{K_1[S]} + \left(1 + \frac{[I]}{K_i}\right) \right]$$

Isolating ES

$$[ES] = \frac{[E_T]}{\frac{(K_2 + K_3)}{K_1[S]} + \left(1 + \frac{[I]}{K_i}\right)}$$

Multiplying both the numerator and denominator by [S]

$$[ES] = \frac{[E_T][S]}{\frac{(K_2 + K_3)}{K_1} + [S] \left(1 + \frac{[I]}{K_i}\right)}$$

But $V_0 = K_3[ES]$ $[ES] = \frac{V}{K_3}$

Substituting for ES

$$V = \frac{K_3[E_T][S]}{\frac{(K_2 + K_3)}{K_1} + [S] \left(1 + \frac{[I]}{K_i}\right)}$$

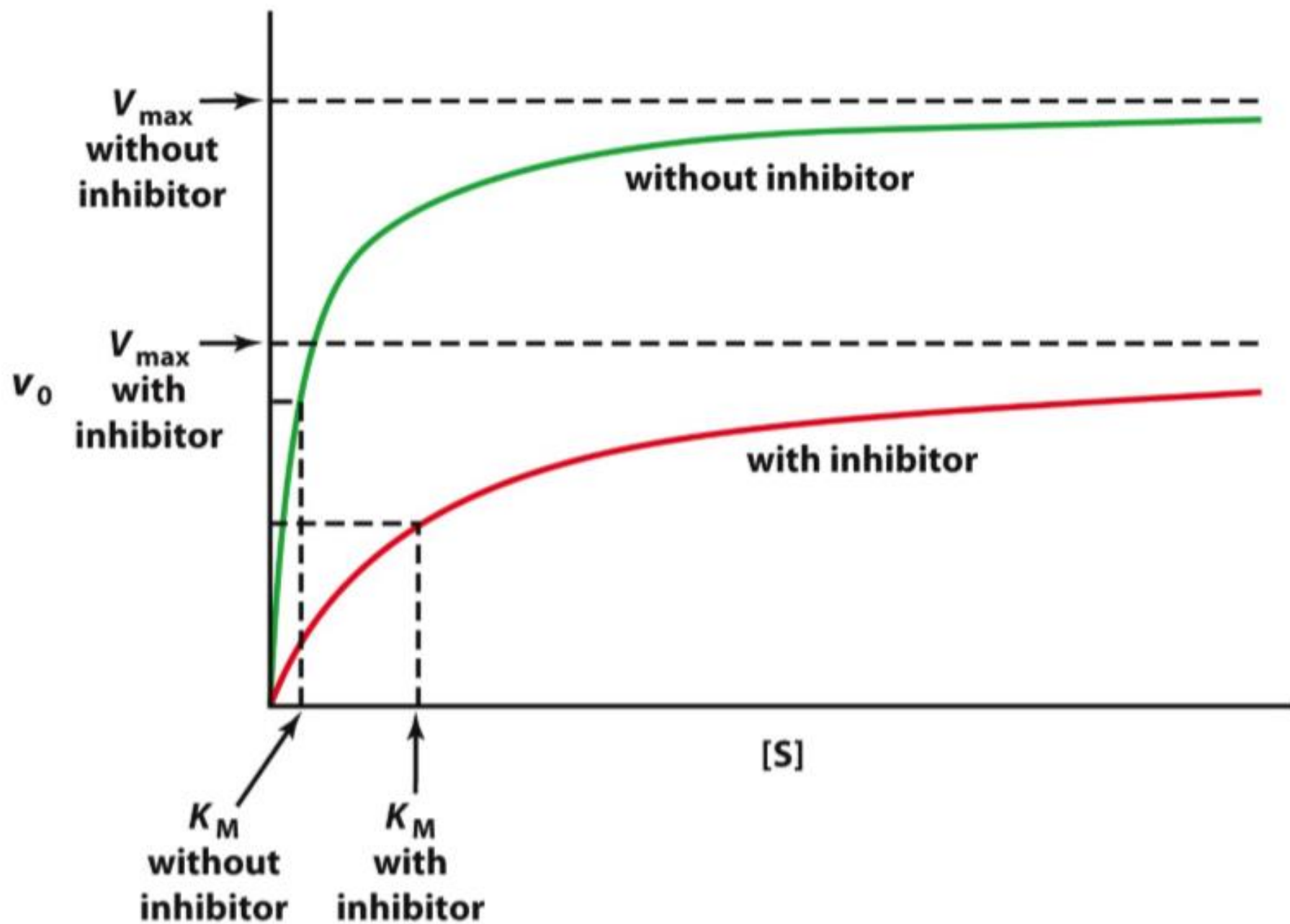
But $K_M = \frac{K_2 + K_3}{K_1}$ and $K_3E_T = V_{\max}$

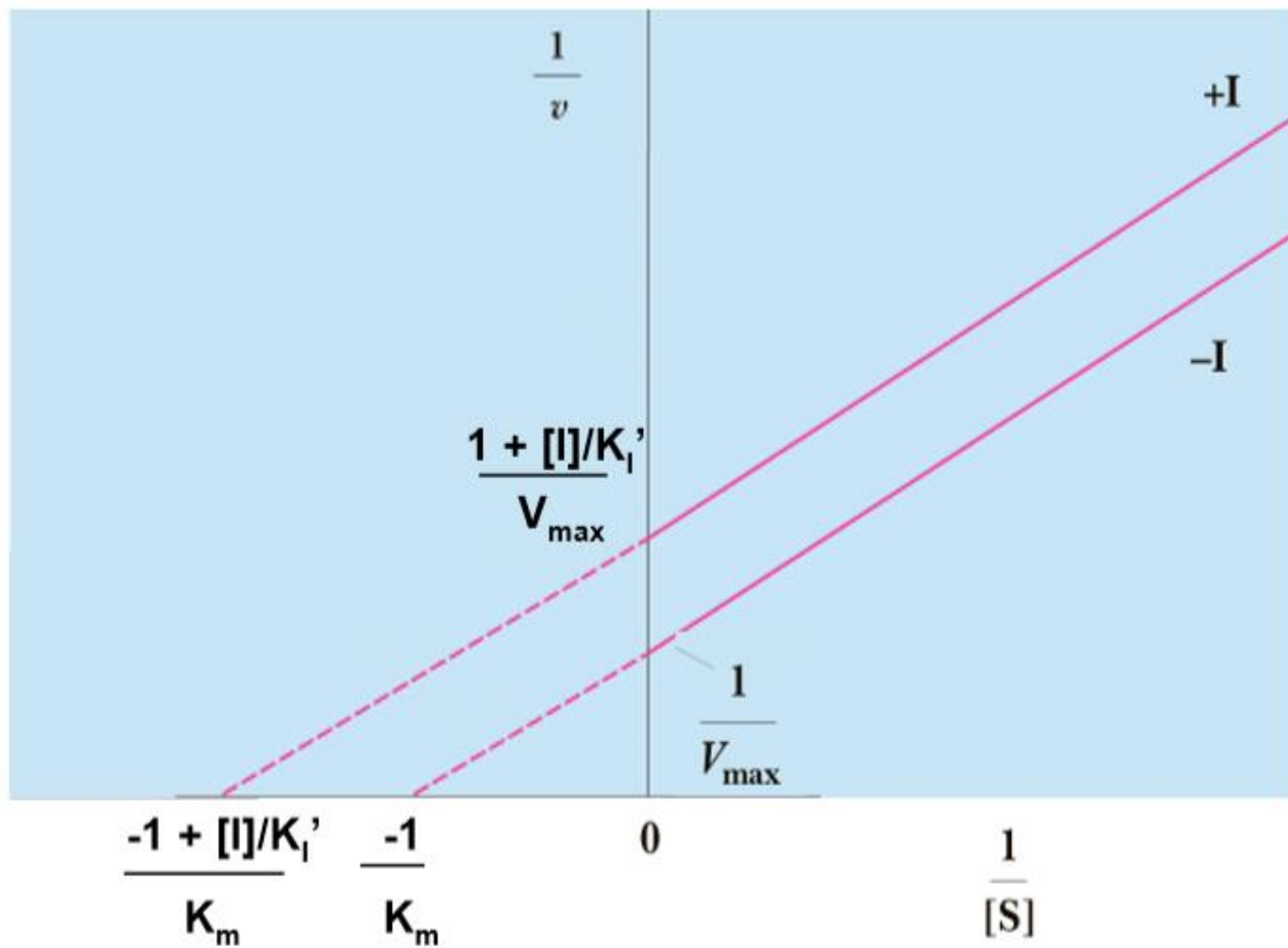
$$V = \frac{V_{\max} [S]}{K_M + [S] \left(1 + \frac{[I]}{K_i}\right)}$$

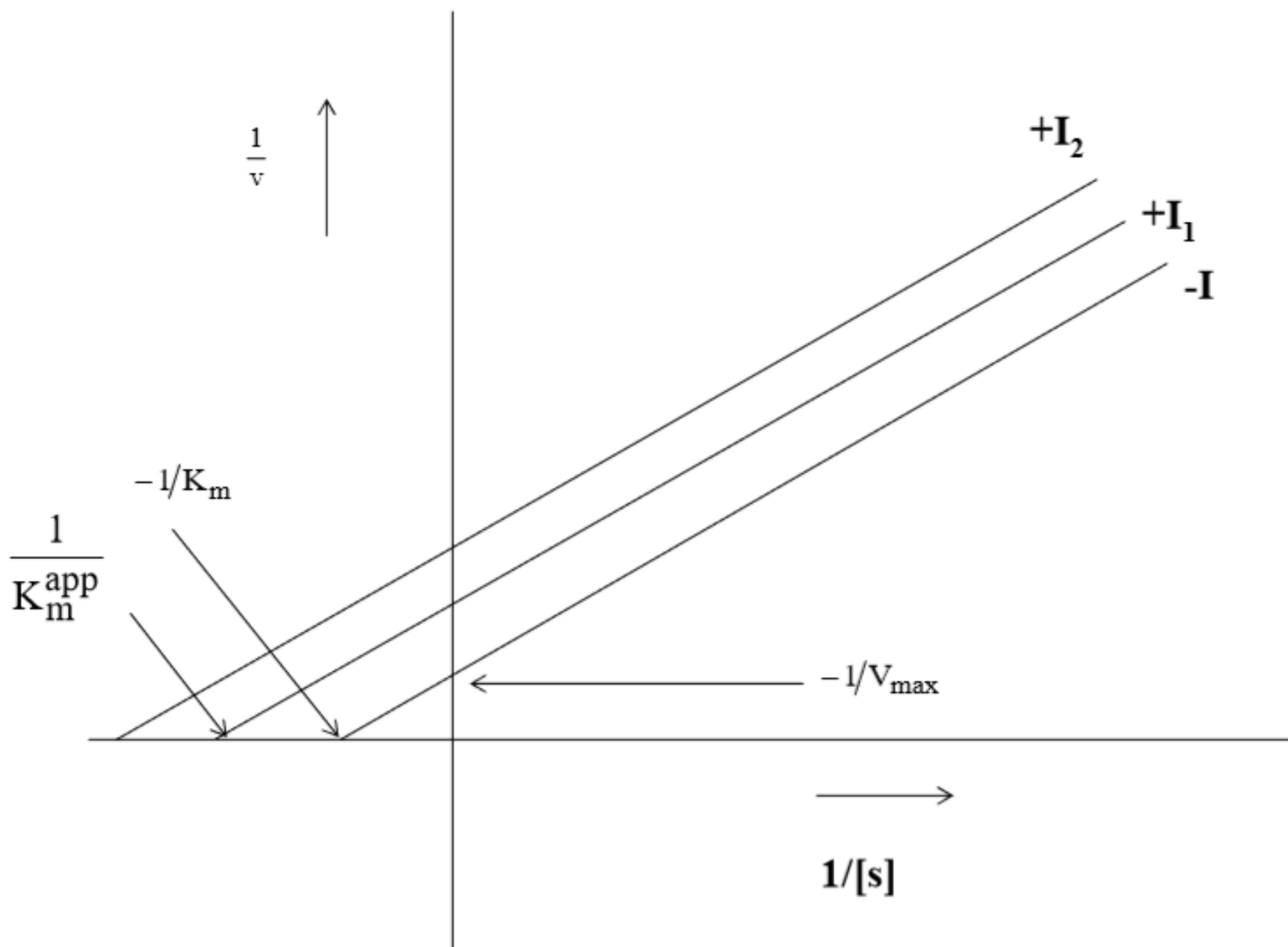
Dividing both the numerator and denominator by $\left(1 + \frac{[I]}{K_i}\right)$

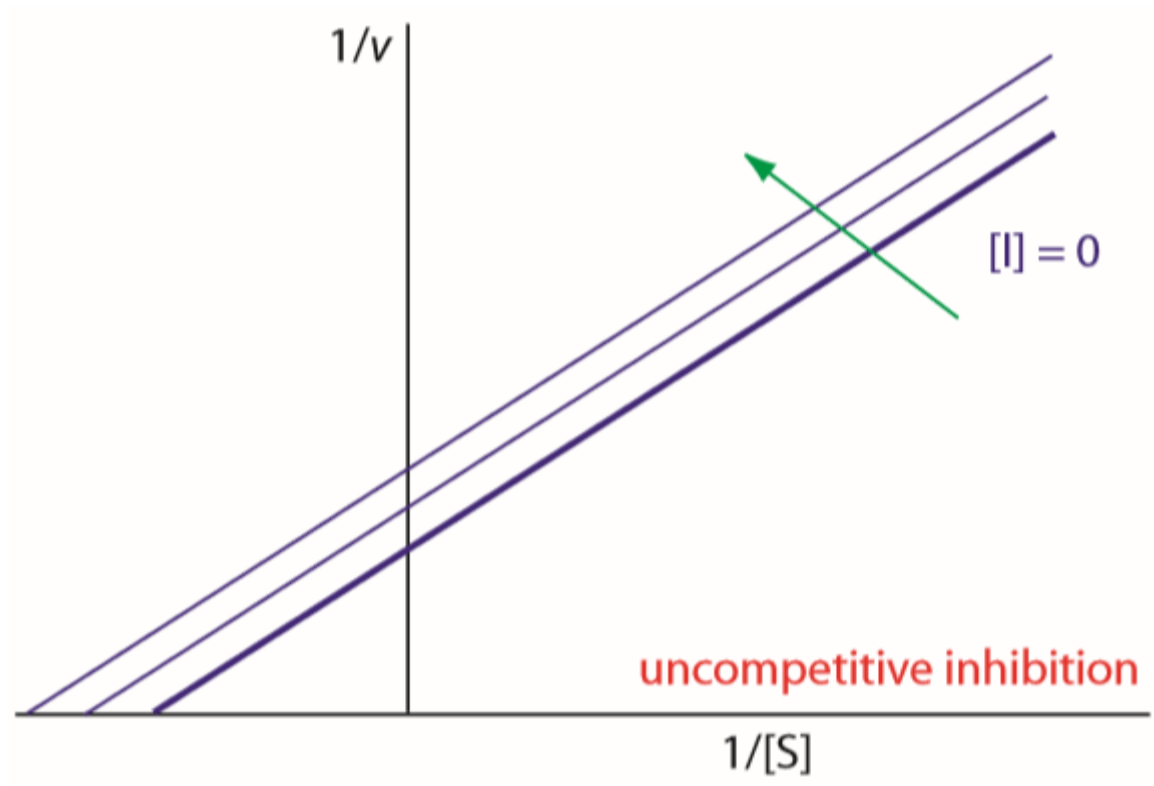
$$v = \frac{\left(V_{\max} / 1 + \frac{[I]}{K_i}\right) [S]}{\left(K_m / 1 + \frac{[I]}{K_i}\right) + [S]}$$

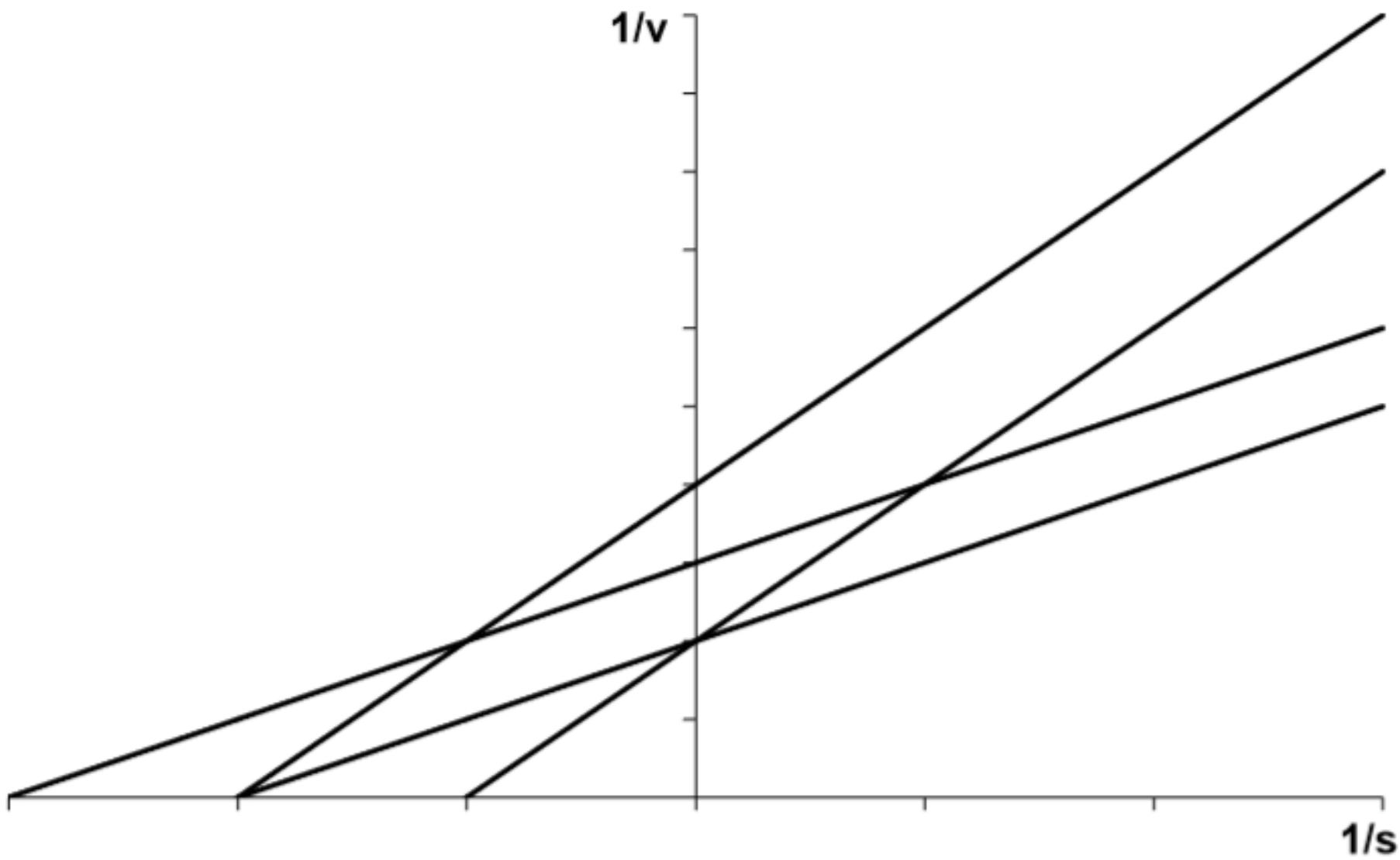
Michaelis-Menten
equation in the
presence of an
uncompetitive
inhibitor











IRREVERSIBLE INHIBITOR

- An irreversible inhibitor binds to the enzyme covalently and cause a permanent loss of catalytic activity.
- An irreversible inhibitor inactivate enzymes by binding covalently to certain R groups of amino acids at the active site of the enzyme.
 - That is this type of enzyme inhibition results in covalent modification of side chain amino acids in the active site of the enzyme.
 - Enzyme activity is destroyed when the inhibitor covalently binds to the active site.
 - The inhibitor changes the shape of the enzyme and this prevents the substrate from binding to the active site.

- It is believe that the inhibitor chemically resembles the substrate and binds in the active site in the same way that the substrate binds but forms an irreversible covalent enzyme-inhibitor complex.
 - This irreversibly inactivate the enzyme.
- The activity of the enzyme is lost and the inhibition is not relieved at high substrate concentration.
 - Inhibitor cannot be removed by dialysis.
- Inhibition increases with time and complete inactivation result if the concentration of the inhibitor exceeds that of the enzyme.

- Michaelis-Menten kinetics are not applied to irreversible inhibition because the inhibitor forms a covalent linkage with the enzyme molecule that cannot be removed.
- Examples of irreversible inhibitors
- CN inhibits xanthine oxidase.
- Nerve gas inhibits cholinesterase.
- Iodoacetamide, heavy metal ions (Hg^{++}), oxidising agents.

- **APPLICATIONS OF ENZYME INHIBITORS**

- Enzymes are normally intracellular and low concentration in blood.
- Enzyme release (leakage) in the blood indicates cell damage (cell –death, hypoxia, intracellular toxicity)
- Quantitative measure of cell/tissue damage
- Injury or death of tissues can cause the release of tissue-specific enzymes into the bloodstream.
- Elevated enzyme levels are often indicators of tissue problems, and are used in the diagnosis of diseases.
- Enzyme activities in the body fluids are altered by pathological processes so, its measurement is used for disease investigation.

- Enzyme inhibitors are widely used:
 - As drugs (in pharmacology)
 - As agricultural pesticides and insecticides.
 - In the study of mechanism of enzyme action.
 - In the diagnosis of diseases

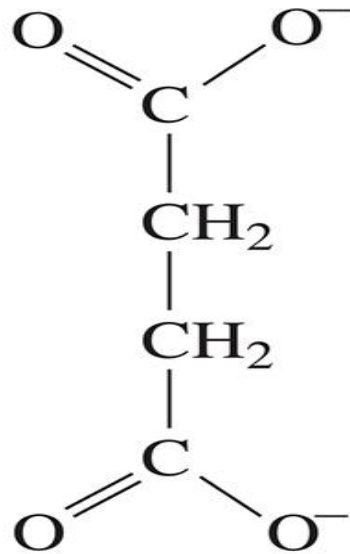
Commonly Used Drugs that are Enzyme Inhibitors

Drug	Therapeutic Use	Target Enzyme	Type of Inhibitor
Mevinolin	Hypercholesterolemia	HMG-CoA reductase	Competitive
5-Fluorouracil	Cancer	Thymidylatesynthase	Suicide
Methotrexate	Cancer	Dihydrofolate reductase	Competitive
Allopurinol	Gout	Xanthineoxidase	Suicide
Coumadin	Anticoagulant	γ -Glutamylcarboxylase	Competitive
Aspirin	anti-inflammatory	Cyclooxygenase	Suicide
Captopril	High blood pressure	Angiotensin-converting enzyme	Competitive

Sulphanilamide

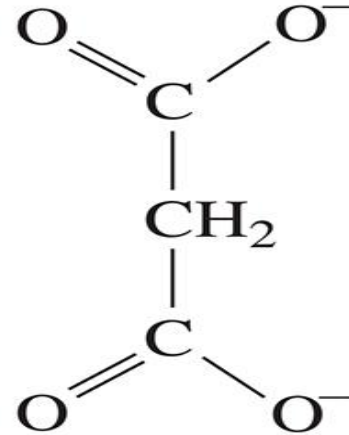
- One of the best-known examples of enzyme inhibitors used as drugs is provided by sulphonamide drugs.
- The synthesis of folic acid from p-aminobenzoic acid (PABA) is a vital metabolic process in bacteria and is controlled by these drugs.
- PABA is an essential metabolite in the growth cycle of bacteria.
- The similarity between PABA and sulphanilamide is obvious that sulphanilamide is used as a competitive inhibitor of Dihydropteroate synthetase, the enzyme which forms folic acid from PABA.
- This is needed for bacterial growth.

Substrate



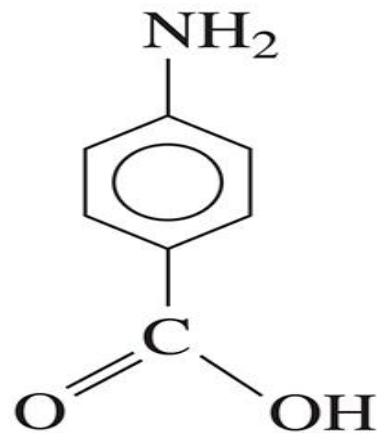
Succinate

Inhibitor



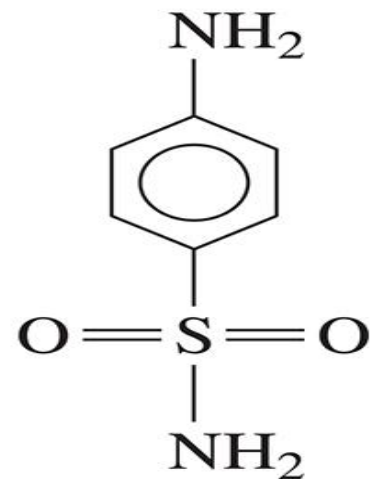
Malonate

Substrate



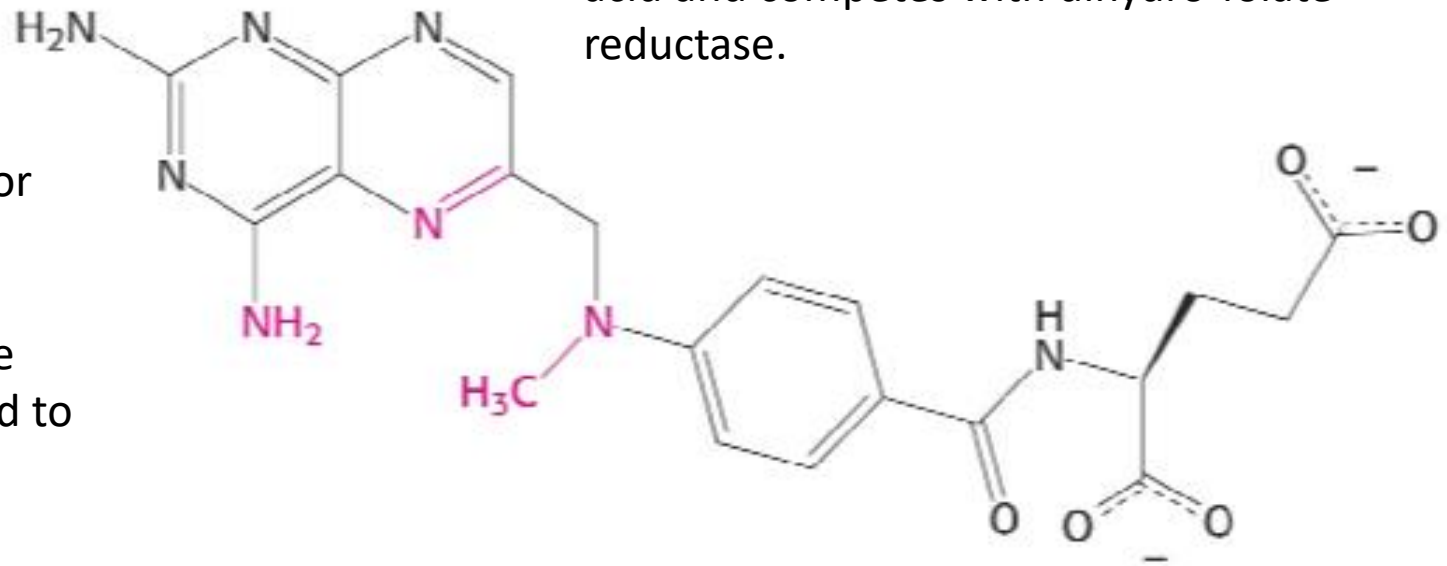
p-Aminobenzoic acid (PABA)

Inhibitor

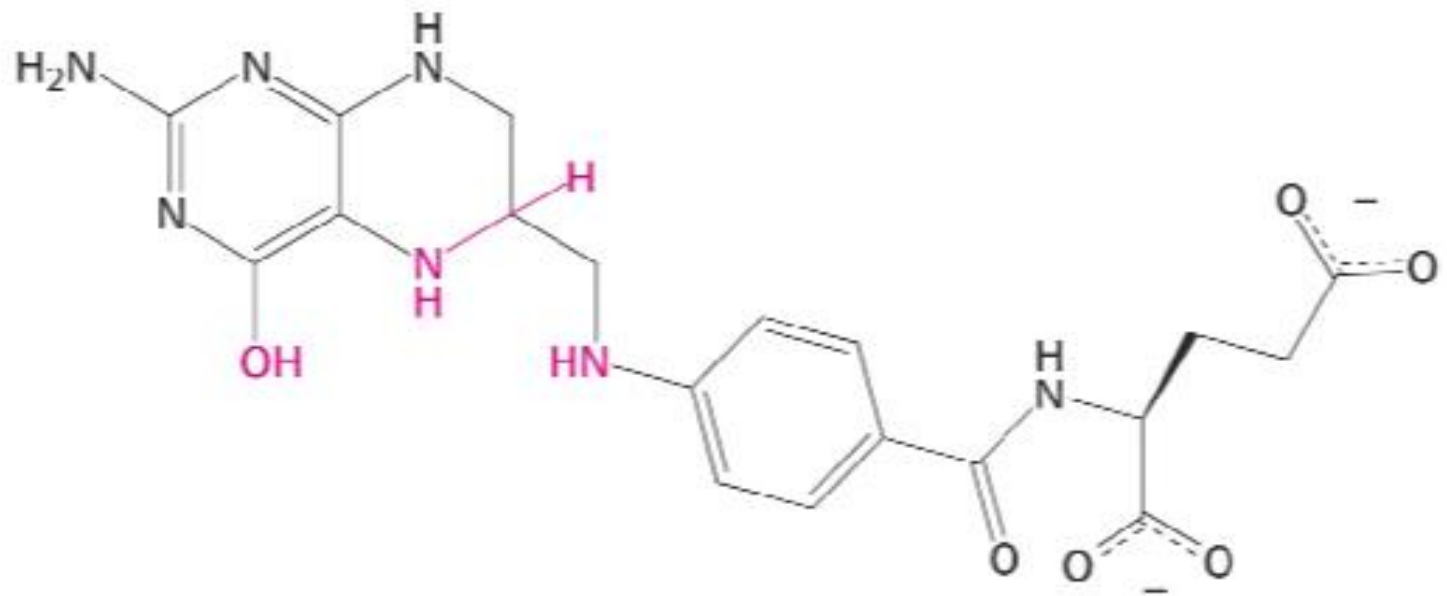


Sulfanilamide

Methotrexate is the structural analog of folic acid and competes with dihydro-folate reductase.



Methotrexate



Tetrahydrofolate

Methotrexate-A competitive inhibitor of dihydrofolate reductase -role in purine & pyrimidine biosynthesis, Is used to treat cancer

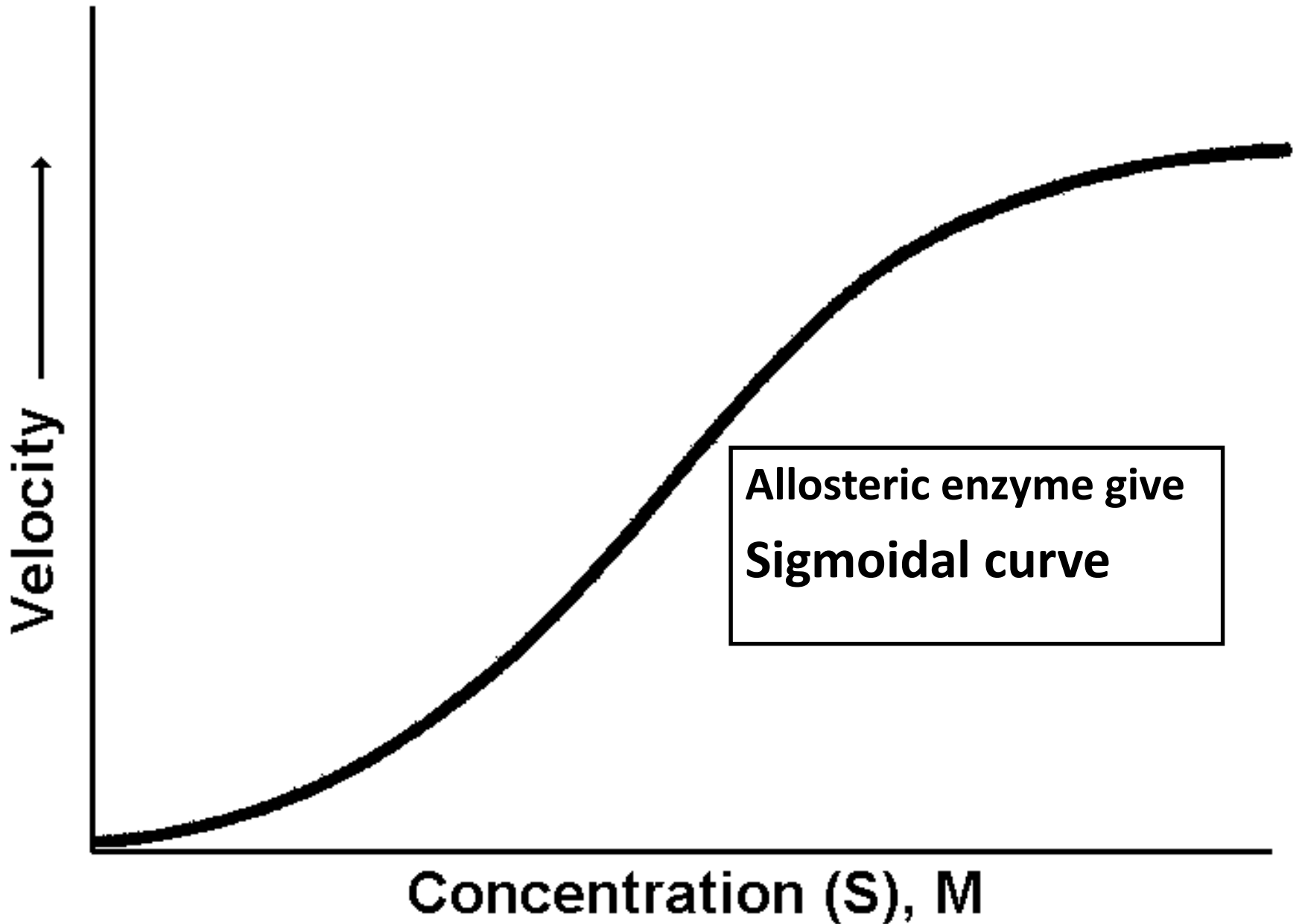
Tetrahydrofolate-Substrate for dihydrofolate reductase

Allosteric enzymes

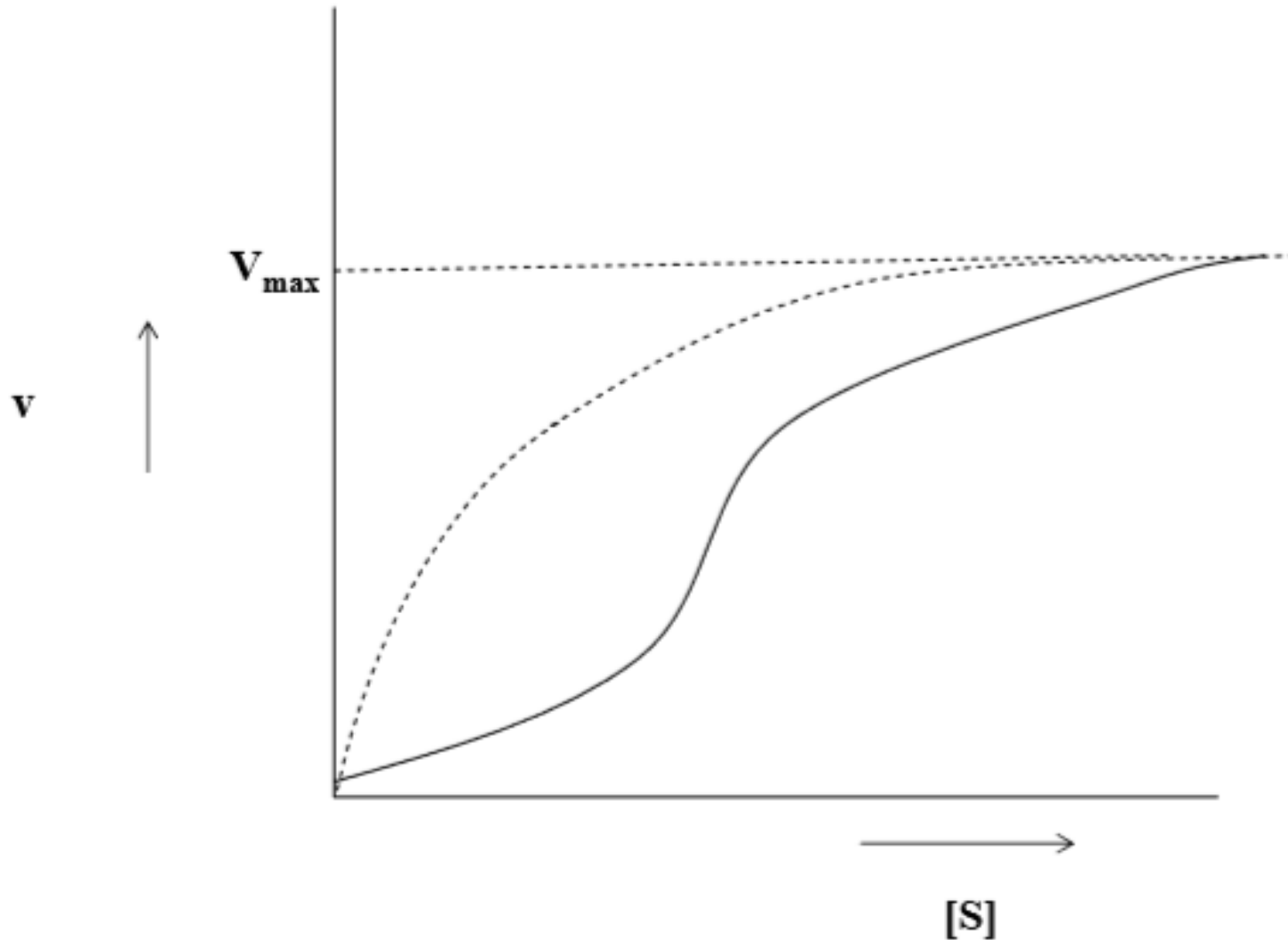
- Allosteric enzymes are oligomeric (consists of multiple subunits) enzymes that change their conformation upon binding of an effector (activator or inhibitor), which results in an apparent change in substrate binding affinity.
- Allosteric enzymes binding sites interact .
 - That is the binding of an effector to one site affects the binding properties of another site on the allosteric enzyme.
 - Binding to one binding site either increase or decrease the binding affinity of the other sites

- Some subunits are catalytic subunits while some are regulatory subunits.
- Catalytic subunits bind with substrate while regulatory subunits bind the allosteric effectors .
- The subunits of allosteric enzymes exhibit cooperative binding of substrate.
- Allosteric enzymes display a **sigmoidal** dependence of reaction rate on the concentration of their substrates.
- This allows most allosteric enzymes to greatly vary catalytic output in response to small changes in effector concentration.
- Plot of V against $[S]$ is distinctly S-shaped (**sigmoidal**) and it cannot be described using the Michaelis model.

- Catalyse an irreversible reaction
- Inhibited by end product (Feed-back inhibition).
- Can be activated by substrate (homotropic) and other positive modulators (heterotropic).
- Do not obey Michaelis Menten Kinetics
- Give sigmoidal curve with S
- Function as regulatory enzymes
- Do not obey Michaelis Menten Kinetics



Plot of V against $[S]$ gives S-shaped or sigmoidal curve for allosteric enzymes while dotted lines describe Michaelis enzyme which gives a hyperbolic curve.



END
OF
THE
LECTURE